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Downsides of face masks and possible mitigation strategies: a systematic review and meta-analysis

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Title page

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Downsides of face masks and possible mitigation strategies: a systematic review and meta-analysis

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Abstract

Objective

To identify, appraise, and synthesise studies evaluating the downsides of wearing facemasks in any setting. We also discuss potential strategies to mitigate these downsides.

Design

Systematic review and meta-analysis

Data sources

PubMed, Embase, CENTRAL, EuropePMC were searched (inception-18/5/2020), and clinical registries were searched via CENTRAL. We also did a forward-backward citation search of the included studies.

Inclusion criteria

We included randomised controlled trials and observational studies comparing facemask use to any active intervention or to control.

Data extraction and analysis

Two author pairs independently screened articles for inclusion, extracted data and assessed the quality of included studies. The primary outcomes were compliance, discomforts, harms, and adverse events of wearing facemasks.

Results

We screened 5471 articles, including 37 (40 references); 11 were meta-analysed. For mask wear adherence, 47% more people wore facemasks in the facemask group compared to control; adherence was significantly higher (26%) in the surgical/medical mask group than in N95/P2 group. The largest number of studies reported on the discomfort and irritation outcome (20-studies); fewest reported on the misuse of masks, and none reported on mask contamination or risk compensation behaviour. Risk of bias was generally high for blinding of participants and personnel and low for attrition and reporting biases.

Conclusions

There are insufficient data to quantify all of the adverse effects that might reduce the acceptability, adherence, and effectiveness of face masks. New research on facemasks should assess and report

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the harms and downsides. Urgent research is also needed on methods and designs to mitigate the downsides of facemask wearing, particularly the assessment of alternatives such as face shields.

Systematic review registration

Open Science Framework website https://osf.io/sa6kf/ (Timestamp 20-05-2020)

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1 Strengths and Limitations of this study

- This is the first systematic review to investigate the downsides of wearing facemasks.
- We included non-randomised study designs in addition to RCTs, as trials frequently underreported or failed to report harms.
- We reported two deviations from the study protocol, relating to the measurement of effect and the sub-group analysis.

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Introduction

Respiratory viruses are predominantly transmitted by aerosol, droplets and fomites.[1] Facemasks – such as surgical masks, N95 masks, and face shields, and substitutes for surgical masks such as home-made cloth masks – are a physical barrier to aerosol and droplet transmission. During the COVID-19 pandemic, some jurisdictions have implemented policies mandating the use of masks in public places, on public transport or in other crowded environments, to prevent people becoming infected, or infecting others.

While most health organisations mandate the use of facemasks by health workers when caring for patients during a pandemic, recommendations for mask wear in the community vary widely – and include use by all, use only in certain situations (e.g. on public transport, or in crowded places where social distancing is not possible), and no specific recommendations about mask use.

Several trials have evaluated the impact on respiratory infections by use of surgical and N95 masks, which may, at best, modestly reduce acute respiratory infection transmission.[2-4] Population observational studies suggest that masks have a more substantial effect.[5] However, the downsides of mask-wearing were either not considered or not reported in most studies. Most trials have focussed on facemasks protecting the wearer, rather than others in the community, are often low powered and include confounding factors resulting in the current evidence for the efficacy of facemasks being less than adequate.[6]

The current controversies, mixed messaging and debate over the use of face masks to prevent the spread of COVID-19 warrants further investigation into their use. Anecdotal evidence, and some studies, suggest that there may be a variety of downsides arising from mask use, including: discomfort, sense of difficulty breathing, and communication problems particularly for those who use lip reading.[6 7] Our aim is to systematically identify and summarise these downsides, to assist policymakers when formulating mask-wearing policies in public settings. We also discuss potential strategies to mitigate downsides of mask-wearing.

Methods

This systematic review is reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[8] We followed the "2 week systematic review" (2weekSR)

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processes for the review.[9] The review protocol was developed prospectively and uploaded to the Open Science Framework data on 20-05-2020 (see: https://osf.io/sa6kf/). Deviations from the protocol are reported in the Methods section and highlighted as a limitation in the Discussion section.

Inclusion criteria

We included studies of people of any age or gender, in any setting. We included studies of any face covering aimed at reducing virus transmission, including surgical masks, N95 masks, cloth masks (both homemade and commercially available). (See Appendix 1 for a summary of included facemasks and their intended purpose). Studies evaluating the use of masks for non-virus transmission purposes (e.g. valved masks) were excluded (see Appendix 2 for a complete list of excluded facemasks).

We included studies comparing the use of facemask to any active intervention (e.g. another mask, or another intervention such as hand-washing etc.), and studies comparing the use of facemask to control (comprising standard practice i.e. face masks not explicitly provided to study group) in situations where their use was not mandatory.

We included only primary studies – i.e., randomised controlled trials (RCTs) and observational studies of any design. We intentionally included this range of study types to identify appropriate data that corresponded with the listed outcomes. For example, surveys are appropriate for identifying frequency of discomfort, and breathing trials are appropriate for understanding physiological changes. We excluded studies that could not provide a quantitative estimate of the size or frequency of adverse effects such as case reports, case series, as well as qualitative studies, and reviews.

Search strategies

We searched PubMed, Embase, Cochrane CENTRAL, EuropePMC (inception-18/5/2020). The search string was designed for PubMed and translated for use in other databases using the Polyglot Search Translator (Appendix 3).[10] Clinical trial registries were searched via Cochrane CENTRAL, which includes the WHO ICTRP and clinicaltrials.gov. On 22/5/2020, we conducted a backwards and forwards citation analysis in Scopus, on all of the included studies.

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No restrictions by language or publication date were imposed. We included publications that were published in full; abstract only publications were included if they had an accompanying record (e.g. trial registry record, or another public report), with additional information.

Study selection and screening

Four authors (MB&NK, AMS&JC) independently screened the titles and abstracts against the inclusion criteria. One author (JC) retrieved full-text, and four authors (MB&NK, AMS&JC) screened the full-texts. Disagreements were resolved by discussion, or a third author (PG or CDM).

Data extraction

A data extraction form was piloted on three studies. Four authors (MB&NK, AMS&JC) extracted the data. (See Box 1)

Box 1. List of extracted information

- **General information:** study authors, location, study design, duration, aim, setting
- **Participants:** health status, disease (if applicable), sample size, age, gender, smoker status, co-morbidities
- **Intervention and Comparator(s):** number of participants, type of face covering, adjunct interventions, number of face coverings used, duration of use, disposal
- **Outcomes:** definition, measurement instrument, number of adverse events or harms reported (The outcomes were discomforts, harms, and adverse events of wearing face masks, Adherence to facemask wearing, Misuse of masks, Discomfort and other physical irritation from masks, Psychological outcomes (e.g. fear, etc.), Dyspnoea (difficulty breathing, shortness of breath) and other physiological impacts, Communication impacts, and mask contamination).

Assessment of risk of bias in included studies

Four authors (MB&NK, AMS&JC) independently assessed the risk of bias using the Cochrane Risk of Bias Tool.[11] Disagreements were resolved by discussion or a third author (CDM or PG). Each potential source of bias was graded as low, high or unclear, and judgements were supported by a quote from the study.

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Measurement of effect and data synthesis

Where feasible (≥2 studies reporting the same outcome), we expressed outcome measures as odds ratios (OR) with 95% confidence intervals, analysed using Review Manager (RevMan 5). Anticipating considerable heterogeneity among the included studies, we used a random effects model. We reported the adherence to facemask wear using risk difference rather than odds ratio, to more clearly convey the differences in adherence between the intervention and control group (not pre-specified in the protocol).

When meta-analysis was not possible or appropriate, we followed the guidance of the Cochrane Collaboration (Cochrane Handbook Section 12.2).[11] When narrative synthesis was required, we reported the results separately for each harm or adverse outcome.

The individual was used as the unit of analysis, where possible; otherwise, we extracted the information as it was presented, e.g. the number of harms in each group. We attempted to contact investigators or study sponsors to provide missing data.

We used the I^2 statistic to measure heterogeneity. Because we included fewer than 10 trials, we did not create a funnel plot.

We did not pre-specify subgroup or sensitivity analyses. However, as data was available, we conducted a subgroup analysis of adherence to mask-wearing by studies which evaluated facemask wear alone, and those evaluating facemask together with handwashing.

Patient and public involvement

Patients and the public were not involved in the design and conduct of this review.

Results

Database searches identified 4,691 publications, supplemented with 2,035 references from forward and backward citation searches and other sources, totalling 6,726. After de-duplication, 5,471 references were screened by title and abstract; we full-text screened 214 references, excluding 174 (see Appendix 4 for a list of excluded studies with reasons). We included 40 articles [12-51] corresponding to 37 studies (Table 1 reports the characteristics of included studies), and meta-analysed 11 studies.[15 21 22 26 31-34 41 45 47] (Figure 1).

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Table 1. Characteristics of included studies

First author (country, year)	Study design	Setting	Study duration	Type of participants				Number of participants	Type of Face Mask intervention**				Type of reported outcomes (adverse effects) analysed						
				Healthy		Sick*													
				A	Ch	A	Ch		Surgical	N95	Any	Other	Discomfort and irritation	Dyspnoea & other physiological	Misuse	Adherence	Psychological impacts	Communication impacts	Mask contamination
Aiello (USA, 2012)	C-RCT	RH	3M	✓				1188; 37 RH	✓				✓			✓			
Allison (USA, 2010)	Single arm study	Sch	1M	✓	✓			20 A, 503 Ch	✓							✓			
Baig (USA, 2010)	S	H	1M	✓				149[1]		✓			✓					✓	
Barasheed (Australia, 2014)	C-RCT	Tents	1W	✓	✓	✓		164; 22 Tents	✓							✓			
Bryce (Canada, 2008)	S	H	1W S	✓				137		✓			✓			✓			
Canini (France, 2010)	C-RCT	HH	1Inf-S	✓	✓		✓	306; 105 HH	✓				✓			✓	✓		
Chen (China, 2016)	Multiple crossover lab-based trials	Lab	NR	✓				15		✓				✓					
Chugtai (China, 2019)[1]	Prevalence	H	2M	✓				148	✓				✓			✓		✓	✓
Cowling (Hong Kong, 2008)	C-RCT	HH	7M	✓	✓	✓	✓	370; 128 HH[3]	✓							✓			
Cowling (Hong Kong, 2009)	C-RCT	HH	8M	✓	✓	✓	✓	1015; 322 HH[3]	✓							✓			
Foo (Singapore 2006)	S	H	2S	✓				340			✓		✓						
Forgie (Canada, 2009)[4]	S	ED	2M	✓	✓			80[5]	✓			✓					✓		
Jacobs (Japan, 2009)	RCT	H	2M	✓				33	✓							✓			
Kao (Taiwan, 2004)[6]	Before-after	H	2M	✓				39		✓			✓						
Larson (USA, 2010) with	C-RCT	HH	19M	✓	✓			2708; 617 HH[3]	✓				✓			✓	✓	✓	

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additional data from (Ferng et al. 2011)																			
Lee (Singapore, 2011)[7]	Single arm study	Lab	NR	✓				14		✓				✓					
Lim (Singapore, 2006)	S	H	1Y	✓				212		✓			✓						
MacIntyre (Australia 2009)	C-RCT	HH	2WS	✓			✓	290 A; 145 HH	✓			✓	✓			✓	✓		
MacIntyre (China, 2011)	C-RCT	H	3M	✓				1441; 15 H	✓	✓			✓			✓		✓	
MacIntyre (China, 2013)	C-RCT	H	3M	✓				1669; 19 H	✓	✓			✓			✓			
MacIntyre (Vietnam, 2015) with additional data from (Chughtai et al. 2016)	C-RCT	H	5W	✓				1607; 14 H	✓			✓	✓			✓			
Martel (Canada, 2013)[8]	Direct observation study	ED	1M	✓				115			✓		✓		✓			✓	
Nickell (Canada, 2004)	S	H	1M	✓				2001	✓	✓			✓				✓	✓	
Ong (Singapore, 2020)	S	H	2M	✓				158	✓				✓						
Or (Hong Kong, 2018)	Lab-based study	Lab	NR	✓				84		✓			✓					✓	
Person (France, 2018)	Randomised crossover	NC[4]	NR	✓				44	✓					✓					
Radonovich (USA, 2009)	Multiple crossover Trial	H	NC	✓				27	✓	✓		✓	✓					✓	
Radonovich (USA, 2019)	C-RCT	H	3.7Y	✓				2862	✓	✓						✓			
Rebmann (USA, 2013)	Multiple Crossover Trial	H	2D	✓				10		✓					✓				
Roberge (USA, 2012)	Crossover	Lab	3M	✓				20	✓					✓					
Shenal (USA, 2012)	Multiple crossover field trial	H	NR	✓				27	✓			✓	✓						
Simmerman (Thailand, 2011)	RCT	HH	16M	✓			✓	465	✓							✓			
Suess (Germany, 2012) with additional data from (Suess et al. 2011)	C-RCT	HH	2Inf-S	✓			✓	111 HH	✓				✓			✓			

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Thomas (USA, 2011)	NC[9]	HB-E	NR	✓				3	✓	✓								✓	
Vanjak (France, 2006)	S	H	2M	✓				238 patients & 210 staff			✓				✓				
Wong (Hong Kong, 2013)	RCT	PCC	2M	✓				1031 patients & 9 doctors			✓						✓		
Yeung (Singapore, 2020)	S	HH	1W	✓				2231 HH		✓					✓				

C-RCT: cluster randomized controlled trial, **RCT**: randomized controlled trial, **S:** Survey, **Sch:** Schools, **RH:** residence halls, **HH:** Households, **H:** Hospitals, **Tel:** Telephone survey of home residents, **Lab:** Laboratory, **ED:** Emergency departments, **HB-E:** Helicopter Bell (used for helicopter emergency medical services), **PCC:** Primary care clinics, **N/A:** Not applicable, **NR:** Not reported, **NC:** Not clear, **Y:** Year, **M:** Months, **W:** Weeks, **D:** Days, **Inf-S:** Influenza season, **WS:** Winter season, **A:** Healthy Adults, **Ch:** Children.

*Sick adults or children, with influenza, influenza-like illness, or upper respiratory tract infections. **Only Face masks arms of the included studies. [1] Authors state 159 surveys returned but 10 were excluded from analysis as ineligible. [2] reports 3 studies: pilot study 1 lab testing; pilot study 2 (small clinical testing); study 3 main study which is extracted here. [3] number of households that were allocated. [4] Participants in the waiting room of ED, could be sick due to an infection. [5] Pairs of adults and children. [6] Patients with End stage renal disease without any influenza or ILI. [7] Data reported here is for the N95 group. [8] 2-part study: part 1) direct observation study [data in this row] + part 2) survey, 4 "in a covered, flat, straight corridor, with a length of 30 meters and marked every 3 meters". [9] In the first part of the study, the pilot recited each word with the aircraft engine off when wearing and not wearing either the surgical facemask or one of the N95 respirators. In the second part of the study, the same procedure occurred, except that the 407-helicopter engine was running at idle.

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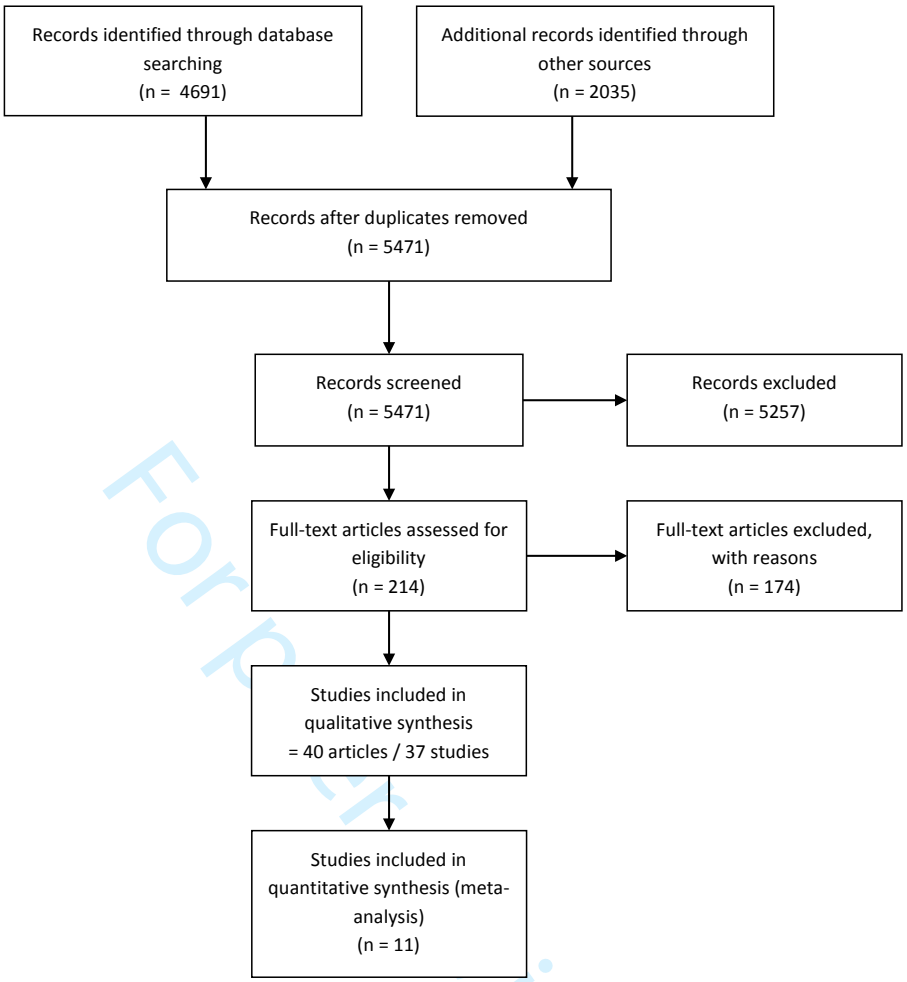


Figure 1. PRISMA flow diagram

Risk of bias

Inclusion of both observational studies and RCTs could explain the high risk of bias across the included studies. Reporting of sequence generation, allocation concealment and blinding of outcome assessment was poor in 20% - 30% of studies. Due to the nature of interventions, blinding of participants was rare. For RCTs, we reported blinding of outcome assessment for the main trial (not the adverse events) – as some outcomes were lab-confirmed and were considered of low risk. We found no evidence of incomplete outcome data or selective reporting of outcomes. Funding statement, funder's role and authors' conflict of interests were adequately reported in most studies (see figure 2, and figure 3 in Appendix 5).

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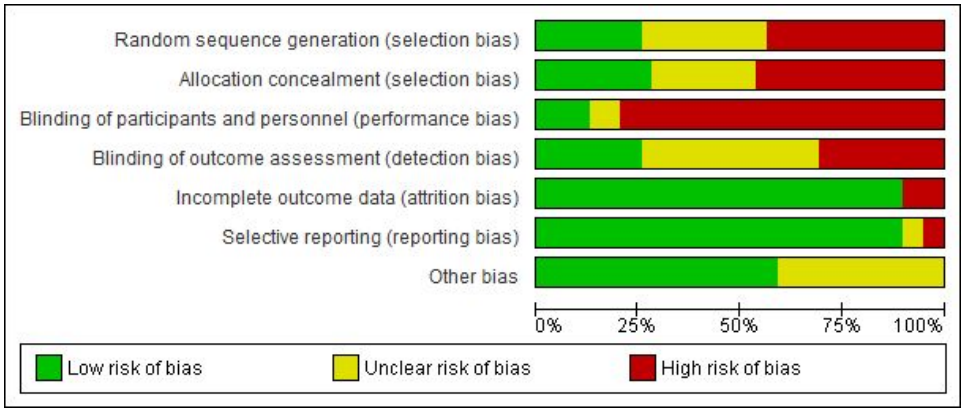


Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

Adherence to facemask wearing

17 studies (14 RCTs, 3 observational [10,13,17]) reported on adherence to facemask wearing; of these 11 had sufficient data to pool and were meta-analysed.

Face mask versus control comparison

Comparison of face masks to control was sub-grouped into studies comparing facemask alone vs control; and studies of facemask plus handwashing vs control. (Figure 3)

Facemasks alone vs control (5 studies) showed the facemask group had a significantly higher facemask wear compared to control (Risk Difference (RD): 0. 46, 95% CI 0. 24 to 0. 67, P < 0.0001). Studies evaluating facemask plus handwashing vs control (n = 3), similarly showed significantly higher facemask wear in the facemask group (RD: 0. 47 (95% CI 0. 07 to 0. 88, P < 0. 0001). Overall, 7 studies (3303 participants) compared facemasks to control. Facemask wear was 47% higher in the facemask group, although heterogeneity was very high (RD: 0. 47, 95% CI 0. 25 to 0. 68, P < 0. 0001, I^2 = 98%.)

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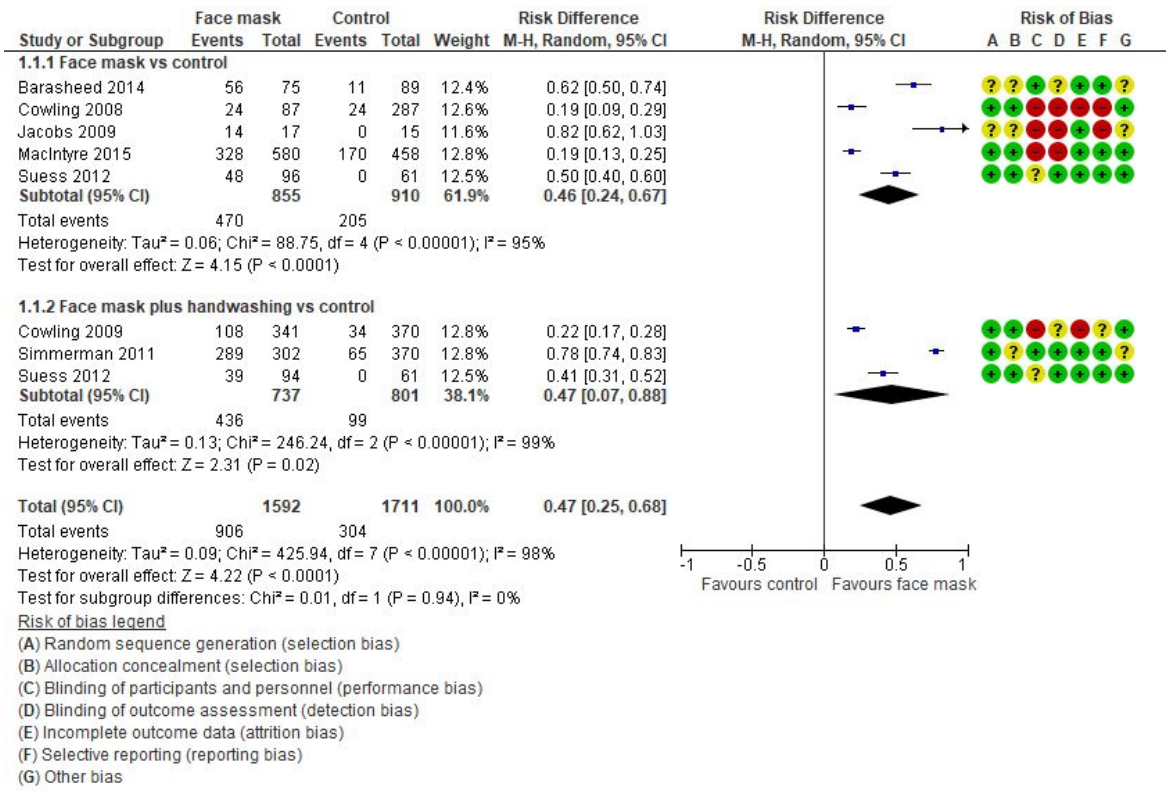


Figure 3. Comparison of adherence to face masks vs control

We explored the possible sources of heterogeneity. Excluding studies with 3 or more domains at high risk of bias did not decrease heterogeneity (I^2 = 96% for facemask vs control; 99% for facemask plus handwashing versus control) (Figure 4, Appendix 5). We excluded study population as the source of heterogeneity, because subgrouping studies by those in a community/household settings (which included both index cases and their contacts) versus those in a hospital setting (which included healthy healthcare workers) likewise did not decrease heterogeneity (I^2 = 99% for community/household studies, and 97% for hospital studies) (Figure 5, Appendix 5). We excluded intervention and control as sources of heterogeneity, since all studies compared medical/surgical masks to control (no mask) – although some mask-wear did occur in the control groups.

We consider the outcome measurement to be the most likely other source of heterogeneity. All studies relied on self-reporting of the outcome; only one verified this by counting the number of masks used.[21] What was considered 'wearing a facemask' varied: it was unclear,[15 21 45] involved wearing a facemask "always or mostly/often",[22 47] included wearing a facemask whilst on hospital property, performing a healthcare worker role,[26] and included facemask wear during a shift for 70% of time or more.[32] The follow-up was very short (5 - 9 days) for 4 studies;[15 21 22 47] it was longer for 3 studies (21 - 77 days).[26 32 45]

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Facemask (surgical/medical) vs facemask (N92/P2 mask)

Four studies (7960 participants) compared adherence for different facemasks.[31 33 34 41] Facemask wear was significantly higher in the surgical/medical facemask group than in the N95/P2 group, OR 1.26 (95% CI 1. 08 to 1. 46, P < 0. 01). Heterogeneity was very low (I^2 = 27%). (Figure 4)

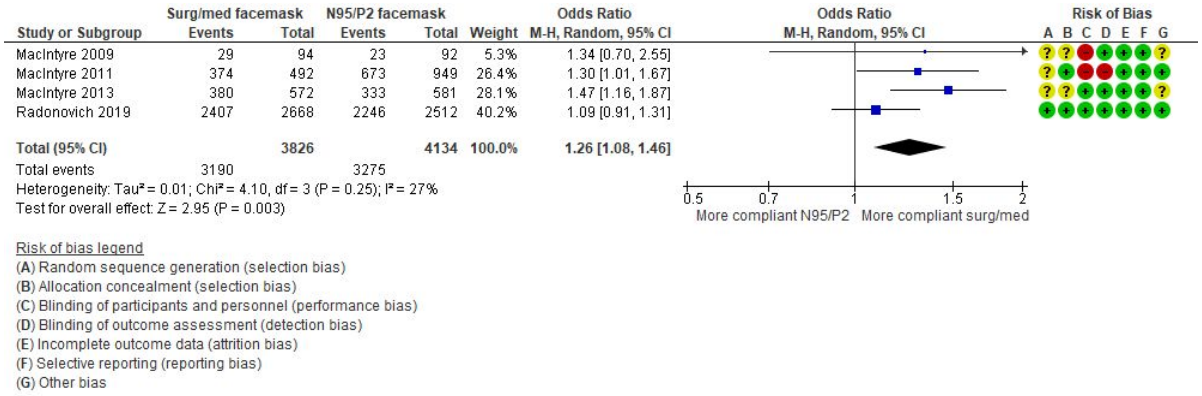


Figure 4. Comparison of adherence to surgical/medical face masks vs N95/P2 masks

Studies not included in the meta-analysis

RCTs (n = 3)

One study in a residential hall setting reported similar duration of facemask wear per day in the facemask alone group vs facemask plus handwashing group (mean of 5.08 and 5.04 hours/day, respectively).[12] Another in a household setting reported that within the facemask group, there were no significant differences between individuals with ILI among contacts vs no ILI among contacts, for facemask use.[17] Finally, 22 of 44 households randomised to the 'education with sanitiser and facemasks' arm, reported having used a mask within 48 hours of episode onset.[28]

Observational studies (n = 3)

In elementary school setting, approximately twice as many teachers as students wore face masks.[13] A mean compliance score with N95 use guidelines was 21.2 (on a 25-point scale) among frequent users of N95 respirators in a hospital setting.[16] Another study among healthcare workers found that majority of survey respondents (91%) wore 1-2 masks per day (range 1 - 4).[20]

Misuse

Mask misuse appears less studied than other harms and discomforts. A study of 10 nurses observed for 10 minutes/hour over 2 shifts found that they touched their face 2-3 times per hour, their mask 5 times per hour, and their eyes once per 2 hours, when observed by students.[42] In a study of health workers, 13 of the 53 who responded (25%), reported wearing masks only covering their mouth, not

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their nose.[49] A study in two hospitals,[35] observed triage nurse behaviour with 118 patients with fever and cough, found that in only 18% of cases the nurses informed patients of the need to wear mask, and in half of those, gave instruction on the need to cover both mouth and nose. A cross-sectional study evaluating the proficiency of the Singaporean public in wearing N95 masks found only 90/714 subjects passed the visual mask fit test; the most common criteria performed incorrectly were: strap placement, leaving a visible gap between the mask and skin, and tightening the nose-clip.[51]

Discomfort and irritation

Several RCTs of specifically measured mask wear discomfort,[17 31-33 47] but most only recorded spontaneously reported events [12 28 34] or did not report any.[15 21 22 26 41 45 50] A trial of household index influenza cases allocated to wear masks or no mask, found the 51 allocated to masks wore them on average 3.8 hours/day and 38 (76%) reported discomfort (Table 2).[17] A study of health care workers in Beijing asked to wear masks for their full shift, found 84% complained of at least one problem (Table 2).[20] In a German household study, 65/172 participants reported problems with mask wearing – most commonly warmth, pain and shortness of breath.[47]

In a trial of healthcare workers comparing surgical and N95 masks to prevent influenza, more workers found the N95 uncomfortable (42%) than the medical mask (10%) when worn an average of 5 hours per day,[33] with significant differences in headaches, difficulty breathing, and pressure on the nose (Table 2). A trial of cloth versus medical masks in healthcare workers found similar rates of discomfort.[32] A community trial comparing surgical and P2 (N95) masks found >50% reporting concerns, primarily discomfort, with similar rates (15% vs. 17%) across groups.[31]

Discomfort increases with duration of mask wearing. A crossover field trial of 27 healthcare workers found increased discomfort over time; half the subjects were unwilling to wear a medical mask for the full 8-hour shift despite regularly wearing them for short periods.[44]

Table 2. Types of discomfort assessed in trials of face masks used to prevent viral transmission.

First author, year (Type of mask)	Population, Number	Difficulty Breathing	Facial irritation or discomfort	Headache	Other
Canini, 2010[17] (surgical masks)	Household, 105 index cases	34%	14%	-	46% warmth
Chughtai 2019[20] (surgical masks)	Health care workers, 148	12%	17%	6%	-

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MacIntyre 2011[33] (Surgical)	Health Care workers, 492	12%	11%	4%	-
MacIntyre 2011[33] (N95)	Health Care workers, 949	19%	52%	13%	-
MacIntyre, 2015[32] (Cloth v Medical)	Health Care workers, 1130	18%	35%	-	-

Two surveys of healthcare staff in Singapore during the SARS epidemic assessed headache and skin reactions.[24 30] In one, 79/212 (37%) reported face-mask-associated headaches, 26 (33%) reported headache frequency exceeding six times/month, and 6 had taken sick leave.[30] Another survey of healthcare workers in Singapore, found that of the 307 staff who used masks regularly, 60% reported acne, 51% facial itch, and 36% rash from N95 mask use.[24] A COVID-19 survey of healthcare workers in Singapore found that 128/158 (81%) developed *de novo* PPE-associated headaches, increasing with duration of use (> 4 hours).[37]

One study (2x2 factorial design) examined the potential of mask fit test and training to mitigate discomfort, but found no clinically or statistically important differences between arms.[38]

Six observational studies reported either general discomfort[33] or spontaneously reported events among participants who wore facemasks.[14 16 27 35 40]

Psychological

6 studies reported on psychological impacts from wearing face masks (4 RCTs, 2 observational).

Fear

A 3-arm RCT in a household setting found significantly higher risk perception scores in the mask group (38/60) than non-mask groups (30/60) ($p < 0.001$); participants in the mask group were more fearful that they and their family would get sick from the flu.[23 28]

In an observational study, children in a paediatric emergency department waiting room (n = 80) were shown pictures of clinicians wearing either a surgical mask or a clear face shield; 18 children (22.5%) reported surgical masks to be more frightening due to an inability to see clinicians faces, and 14 children (17.5%) reported face shields to be more frightening. However, 47 children (59%) reported that neither were frightening.[25]

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Stigma

In a 2-arm cluster-RCT, 15 (29%) patients wearing masks reported they did not like being seen wearing a mask.[17] In a 3-arm RCT, more children reported disliking their parents wearing a P2 mask than a surgical mask (8/92 vs 6/94), however the difference was not significant.[31]

Loneliness

One observational study reported on the loneliness outcome. In a survey investigating the psychosocial effects associated with working in a hospital during the SARS outbreak, 222 (13%) respondents reported a sense of isolation as one reason masks were perceived as bothersome.[36]

Empathy

One RCT reported that the wearing of a face mask by doctors had a negative effect on patient perceptions of the doctors' empathy during consultations, with a mean Consultation and Relational Empathy (CARE) score in the mask group of 33.93 (SD = 7.65, n = 514), and 34.91 (SD = 7.84, n = 516) in the no mask group (p = 0.04).[50]

Dyspnoea and other physiological consequences

Studies of physiological impacts were generally done on masks designed for dust, vapours, and other non-transmission purposes; few studied surgical or N95 masks.

A French crossover study (44 subjects) found surgical masks had no impact on 6-minute walking time, but subjects had an increased sense of dyspneal with a mask: 5.6 vs 4.6cm on a 10cm visual analogue scale (p<0.001),[39] which may come from the increased effort required. A study in 14 adults found that N95 masks increased respiratory resistance in 30 seconds of breathing by over 100%, resulting in average reduction in nasal spirometry of 37%.[29] A study of 20 subjects on a treadmill found the surgical masks increased respiratory rate by 1.6 breaths/minute (p = 0.02), heart rate by 9.5 beats/minute (p<0.001), and transcutaneous CO2 levels of 2.2 mmHg (p<0.001).[43] Finally, a Chinese study of 15 subjects wearing a monitoring garment for respiratory signals found that N95 masks increased both subjective breathing resistance (from none to mild) and increased respiratory rate, the muscle activity of sternomastoid, scalene, diaphragm and abdominal, the fatigue of scalene and intercostal.[18]

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Communication

Nine studies (2 RCT, 7 observational) reported on communication difficulties while wearing facemasks.

A trial comparing the use of surgical and N95 masks by healthcare workers to prevent influenza, found more workers in the N95 mask group than the surgical mask group reported mask causing trouble with patient communication (8% vs 3%).[33] Another household-based trial of 15 participants who wore a surgical mask for approximately 60 minutes while performing various tasks,[23 28] found that participants did not report any interference with communication while answering the phone.

In a cross-over trial of 27 healthcare workers,[40] more participants in the surgical mask group reported diminished communication acuity (visual, auditory or vocal) as the reason for discontinuing mask use before the end of an 8 hour shift (7 complaints compared to 4 complaints among N95 mask wearers).

Of 2001 healthcare workers in Toronto responding to a survey during the SARS outbreak, difficulty communicating (47%), and difficulty recognizing people (24%) were identified as key reasons masks (surgical or N95) were perceived as being particularly bothersome.[36] In a survey of 149 healthcare workers,[14] 41 (27.5%) of respondents reported a difficulty 'always'/'most of time' in verbally communicating with patients while wearing a mask.

In another Canadian survey (115 healthcare workers),[35] 26 (23%) respondents reported that wearing masks interfered with their relationships with their patients. Among 148 healthcare workers asked to wear a mask during a 6-8 hour shift, 11 (7.4%) reported trouble communicating with patients.[20]

In a study of 3 participants evaluating the impact of wearing a surgical or N95 mask on radio reception, all participants were able to accurately record all pilot-recited words regardless of the type of mask worn by the pilot. However, when the aircraft engine was turned on, the accuracy decreased for the N95 mask, compared to surgical or no mask.[48]

In another lab-based study, the performance or absence of fit testing prior to mask use did not affect communication, as 2 participants (out of 21) in each group reported ease of talking to be unsatisfactory.[38]

Mask contamination and other issues

One concern about mask use is the potential for contamination of the mask surface, and subsequent self-inoculation to the wearer's eyes or when demasking. No studies examined that directly, but one study of the health care workers found on average 10% of masks had viral contamination after

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usage, and that was higher for masks worn > 6 hours (OR 7.9) or > 25 patients seen (OR 5.0).[20] Given the rates of misuse (see Misuse section above) this contamination raises concerns about self-inoculation.

Several authors have raised concerns about "risk compensation" – non-adherence to other precautions because of the sense of protection – but we found no studies that quantify its extent.

Discussion

We identified 37 studies reporting downsides, harms and adverse events associated with the wearing of facemasks – 15 RCTs and 22 observational studies. The largest number of studies reported on the discomfort and irritation outcome (20 studies), fewest on misuse of mask (4 studies), with no studies directly investigating or quantifying mask contamination or risk compensation behaviour. The only meta-analysable outcome was adherence to facemask wear (17 studies, 11 meta-analysed). 47% more people wore facemasks in the facemask group compared to control, although the percentage of people wearing facemasks in the control group was non-zero in 5 studies; facemask wear adherence was also significantly higher (26%) in the surgical/medical mask group than the N95/P2 group. Risk of bias was generally high for blinding of participants and personnel, and selection bias, and low for attrition and reporting biases.

This is the first systematic review to investigate the downsides of wearing facemasks and forms an important step as a bridge between research and action. The review aligns with the aims of the BESSI (behavioural, environmental, social and systems interventions) collaboration, and addresses important evidence gaps in the appropriate use of facemasks. The review's strength lies in it's inclusion of non-randomised study designs in addition to RCTs, as trials frequently underreport or fail to report harms.[52] Additionally, the inclusion and exclusion criteria were tested and refined on a test library of 98 references, prior to screening the full search results. The key limitation includes the hospital-setting of most of the included studies: as hospital workers are accustomed to wearing masks, the conclusions may not be fully generalisable to the community. Although this varies among the studies that reported mask use in hospital setting, as there are different confounding factors that may contribute to increased reporting of irritation (e.g. length of shift, air-conditioning on the wards and whether the staff were wearing the full PPE which adds to the full discomfort). We report two differences between the protocol and the review: first, the comparison of facemask to control in the adherence outcome was reported using risk difference (rather than pre-planned odds ratio) to more clearly convey the differences between the two groups (odds ratio for compliance with

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facemask wear was reported for the facemask versus facemask comparison, however). Second, not having anticipated data availability, we did not prespecify a subgroup analysis of the intervention (facemask wearing) by studies which evaluated facemask wear alone, and studies evaluating facemask with handwashing. Potential behavioural or cultural bias may have been introduced into the findings due to some cultural groups already being more accustomed to wearing face masks when coughing or sick. Furthermore, the use of face masks/coverings in the various study settings compared to their use in a real-life pandemic may differ from a behavioural perspective and this may affect findings related to adherence. However, other reported harms, such as physiological changes, may be less affected by the differences in circumstance.

Several recent systematic reviews have focused on the effectiveness of masks in preventing or reducing viral transmission; some of these reviews reported on harms in the included studies.[8 18 42 53] However, none specifically focused on the wider set of studies examining the physiological, psychological and other adverse effects addressed in this review. The Cochrane review on physical barriers noted the impact of masks on discomfort and communication in some of the randomised trials, and its findings are consistent with this review, but did not extend to studies with outcomes other than viral transmission or non-randomised study design.[3]

The downsides identified in this review should aid in designing strategies to mitigate problems, and guide the situations where the benefits of masks might outweigh the downsides. As suggested by the higher adherence to surgical masks than to the N95 masks, mitigation of discomforts may also increase adherence to facemask wear, and hence their effectiveness – whether for preventing transmission of the virus by the wearer (e.g. surgical masks) or for preventing inhalation of viral particles in the environment (e.g. N95 masks). Mitigation might be achieved by considering of the when, where, and how of mask wearing (including the fitting process required for some masks like FFP and N95) or by mask redesign or substitution with alternatives (e.g. face shields).

Potential mitigation strategies

Limiting circumstances: Use of facemasks should be restricted to higher risk circumstances, including crowded, indoor spaces, where physical distancing is not possible, e.g. public transport. This recommendation corresponds with suggestions by Chu et.al. who reported that optimum face mask use in public settings could result in a large reduction in infection.[5] Conversely, exercising outdoors is

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both low risk and has higher downside of wearing masks, because of the increased perceived dyspnoea.

Limiting duration of facemask wear: Duration increases both discomfort and non-adherence. Duration might be decreased by demasking during breaks or scheduling mask breaks. Changing masks more often will help with adherence and the contamination risks, but will increase costs and environmental problems with waste disposal, as well as lead to potential contamination/transmission risks if not performed appropriately.

Modification for specific groups. Some groups are likely to have greater difficulty with mask wearing adherence and correct usage – including children, some patients with mental illnesses, those with cognitive impairment, or respiratory disorders such as asthma or chronic airways disease, and patients with recent facial trauma or oromaxillofacial surgery.[54-56]

Substitution. Face shields may provide an alternative to facemasks, which may mitigate several of the downsides (e.g. reducing the communication difficulties and breathing resistance), while also providing eye protection. However, there is little evidence on the discomforts of wearing face shields, and on the degree of protection provided. Other innovative mask designs currently being developed, require discomfort and adherence evaluations in addition to the droplet penetration.

Currently, existing research does not allow firm conclusions as there are insufficient data to quantify all of the adverse effects that might reduce the acceptability, adherence, and effectiveness of facemasks. Any new research on facemasks should assess and report the harms and downsides, including behavioural issues (i.e. risk compensation behaviour) and the psychological impact of mandated facemask wear. There is an urgent need for high quality research on methods and designs to mitigate downsides of facemask wearing, particularly assessment of alternatives such as face shields.

Statements

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Contributors

CDM, and PG conceived the study. MB, AMS, NK, PG, and CDM drafted the protocol. JC designed the search strategy and performed the literature search. MB, NK, AMS and JC screened studies for eligibility, and assessed the risk of bias. All authors performed data extraction. AMS performed the meta-analysis. MB, NK, AMS, PG and CDM performed the narrative analysis for non-meta-analysed studies. All authors critically reviewed the manuscript, contributed to its revision, and approved the final version submitted. The corresponding author confirms that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted. CDM is the guarantor.

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Competing interests

All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: no support from any organisation for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

Ethical approval

Not needed.

Data sharing

No additional data available.

Transparency

The corresponding author affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

This manuscript has been deposited as a preprint. Link: https://www.medrxiv.org/content/10.1101/2020.06.16.20133207v1

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Patient and public involvement

No patient or public involvement in the study.

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Appendix 1. Summary of the different types of facemasks included in the review and their intended uses

Type of face mask/covering	Description and Intended use
Surgical mask[1, 2]	- Loose-fitting, disposable device, made in different thicknesses - Creates a physical, protective barrier between the mouth and nose of the wearer and potential contaminants in the immediate environment i.e. splashes, sprays, large droplets or splatter. - Primarily intended to prevent the wearer from spreading respiratory droplets - Intended for single use - Does not filter or block very small particles, gases or chemicals in the air that may be transmitted by coughs, sneezes, or certain medical procedures. - Do not provide complete protection from germs and other contaminants because of the loose fit between the surface of the mask and the face.
N95 mask[2, 3]	- Respiratory protective device designed to achieve a very close facial fit. The edges of the respirator are designed to form a seal around the nose and mouth. - Require fit-testing and must be adjusted to the face. - Intended to help reduce wearer exposure to airborne particles as well as larger droplets and protect the wearer from splashes and sprays. - Filters out very fine particles. - Commonly used in healthcare settings. - Should not be shared or reused.
Non-medical face masks/covering[4]	- Cloth face coverings are only intended to help contain the wearer's respiratory droplets from being spread. - At least 2-3 layers needed to block respiratory droplets. - Not intended for use in a clinical setting or explicitly to prevent the transmission of disease between persons. - Some are reusable, and need to be washed after each use.

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2. U.S. Food and Drug Administration (2020). "N95 Respirators, Surgical Masks, and Face Masks." Retrieved 17/09/20, from https://www.fda.gov/medical-devices/personal-protective-equipment-infection-control/n95-respirators-surgical-masks-and-face-masks.
3. N95) Victoria State Government (2020). Coronavirus (COVID-19) – PPE and levels of protection. Department of Health and Human Services.
4. U.S. Food and Drug Administration (2020). "Use of Respirators, Facemasks, and Cloth Face Coverings in the Food and Agriculture Sector During Coronavirus Disease (COVID-19) Pandemic." Retrieved 17/09/20, from https://www.fda.gov/food/food-safety-during-emergencies/use-respirators-facemasks-and-cloth-face-coverings-food-and-agriculture-sector-during-coronavirus#:~:text=Cloth%20face%20coverings%20are%20only,from%20spreading%20it%20to%20others.

Appendix 2. Database search strings

Searches conducted 18 May 2020:

PubMed

(("Masks/adverse effects"[MAJR])

OR

((Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR “Face shield”[tiab] OR “Face shields”[tiab] OR Visors[tiab] OR “Face covering”[tiab])

AND

("Adverse effects"[sh] OR “Adverse effect”[tiab] OR “Adverse effects”[tiab] OR “Adversely affect”[tiab] OR “Adverse event”[tiab] OR “Adverse events”[tiab] OR Discomfort[tiab] OR Fatigue[tiab] OR Pain[tiab] OR “Difficulty breathing”[tiab] OR “Breathing difficulty”[tiab] OR “Shortness of breath”[tiab] OR “Risk compensation”[tiab] OR Fomite[tiab] OR Fomites[tiab] OR Dislike[tiab] OR “Physiological impacts”[tiab] OR “Physiological impact”[tiab] OR Sweating[tiab] OR Irritation[tiab] OR “Psychological reactions”[tiab] OR Fear[tiab] OR Communication[tiab] OR “Deaf persons”[tiab] OR Lip-reading[tiab] OR “Lip reading”[tiab] OR Vitiligo[tiab] OR Humidity[tiab] OR “Face dermatitis”[tiab] OR Acne[tiab] OR Redness[tiab] OR Ulcer[tiab] OR “Speech understanding”[tiab] OR “Pinching of skin”[tiab] OR “Moisture buildup”[tiab] OR “Facial warmth”[tiab])

AND

(Relieve[tiab] OR Attitudes[tiab] OR Induced[tiab] OR Modified[tiab] OR Decreased[tiab] OR Decrease[tiab] OR Associated[tiab] OR Impair[tiab] OR Reduce[tiab] OR Reducing[tiab] OR Reduced[tiab] OR Duration[tiab])

AND

(“randomized controlled trial”[pt] OR “controlled clinical trial”[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR “case-control studies”[Mesh] OR “Cohort Studies”[Mesh] OR “case control”[tiab] OR Cohort[tiab] OR “Follow up”[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR Retrospective[tiab] OR “cross sectional”[tiab] OR “Cross-Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh] OR Etiology[sh] OR Effect[tiab] OR Effects[tiab] OR Affect[tiab])))

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR

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"Oxygen mask"[tiab] OR OxyMask[tiab] OR "Nasal masks"[tiab] OR "Patil-Syracuse Mask"[tiab] OR "Ventilation masks"[tiab] OR "Venti-Mask"[tiab] OR "Venturi mask"[tiab] OR "Diving masks"[tiab] OR "Thermoplastic masks"[tiab] OR Bag-valve-mask[tiab] OR "Laryngeal mask"[tiab] OR "Laryngeal masks"[tiab] OR "Oronasal masks"[tiab] OR "Sleep mask"[tiab] OR Editorial[pt] OR Letter[pt] OR "Case reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

OR

(Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR "Face shield"[tiab] OR "Face shields"[tiab] OR Visors[tiab] OR "Face covering"[tiab])

AND

("Respiratory Tract Infections"[Mesh] OR Influenza[tiab] OR "Respiratory Tract Diseases"[Mesh] OR Influenzas[tiab] OR "Influenza-like"[tiab] OR ILI[tiab] OR "common cold"[tiab] OR colds[tiab] OR coryza[tiab] OR coronavirus[Mesh] OR "sars virus"[Mesh] OR coronavirus[tiab] OR Coronaviruses[tiab] OR "coronavirus infections"[Mesh] OR "severe acute respiratory syndrome"[Mesh] OR "severe acute respiratory syndrome"[tiab] OR "severe acute respiratory syndromes"[tiab] OR sars[tiab] OR "respiratory syncytial viruses"[Mesh] OR "respiratory syncytial virus, human"[Mesh] OR "Respiratory Syncytial Virus Infections"[Mesh] OR "respiratory syncytial virus"[tiab] OR "respiratory syncytial viruses"[tiab] OR rsv[tiab] OR parainfluenza[tiab] OR "Respiratory illness"[tiab] OR ((Transmission[tiab]) AND (Coughing[tiab] OR Sneezing[tiab])) OR ((respiratory[tiab] AND Tract[tiab]) AND (infection[tiab] OR Infections[tiab] OR illness[tiab])) OR "Communicable Disease Control"[Mesh] OR "Disease Outbreaks"[Mesh] OR "Disease Transmission, Infectious"[Mesh] OR "Infection Control"[Mesh] OR Transmission[sh] OR "Communicable Disease Control"[tiab] OR "Secondary transmission"[tiab] OR ((Reduced[tiab] OR Reduce[tiab] OR Reduction[tiab] OR Reducing[tiab] OR Lower[tiab]) AND (Incidence[tiab] OR Occurrence[tiab] OR Transmission[tiab] OR Secondary[tiab])) OR Physiological[tiab])

AND

("Adverse effects"[sh] OR "Adverse effect"[tiab] OR "Adverse effects"[tiab] OR "Adverse event"[tiab] OR "Adverse events"[tiab] OR "Index cases"[tiab] OR "Attack rates"[tiab] OR ((Reduce[tiab] OR Reduction[tiab]) AND (Rate[tiab] OR Rates[tiab])) OR Perception[tiab] OR Prevent[tiab] OR Adherence[tiab])

AND

("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR "case-control studies"[Mesh] OR "Cohort Studies"[Mesh] OR "case control"[tiab] OR Cohort[tiab] OR "Follow up"[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR retrospective[tiab] OR "cross sectional"[tiab] OR "Cross-

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Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh])

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR “Oxygen mask”[tiab] OR OxyMask[tiab] OR “Goldman nasal masks”[tiab] OR “Patil-Syracuse Mask”[tiab] OR “Ventilation masks”[tiab] OR “Venti-Mask”[tiab] OR “Venturi mask”[tiab] OR “Hockey mask”[tiab] OR “Diving masks”[tiab] OR “Thermoplastic masks”[tiab] OR Bag-valve-mask[tiab] OR “Laryngeal mask”[tiab] OR “Laryngeal masks”[tiab] OR “Oronasal masks”[tiab] OR “Sleep mask”[tiab] OR "Case Reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

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(([mh Masks/AE])

OR

(([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)))

NOT

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([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Nasal masks":ti,ab OR "Patil Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

OR

([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

([mh "Respiratory Tract Infections"] OR Influenza:ti,ab OR [mh "Respiratory Tract Diseases"] OR Influenzas:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR [mh coronavirus] OR [mh "sars virus"] OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR [mh "coronavirus infections"] OR [mh "severe acute respiratory syndrome"] OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR [mh "respiratory syncytial viruses"] OR [mh "respiratory syncytial virus, human"] OR [mh "Respiratory Syncytial Virus Infections"] OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR [mh "Communicable Disease Control"] OR [mh "Disease Outbreaks"] OR [mh "Disease Transmission, Infectious"] OR [mh "Infection Control"] OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

NOT

([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

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Embase

('Masks'/exp OR 'Respiratory protective devices'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR Lip-reading:ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab OR Etiology:ti,ab OR Effect:ti,ab OR Effects:ti,ab OR Affect:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR 'letter'/exp OR 'editorial'/exp OR 'case report'/exp)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

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[embase]/lim

OR

('Mask'/exp OR 'gas mask'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

('respiratory tract infection'/exp OR Influenza:ti,ab OR 'respiratory tract disease'/exp OR Influenzas:ti,ab OR Influenza-like:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR 'Coronavirinae'/exp OR 'SARS coronavirus'/exp OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR 'Coronavirus infection'/exp OR 'severe acute respiratory syndrome'/exp OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR 'Pneumovirus'/exp OR 'Human respiratory syncytial virus'/exp OR 'respiratory syncytial virus infection'/exp OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR 'communicable disease control'/exp OR 'epidemic'/exp OR 'disease transmission'/exp OR 'Infection Control'/exp OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

(‘adverse event’/exp OR "Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal

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mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR "Case Reports":it)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

[embase]/lim

Europe PMC

https://europepmc.org/search?query=%28TITLE%3A"masks"%20AND%20%28%222019-nCoV%22%20OR%20%222019nCoV%22%20OR%20%22COVID-19%22%20OR%20%22SARS-CoV-2%22%20OR%20%28%22wuhan%22%20AND%20%22coronavirus%22%29%20OR%20%22Coronavirus%22%20OR%20%22Corona%20virus%22%20OR%20%22corona-virus%22%20OR%20%22corona%20viruses%22%20OR%20%22coronaviruses%22%20OR%20%22SARS-CoV%22%20OR%20%22Orthocoronavirinae%22%20OR%20%22MERS-CoV%22%20OR%20%22Severe%20Acute%20Respiratory%20Syndrome%22%20OR%20%22Middle%20East%20Respiratory%20Syndrome%22%20OR%20%28%22SARS%22%20AND%20%22virus%22%29%20OR%20%22soluble%20ACE2%22%20OR%20%28%22ACE2%22%20AND%20%22virus%22%29%20OR%20%28%22ARDS%22%20AND%20%22virus%22%29%20or%20%28%22angiotensin-converting%20enzyme%202%22%20AND%20%22virus%22%29%29%29%20AND%20%28SRC%3APPR%29&page=1

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Appendix 3. List of excluded studies

Excluded studies from database search

No.	Reference	Reason for exclusion
1.	Ahmad, E. F., et al. (2001). "The effect of wearing the veil by Saudi ladies on the occurrence of respiratory diseases." J Asthma 38(5): 423-426.	Study type
2.	Ahmad, I., et al. (2017). "Assessment of knowledge, practice and barrier in use of facemask among university students." Pakistan Journal of Medical and Health Sciences 11(4): 1657-1658.	Reported outcomes
3.	Aiello, A. E., et al. (2010). "A randomized intervention trial of mask use and hand hygiene to reduce seasonal influenza-like illness and influenza infections among young adults in a university setting." International Journal of Infectious Diseases 14: e320.	Duplicate
4.	Aiello, A. E., et al. (2010). "Mask use, hand hygiene, and seasonal influenza-like illness among young adults: a randomized intervention trial." J Infect Dis 201(4): 491-498.	No extractable data
5.	Akyuz, N., et al. (2014). "How and how much do endoscopy professionals protect themselves against infection?" Int J Surg 12(7): 720-724.	Reported outcomes
6.	Albarrak, A. I., et al. (2019). "Middle East Respiratory Syndrome (MERS): Comparing the knowledge, attitude and practices of different health care workers." J Infect Public Health.	Reported outcomes
7.	AlGhamri, A. A., et al. (2013). "The effects of wearing respirators on human fine motor, visual, and cognitive performance." Ergonomics 56(5): 791-802.	Intervention
8.	Allsopp, J., et al. (1997). "Survey of the use of personal protective equipment and prevalence of work related symptoms among dental staff." Occup Environ Med 54(2): 125-134.	Reported outcomes
9.	Alqahtani, A. S., et al. (2016). "Exploring barriers to and facilitators of preventive measures against infectious diseases among Australian Hajj pilgrims: cross-sectional studies before and after Hajj." Int J Infect Dis 47: 53-59.	No extractable data
10.	Alsarheed, M. (2011). "Children's Perception of Their Dentists." Eur J Dent 5(2): 186-190.	Study type
11.	Aydin, O., et al. (24 April 2020). "Performance of fabrics for home-	Reported

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	made masks against spread of respiratory infection through droplets: a quantitative mechanistic study." PREPRINT (Version 1 - 2020.04.19.20071779), Available at: medRxix: https://doi.org/10.1101/2020.04.19.20071779.	outcomes
12.	Badin, A. (1969). "[The surgeon's fatigue and discomfort behind the mask]." Gynaecologia 167(5): 430-433.	Study type
13.	Bansal, S., et al. (2009). "Respirator physiological effects under simulated work conditions." J Occup Environ Hyg 6(4): 221-227.	Intervention
14.	Beuvelet, M., et al. (2017). "Healthcare workers' perception towards the systematic use of mask during a seasonal influenza outbreak in a French University Hospital: A descriptive study." Antimicrobial Resistance and Infection Control 6.	Unable to access full text
15.	Bhoyrul, B., et al. (2019). "A review of non-glove personal protective equipment-related occupational dermatoses reported to EPIDERM between 1993 and 2013." Contact Dermatitis 80(4): 217-221.	Study type
16.	Bischoff, W., et al. (2017). "Evaluation of a novel powered air-purifying respirator (PAPR) vs. a N95 respirator mask for the protection against influenza in a human exposure model." Open Forum Infectious Diseases 4: S168.	Reported outcomes
17.	Brandt, C., et al. (2011). "Attitudes of influenza-vaccinated health care workers toward masks to prevent nosocomial transmission of influenza." Influenza Other Respir Viruses 5(1): 61-66.	Study type
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100.	Radonovich Jr, L. J., et al. (2009). "Diminished speech intelligibility associated with certain types of respirators worn by healthcare workers." Journal of Occupational and Environmental Hygiene 7(1): 63-70.	Intervention
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103.	Raven, P. B., et al. (1977). "Maximal stress test performance while wearing a self-contained breathing apparatus." J Occup Med 19(12): 802-806.	Intervention
104.	Rengasamy, S. and B. C. Eimer (2013). "N95-companion measurement of cout/cin ratios for two n95 filtering facepiece respirators and one surgical mask." J Occup Environ Hyg 10(10): 527-532.	Population
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112.	Seliga, R., et al. (1991). "Effect of work load and respirator wear on postural stability, heart rate, and perceived exertion." Am Ind Hyg Assoc J 52(10): 417-422.	Intervention
113.	Seng, M., et al. (2018). "Comfort and exertion while using filtering facepiece respirators with exhalation valve and an active venting system among male military personnel." Singapore Med J 59(6): 327-334.	Purpose of study
114.	Smith, C. L., et al. (2013). "Carbon dioxide rebreathing in respiratory protective devices: influence of speech and work rate in full-face masks." Ergonomics 56(5): 781-790.	Intervention
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117.	Stockwell, R. E., et al. (2018). "Prolonged wear of face masks prevents the release of pseudomonas aeruginosa aerosols during coughing in people with cystic fibrosis." Pediatric Pulmonology 53: 269.	Unable to access full text
118.	Strauch, A. L., et al. (2016). "Assessing the efficacy of tabs on filtering facepiece respirator straps to increase proper doffing techniques while reducing contact transmission of pathogens." J Occup Environ Hyg 13(10): 794-801.	Intervention
119.	Sullivan, K. (1993). "Surgical face masks: protection of self or patient?" N Z Dent J 89(398): 137.	Study type

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127.	Wong, S. Y., et al. (2012). "Primary care physicians' response to pandemic influenza in Hong Kong: a mixed quantitative and qualitative study." Int J Infect Dis 16(9): e687-691.	Reported outcomes
128.	Yang, P., et al. (2011). "Mask-wearing and respiratory infection in healthcare workers in Beijing, China." Braz J Infect Dis 15(2): 102-108.	Study type

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131.	Chughtai, A. A., et al. (2015). "Practices around the use of masks and	Reported

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132.	Enerson, D. M., et al. (1967). "Heat and moisture trapping beneath surgical face masks: a consideration of factors affecting the surgeon's discomfort and performance." Surgery 62(6): 1007-1016.	Reported outcomes
133.	Goh, D. Y. T., et al. (2019). "A randomised clinical trial to evaluate the safety, fit, comfort of a novel N95 mask in children." Scientific Reports 9(1).	Purpose of study
134.	Harber P, Bansal S, Santiago S, et al. Multidomain subjective response to respirator use during simulated work. Journal of Occupational and Environmental Medicine. 2009;51(1):38-45.	Reported outcomes
135.	Harber, P., et al. (2010). "Respirator physiologic impact in persons with mild respiratory disease." Journal of Occupational and Environmental Medicine 52(2): 155-162.	Intervention
136.	Harber, P., et al. (2011). "Respirator impact on work task performance." Journal of Occupational and Environmental Medicine 53(1): 22-26.	Reported outcomes
137.	Honarbakhsh, M., et al. (2018). "Knowledge, perceptions and practices of healthcare workers regarding the use of respiratory protection equipment at Iran hospitals." Journal of Infection Prevention 19(1): 29-36.	Reported outcomes
138.	Jacobs, J. L., et al. (2009). "Use of surgical face masks to reduce the incidence of the common cold among health care workers in Japan: A randomized controlled trial." American Journal of Infection Control 37(5): 417-419.	Duplicate
139.	Jones, J. G. (1991). "The physiological cost of wearing a disposable respirator." American Industrial Hygiene Association Journal 52(6): 219-225.	Intervention
140.	Kim, J. H., et al. (2013). "Pulmonary and heart rate responses to wearing N95 filtering facepiece respirators." American Journal of Infection Control 41(1): 24-27.	Reported outcomes
141.	Kim, J. H., et al. (2015). "Pressure drop of filtering facepiece respirators: How low should we go?" International Journal of Occupational Medicine and Environmental Health 28(1): 71-80.	Intervention
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147.	Luximon, Y., et al. (2016). "Time dependent infrared thermographic evaluation of facemasks." Work 54(4): 825-835.	Reported outcomes
148.	MacIntyre, C. R., et al. (2014). "Efficacy of face masks and respirators in preventing upper respiratory tract bacterial colonization and co-infection in hospital healthcare workers." Preventive Medicine 62: 1-7.	Reported outcomes
149.	MacIntyre, C. R., et al. (2013). "A randomized clinical trial of three options for N95 respirators and medical masks in health workers." American Journal of Respiratory and Critical Care Medicine 187(9): 960-966.	Duplicate
150.	Mauritzson-Sandberg, E. V. A. (1991). "Psychological effects on prolonged use of respiratory protective devices in children." Ergonomics 34(3): 313-319.	Reported outcomes
151.	McDiarmid, M., et al. (2020). "N95 Respirators vs Medical Masks in Outpatient Settings." JAMA - Journal of the American Medical Association 323(8): 789.	Study type
152.	Morishima, M., et al. (2014). "Experiences and problems with hygiene masks reported by japanese hay fever sufferers." International Journal of Clothing Science and Technology 26(4): 262-273.	Intervention
153.	Morishima, M., et al. (2017). "Problems related to mask use in hay fever sufferers by repeated surveys." International Journal of Clothing Science and Technology 29(1): 123-134.	Purpose of study

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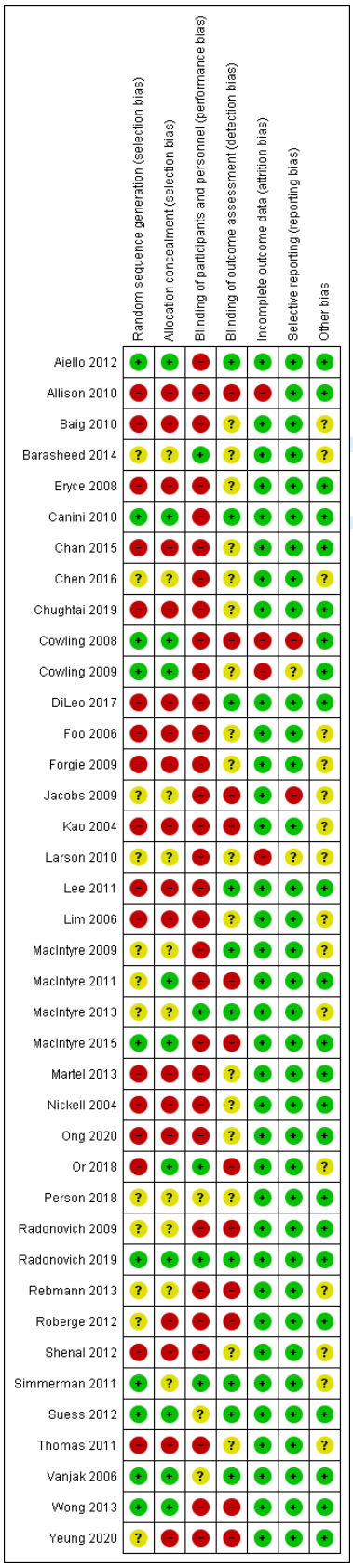
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155.	Phin NF, Rylands AJ, Allan J, Edwards C, Enstone JE, Nguyen-Van-Tam JS. Personal protective equipment in an influenza pandemic: a UK simulation exercise. Journal of Hospital Infection. 2009;71(1):15-21.	Study design
156.	Radonovich Jr, L. J., et al. (2010). "Diminished speech intelligibility associated with certain types of respirators worn by healthcare workers." Journal of Occupational and Environmental Hygiene 7(1): 63-70.	Intervention
157.	Radonovich, L. J., et al. (2019). "A tolerability assessment of new respiratory protective devices developed for health care personnel: A randomized simulated clinical study." PLoS ONE 14(1).	No extractable data
158.	Roberge, R., et al. (2012). "Thermal burden of N95 filtering facepiece respirators." Annals of Occupational Hygiene 56(7): 808-814.	Reported outcomes
159.	Roberge, R. J. (2008). "Effect of surgical masks worn concurrently over N95 filtering facepiece respirators: Extended service life versus increased user burden." Journal of Public Health Management and Practice 14(2): E19-E26.	Study type
160.	Roberge RJ, Coca A, Williams WJ, Powell JB, Palmiero AJ. Physiological impact of the n95 filtering facepiece respirator on healthcare workers. Respiratory Care. 2010;55(5):569-577.	Intervention
161.	Roberge, R. J., et al. (2010). "Effect of exhaled moisture on breathing resistance of N95 filtering facepiece respirators." Annals of Occupational Hygiene 54(6): 671-677.	Study type
162.	Roberge, R. J., et al. (2012). "N95 filtering facepiece respirator deadspace temperature and humidity." Journal of Occupational and Environmental Hygiene 9(3): 166-171.	Reported outcomes
163.	Roberge, R. J., et al. (2013). "Impact of low filter resistances on subjective and physiological responses to filtering facepiece respirators." PLoS ONE 8(12).	Intervention
164.	Roberge RJ, Kim JH, Powell JB. N95 respirator use during advanced pregnancy. American Journal of Infection Control. 2014;42(10):1097-1100.	No extractable data
165.	Sakaguchi, H., et al. (2010). "Maintenance of influenza virus infectivity on the surfaces of personal protective equipment and clothing used in healthcare settings." Environmental Health and Preventive Medicine	Purpose of study

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	15(6): 344-349.	
166.	Shi, J., et al. (2017). "Cardiovascular benefits of wearing particulate-filtering respirators: A randomized crossover trial." Environmental Health Perspectives 125(2): 175-180.	Purpose of study
167.	Shigayeva, A., et al. (2007). "Factors associated with critical-care healthcare workers' adherence to recommended barrier precautions during the Toronto severe acute respiratory syndrome outbreak." Infection Control and Hospital Epidemiology 28(11): 1275-1283.	No extractable data
168.	Skaria, S. D. and G. C. Smaldone (2014). "Respiratory source Control using surgical masks with nanofiber media." Annals of Occupational Hygiene 58(6): 771-781.	Purpose of study
169.	Tamura, E., et al. (2013). "Effect of pharmacists wearing a mask on patient consultation intention." Yakugaku Zasshi 133(6): 737-745.	Study type
170.	Tong, P. S. Y., et al. (2015). "Respiratory consequences of N95-type Mask usage in pregnant healthcare workers-A controlled clinical study." Antimicrobial Resistance and Infection Control 4(1).	Intervention
171.	Turnberg, W., et al. (2009). "Personal healthcare worker (HCW) and work-site characteristics that affect HCWs' use of respiratory-infection control measures in ambulatory healthcare settings." Infection Control and Hospital Epidemiology 30(1): 47-52.	Reported outcomes
172.	Wada, K., et al. (2012). "Wearing face masks in public during the influenza season may reflect other positive hygiene practices in Japan." BMC Public Health 12(1).	Reported outcomes
173.	Wu, S., et al. (2011). "Anxiety during respirator use: Comparison of two respirator types." Journal of Occupational and Environmental Hygiene 8(3): 123-128.	Study type
174.	Yip, W. L., et al. (2005). "The effect of wearing a face mask on body temperature." Hong Kong Journal of Emergency Medicine 12(1): 23-27.	Reported outcomes

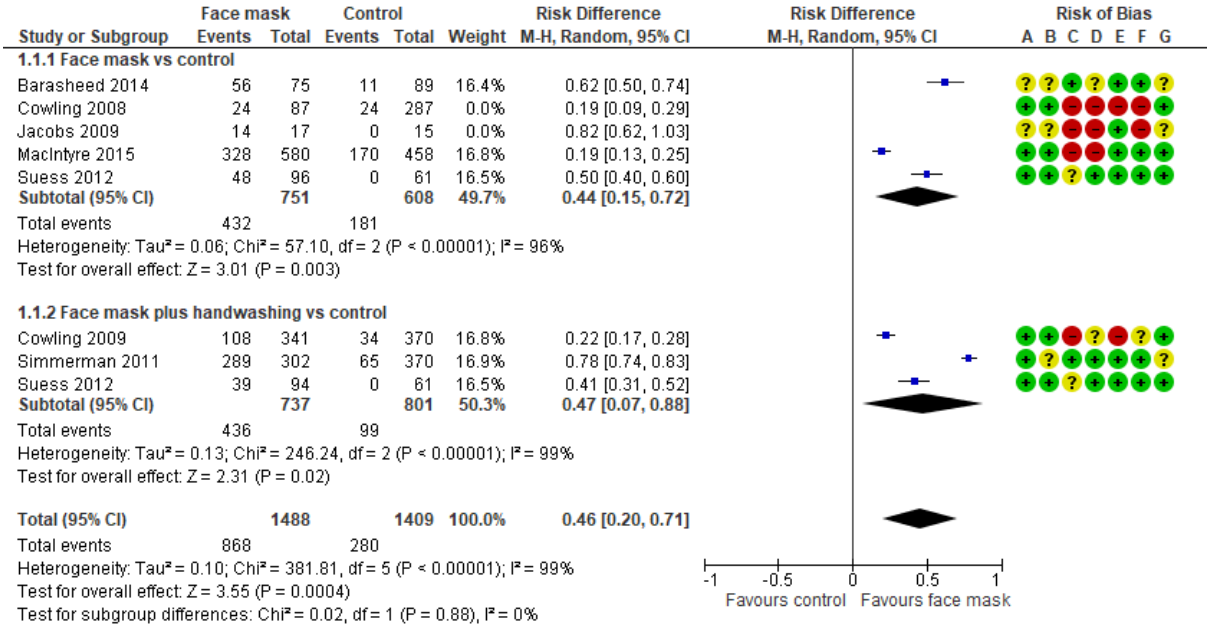
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Appendix 4. Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

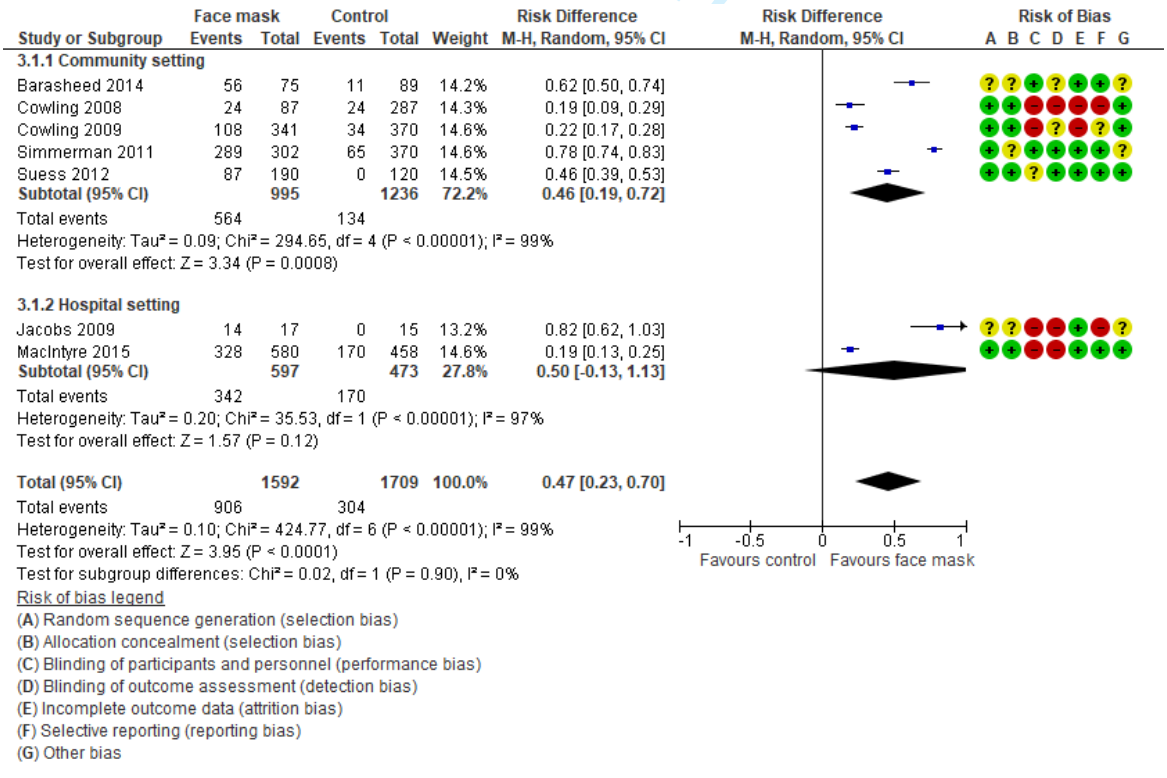


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Appendix 4. Figure 4. Analysis with studies rated at high risk of bias in more than 2 domains excluded



Appendix 4. Figure 5. Analysis with studies sub-grouped by setting (community setting, hospital setting)



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5, 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-17
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11, 12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11, 12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Title page

Title

Downsides of face masks and possible mitigation strategies: a systematic review and meta-analysis

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Abstract

Objective

To identify, appraise, and synthesise studies evaluating the downsides of wearing facemasks in any setting. We also discuss potential strategies to mitigate these downsides.

Design

Systematic review and meta-analysis

Data sources

PubMed, Embase, CENTRAL, EuropePMC were searched (inception-18/5/2020), and clinical registries were searched via CENTRAL. We also did a forward-backward citation search of the included studies.

Inclusion criteria

We included randomised controlled trials and observational studies comparing facemask use to any active intervention or to control.

Data extraction and analysis

Two author pairs independently screened articles for inclusion, extracted data and assessed the quality of included studies. The primary outcomes were compliance, discomforts, harms, and adverse events of wearing facemasks.

Results

We screened 5471 articles, including 37 (40 references); 11 were meta-analysed. For mask wear adherence, 47% (95%CI 25%-68%, P< 0.0001) more people wore facemasks in the facemask group compared to control; adherence was significantly higher (26%, 95%CI 8%-46%, P< 0.01) in the surgical/medical mask group than in N95/P2 group. The largest number of studies reported on the discomfort and irritation outcome (20-studies); fewest reported on the misuse of masks, and none reported on mask contamination or risk compensation behaviour. Risk of bias was generally high for blinding of participants and personnel and low for attrition and reporting biases.

Conclusions

There are insufficient data to quantify all of the adverse effects that might reduce the acceptability, adherence, and effectiveness of face masks. New research on facemasks should assess and report

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the harms and downsides. Urgent research is also needed on methods and designs to mitigate the downsides of facemask wearing, particularly the assessment of possible alternatives.

Systematic review registration

Open Science Framework website https://osf.io/sa6kf/ (Timestamp 20-05-2020)

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Strengths and Limitations of this study

- This is the first systematic review to investigate the downsides of wearing facemasks.
- We included non-randomised study designs in addition to RCTs, as trials frequently underreported or failed to report harms.
- We reported two deviations from the study protocol, relating to the measurement of effect and the sub-group analysis.

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Introduction

Respiratory viruses are predominantly transmitted by aerosol, droplets and fomites.[1] Facemasks – such as surgical masks, N95 masks, and face shields, and substitutes for surgical masks such as home-made cloth masks – are a physical barrier to aerosol and droplet transmission. During the COVID-19 pandemic, some jurisdictions have implemented policies mandating the use of masks in public places, on public transport or in other crowded environments, to prevent people becoming infected, or infecting others.

While most health organisations mandate the use of facemasks by health workers when caring for patients during a pandemic, recommendations for mask wear in the community vary widely – and include use by all, use only in certain situations (e.g. on public transport, or in crowded places where social distancing is not possible), and no specific recommendations about mask use.

Several trials have evaluated the impact on respiratory infections by use of surgical and N95 masks, which may, at best, modestly reduce acute respiratory infection transmission.[2-4] Population observational studies suggest that masks have a more substantial effect.[5] However, the downsides of mask-wearing were either not considered or not reported in most studies. Most trials have focussed on facemasks protecting the wearer, rather than others in the community, are often low powered and include confounding factors resulting in the current evidence for the efficacy of facemasks being less than adequate.[6]

The current controversies, mixed messaging and debate over the use of face masks to prevent the spread of COVID-19 warrants further investigation into their use. Anecdotal evidence, and some studies, suggest that there may be a variety of downsides arising from mask use, including: discomfort, sense of difficulty breathing, and communication problems particularly for those who use lip reading.[6 7] Our aim is to systematically identify and summarise these downsides, to assist policymakers when formulating mask-wearing policies in public settings. We also discuss potential strategies to mitigate downsides of mask-wearing.

Methods

This systematic review is reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[8] We followed the "2 week systematic review" (2weekSR)

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processes for the review.[9] The review protocol was developed prospectively and uploaded to the Open Science Framework data on 20-05-2020 (see: https://osf.io/sa6kf/). Deviations from the protocol are reported in the Methods section and highlighted as a limitation in the Discussion section.

Inclusion criteria

We included studies of people of any age or gender, in any setting. We included studies of any face covering aimed at reducing virus transmission, including surgical masks, N95 masks, cloth masks (both homemade and commercially available). (See Appendix 1 for a summary of included facemasks and their intended purpose). Studies evaluating the use of masks for non-virus transmission purposes (e.g. valved masks for preventing inhalation of airborne contaminants such as particles, gases, or vapours) were excluded (see Appendix 2 for a complete list of excluded facemasks).

We included studies comparing the use of facemask to any active intervention (e.g. another mask, or another intervention such as hand-washing etc.), and studies comparing the use of facemask to control (comprising standard practice i.e. face masks not explicitly provided to study group) in situations where their use was not mandatory.

We included only primary studies – i.e., randomised controlled trials (RCTs) and observational studies of any design. We intentionally included this range of study types to identify appropriate data that corresponded with the listed outcomes. For example, surveys are appropriate for identifying frequency of discomfort, and breathing trials are appropriate for understanding physiological changes. We excluded studies that could not provide a quantitative estimate of the size or frequency of adverse effects such as case reports, case series, as well as qualitative studies, and reviews.

Search strategies

We searched PubMed, Embase, Cochrane CENTRAL, EuropePMC (inception-18/5/2020). The search string was designed for PubMed and translated for use in other databases using the Polyglot Search Translator (Appendix 3).[10] Clinical trial registries were searched via Cochrane CENTRAL, which includes the WHO ICTRP and clinicaltrials.gov. On 22/5/2020, we conducted a backwards and forwards citation analysis in Scopus, on all of the included studies.

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No restrictions by language or publication date were imposed. We included publications that were published in full; abstract only publications were included if they had an accompanying record (e.g. trial registry record, or another public report), with additional information.

Study selection and screening

Four authors (MB&NK, AMS&JC) independently screened the titles and abstracts against the inclusion criteria. One author (JC) retrieved full-text, and four authors (MB&NK, AMS&JC) screened the full-texts. Disagreements were resolved by discussion, or a third author (PG or CDM).

Data extraction

A data extraction form was piloted on three studies. Four authors (MB&NK, AMS&JC) extracted the data. (See Box 1)

Box 1. List of extracted information

- **General information:** study authors, location, study design, duration, aim, setting
- **Participants:** health status, disease (if applicable), sample size, age, gender, smoker status, co-morbidities
- **Intervention and Comparator(s):** number of participants, type of face covering, adjunct interventions, number of face coverings used, duration of use, disposal
- **Outcomes:** definition, measurement instrument, number of adverse events or harms reported (The outcomes were discomforts, harms, and adverse events of wearing face masks, Adherence to facemask wearing, Misuse of masks, Discomfort and other physical irritation from masks, Psychological outcomes (e.g. fear, etc.), Dyspnoea (difficulty breathing, shortness of breath) and other physiological impacts, Communication impacts, and mask contamination).

Assessment of risk of bias in included studies

Four authors (MB&NK, AMS&JC) independently assessed the risk of bias using the Cochrane Risk of Bias Tool.[11] Disagreements were resolved by discussion or a third author (CDM or PG). Each potential source of bias was graded as low, high or unclear, and judgements were supported by a quote from the study.

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Measurement of effect and data synthesis

Where feasible (≥2 studies reporting the same outcome), we expressed outcome measures as odds ratios (OR) with 95% confidence intervals, analysed using Review Manager (RevMan 5). Anticipating considerable heterogeneity among the included studies, we used a random effects model. We reported the adherence to facemask wear using risk difference rather than odds ratio, to more clearly convey the differences in adherence between the intervention and control group (not pre-specified in the protocol).

When meta-analysis was not possible or appropriate, we followed the guidance of the Cochrane Collaboration (Cochrane Handbook Section 12.2).[11] When narrative synthesis was required, we reported the results separately for each harm or adverse outcome.

The individual was used as the unit of analysis, where possible; otherwise, we extracted the information as it was presented, e.g. the number of harms in each group. We attempted to contact investigators or study sponsors to provide missing data.

We used the I^2 statistic to measure heterogeneity. Because we included fewer than 10 trials, we did not create a funnel plot.

We did not pre-specify subgroup or sensitivity analyses. However, as data was available, we conducted a subgroup analysis of adherence to mask-wearing by studies which evaluated facemask wear alone, and those evaluating facemask together with handwashing.

Patient and public involvement

Patients and the public were not involved in the design and conduct of this review.

Results

Database searches identified 4,691 publications, supplemented with 2,035 references from forward and backward citation searches and other sources, totalling 6,726. After de-duplication, 5,471 references were screened by title and abstract; we full-text screened 214 references, excluding 174 (see Appendix 4 for a list of excluded studies with reasons). We included 40 articles [12-51] corresponding to 37 studies (Table 1 reports the characteristics of included studies), and meta-analysed 11 studies.[15 21 22 26 31-34 41 45 47] (Figure 1).

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Table 1. Characteristics of included studies

First author (country, year)	Study design	Setting	Study duration	Type of participants				Number of participants	Type of Face Mask intervention**				Type of reported outcomes (adverse effects) analysed						
				Healthy		Sick*													
				A	Ch	A	Ch		Surgical	N95	Any	Other	Discomfort and irritation	Dyspnoea & other physiological	Misuse	Adherence	Psychological impacts	Communication impacts	Mask contamination
Aiello (USA, 2012)	C-RCT	RH	3M	✓				1188; 37 RH	✓				✓			✓			
Allison (USA, 2010)	Single arm study	Sch	1M	✓	✓			20 A, 503 Ch	✓							✓			
Baig (USA, 2010)	S	H	1M	✓				149[1]		✓			✓					✓	
Barasheed (Australia, 2014)	C-RCT	Tents	1W	✓	✓	✓		164; 22 Tents	✓							✓			
Bryce (Canada, 2008)	S	H	1W S	✓				137		✓			✓			✓			
Canini (France, 2010)	C-RCT	HH	1Inf-S	✓	✓		✓	306; 105 HH	✓				✓			✓	✓		
Chen (China, 2016)	Multiple crossover lab-based trials	Lab	NR	✓				15		✓				✓					
Chugtai (China, 2019)[1]	Prevalence	H	2M	✓				148	✓				✓			✓		✓	✓
Cowling (Hong Kong, 2008)	C-RCT	HH	7M	✓	✓	✓	✓	370; 128 HH[3]	✓							✓			
Cowling (Hong Kong, 2009)	C-RCT	HH	8M	✓	✓	✓	✓	1015; 322 HH[3]	✓							✓			
Foo (Singapore 2006)	S	H	2S	✓				340			✓		✓						
Forgie (Canada, 2009)[4]	S	ED	2M	✓	✓			80[5]	✓			✓					✓		
Jacobs (Japan, 2009)	RCT	H	2M	✓				33	✓							✓			
Kao (Taiwan, 2004)[6]	Before-after	H	2M	✓				39		✓			✓						
Larson (USA, 2010) with	C-RCT	HH	19M	✓	✓			2708; 617 HH[3]	✓				✓			✓	✓	✓	

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additional data from (Ferng et al. 2011)																			
Lee (Singapore, 2011)[7]	Single arm study	Lab	NR	✓				14		✓				✓					
Lim (Singapore, 2006)	S	H	1Y	✓				212		✓			✓						
MacIntyre (Australia 2009)	C-RCT	HH	2WS	✓			✓	290 A; 145 HH	✓			✓	✓			✓	✓		
MacIntyre (China, 2011)	C-RCT	H	3M	✓				1441; 15 H	✓	✓			✓			✓		✓	
MacIntyre (China, 2013)	C-RCT	H	3M	✓				1669; 19 H	✓	✓			✓			✓			
MacIntyre (Vietnam, 2015) with additional data from (Chughtai et al. 2016)	C-RCT	H	5W	✓				1607; 14 H	✓			✓	✓			✓			
Martel (Canada, 2013)[8]	Direct observation study	ED	1M	✓				115			✓		✓		✓			✓	
Nickell (Canada, 2004)	S	H	1M	✓				2001	✓	✓			✓				✓	✓	
Ong (Singapore, 2020)	S	H	2M	✓				158	✓				✓						
Or (Hong Kong, 2018)	Lab-based study	Lab	NR	✓				84		✓			✓					✓	
Person (France, 2018)	Randomised crossover	NC[4]	NR	✓				44	✓					✓					
Radonovich (USA, 2009)	Multiple crossover Trial	H	NC	✓				27	✓	✓		✓	✓					✓	
Radonovich (USA, 2019)	C-RCT	H	3.7Y	✓				2862	✓	✓						✓			
Rebmann (USA, 2013)	Multiple Crossover Trial	H	2D	✓				10		✓					✓				
Roberge (USA, 2012)	Crossover	Lab	3M	✓				20	✓					✓					
Shenal (USA, 2012)	Multiple crossover field trial	H	NR	✓				27	✓			✓	✓						
Simmerman (Thailand, 2011)	RCT	HH	16M	✓			✓	465	✓							✓			
Suess (Germany, 2012) with additional data from (Suess et al. 2011)	C-RCT	HH	2Inf-S	✓			✓	111 HH	✓				✓			✓			

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Thomas (USA, 2011)	NC[9]	HB-E	NR	✓				3	✓	✓								✓	
Vanjak (France, 2006)	S	H	2M	✓				238 patients & 210 staff			✓				✓				
Wong (Hong Kong, 2013)	RCT	PCC	2M	✓				1031 patients & 9 doctors			✓						✓		
Yeung (Singapore, 2020)	S	HH	1W	✓				2231 HH		✓					✓				

C-RCT: cluster randomized controlled trial, **RCT**: randomized controlled trial, **S:** Survey, **Sch:** Schools, **RH:** residence halls, **HH:** Households, **H:** Hospitals, **Tel:** Telephone survey of home residents, **Lab:** Laboratory, **ED:** Emergency departments, **HB-E:** Helicopter Bell (used for helicopter emergency medical services), **PCC:** Primary care clinics, **N/A:** Not applicable, **NR:** Not reported, **NC:** Not clear, **Y:** Year, **M:** Months, **W:** Weeks, **D:** Days, **Inf-S:** Influenza season, **WS:** Winter season, **A:** Healthy Adults, **Ch:** Children.

*Sick adults or children, with influenza, influenza-like illness, or upper respiratory tract infections. **Only Face masks arms of the included studies. [1] Authors state 159 surveys returned but 10 were excluded from analysis as ineligible. [2] reports 3 studies: pilot study 1 lab testing; pilot study 2 (small clinical testing); study 3 main study which is extracted here. [3] number of households that were allocated. [4] Participants in the waiting room of ED, could be sick due to an infection. [5] Pairs of adults and children. [6] Patients with End stage renal disease without any influenza or ILI. [7] Data reported here is for the N95 group. [8] 2-part study: part 1) direct observation study [data in this row] + part 2) survey, 4 "in a covered, flat, straight corridor, with a length of 30 meters and marked every 3 meters". [9] In the first part of the study, the pilot recited each word with the aircraft engine off when wearing and not wearing either the surgical facemask or one of the N95 respirators. In the second part of the study, the same procedure occurred, except that the 407-helicopter engine was running at idle.

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[Insert Figure 1 here]

Risk of bias

Inclusion of both observational studies and RCTs could explain the high risk of bias across the included studies. Reporting of sequence generation, allocation concealment and blinding of outcome assessment was poor in 20% - 30% of studies. Due to the nature of interventions, blinding of participants was rare. For RCTs, we reported blinding of outcome assessment for the main trial (not the adverse events) – as some outcomes were lab-confirmed and were considered of low risk. We found no evidence of incomplete outcome data or selective reporting of outcomes. Funding statement, funder's role and authors' conflict of interests were adequately reported in most studies (see Figure 2, and figure 3 in Appendix 5).

[Insert Figure 2 here]

Adherence to facemask wearing

17 studies (14 RCTs, 3 observational [10,13,17]) reported on adherence to facemask wearing; of these 11 had sufficient data to pool and were meta-analysed.

Face mask versus control comparison

Comparison of face masks to control was sub-grouped into studies comparing facemask alone vs control; and studies of facemask plus handwashing vs control. (Figure 3)

Facemasks alone vs control (5 studies) showed the facemask group had a significantly higher facemask wear compared to control (Risk Difference (RD): 0.46, 95% CI 0.24 to 0.67, P< 0.0001). Studies evaluating facemask plus handwashing vs control (n= 3), similarly showed significantly higher facemask wear in the facemask group (RD: 0.47 (95% CI 0.07 to 0.88, P< 0.0001).
Overall, 7 studies (3303 participants) compared facemasks to control. Facemask wear was 47% higher in the facemask group, although heterogeneity was very high (RD: 0.47, 95% CI 0.25 to 0.68, P< 0.0001, I^2 = 98%.)

[Insert Figure 3 here]

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We explored the possible sources of heterogeneity. Excluding studies with 3 or more domains at high risk of bias did not decrease heterogeneity (I^2 = 96% for facemask vs control; 99% for facemask plus handwashing versus control) (Figure 4, Appendix 5). We excluded study population as the source of heterogeneity, because subgrouping studies by those in a community/household settings (which included both index cases and their contacts) versus those in a hospital setting (which included healthy healthcare workers) likewise did not decrease heterogeneity (I^2 = 99% for community/household studies, and 97% for hospital studies) (Figure 5, Appendix 5). We excluded intervention and control as sources of heterogeneity, since all studies compared medical/surgical masks to control (no mask) – although some mask-wear did occur in the control groups.

We consider the outcome measurement to be the most likely other source of heterogeneity. All studies relied on self-reporting of the outcome; only one verified this by counting the number of masks used.[21] What was considered 'wearing a facemask' varied: it was unclear,[15 21 45] involved wearing a facemask "always or mostly/often",[22 47] included wearing a facemask whilst on hospital property, performing a healthcare worker role,[26] and included facemask wear during a shift for 70% of time or more.[32] The follow-up was very short (5 - 9 days) for 4 studies;[15 21 22 47] it was longer for 3 studies (21 - 77 days).[26 32 45]

Facemask (surgical/medical) vs facemask (N92/P2 mask)

Four studies (7960 participants) compared adherence for different facemasks.[31 33 34 41] Facemask wear was significantly higher in the surgical/medical facemask group than in the N95/P2 group, OR= 1.26 (95% CI 1.08 to 1.46, P< 0.01). Heterogeneity was very low (I^2 = 27%). (Figure 4)

[Insert Figure 4 here]

Studies not included in the meta-analysis

RCTs (n = 3)

One study in a residential hall setting reported similar duration of facemask wear per day in the facemask alone group vs facemask plus handwashing group (mean of 5.08 and 5.04 hours/day, respectively).[12] Another in a household setting reported that within the facemask group, there were no significant differences between individuals with ILI among contacts vs no ILI among contacts, for facemask use.[17] Finally, 22 of 44 households randomised to the 'education with sanitiser and facemasks' arm, reported having used a mask within 48 hours of episode onset.[28]

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Observational studies (n = 3)

In elementary school setting, approximately twice as many teachers as students wore face masks.[13] A mean compliance score with N95 use guidelines was 21.2 (on a 25-point scale) among frequent users of N95 respirators in a hospital setting.[16] Another study among healthcare workers found that majority of survey respondents (91%) wore 1-2 masks per day (range 1 - 4).[20]

Misuse

Mask misuse appears less studied than other harms and discomforts. A study of 10 nurses observed for 10 minutes/hour over 2 shifts found that they touched their face 2-3 times per hour, their mask 5 times per hour, and their eyes once per 2 hours, when observed by students.[42] In a study of health workers, 13 of the 53 who responded (25%), reported wearing masks only covering their mouth, not their nose.[49] A study in two hospitals,[35] observed triage nurse behaviour with 118 patients with fever and cough, found that in only 18% of cases the nurses informed patients of the need to wear mask, and in half of those, gave instruction on the need to cover both mouth and nose. A cross-sectional study evaluating the proficiency of the Singaporean public in wearing N95 masks found only 90/714 subjects passed the visual mask fit test; the most common criteria performed incorrectly were: strap placement, leaving a visible gap between the mask and skin, and tightening the nose-clip.[51]

Discomfort and irritation

Several RCTs of specifically measured mask wear discomfort,[17 31-33 47] but most only recorded spontaneously reported events [12 28 34] or did not report any.[15 21 22 26 41 45 50] A trial of household index influenza cases allocated to wear masks or no mask, found the 51 allocated to masks wore them on average 3.8 hours/day and 38 (76%) reported discomfort (Table 2).[17] A study of health care workers in Beijing asked to wear masks for their full shift, found 84% complained of at least one problem (Table 2).[20] In a German household study, 65/172 participants reported problems with mask wearing – most commonly warmth, pain and shortness of breath.[47]

In a trial of healthcare workers comparing surgical and N95 masks to prevent influenza, more workers found the N95 uncomfortable (42%) than the medical mask (10%) when worn an average of 5 hours per day,[33] with significant differences in headaches, difficulty breathing, and pressure on the nose (Table 2). A trial of cloth versus medical masks in healthcare workers found similar rates of discomfort.[32] A community trial comparing surgical and P2 (N95) masks found >50% reporting concerns, primarily discomfort, with similar rates (15% vs. 17%) across groups.[31]

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Discomfort increases with duration of mask wearing. A crossover field trial of 27 healthcare workers found increased discomfort over time; half the subjects were unwilling to wear a medical mask for the full 8-hour shift despite regularly wearing them for short periods.[44]

Table 2. Types of discomfort assessed in trials of face masks used to prevent viral transmission.

First author, year (Type of mask)	Population, Number	Difficulty Breathing	Facial irritation or discomfort	Headache	Other
Canini, 2010[17] (surgical masks)	Household, 105 index cases	34%	14%	-	46% warmth
Chughtai 2019[20] (surgical masks)	Health care workers, 148	12%	17%	6%	-
MacIntyre 2011[33] (Surgical)	Health Care workers, 492	12%	11%	4%	-
MacIntyre 2011[33] (N95)	Health Care workers, 949	19%	52%	13%	-
MacIntyre, 2015[32] (Cloth v Medical)	Health Care workers, 1130	18%	35%	-	-

Two surveys of healthcare staff in Singapore during the SARS epidemic assessed headache and skin reactions.[24 30] In one, 79/212 (37%) reported face-mask-associated headaches, 26 (33%) reported headache frequency exceeding six times/month, and 6 had taken sick leave.[30] Another survey of healthcare workers in Singapore, found that of the 307 staff who used masks regularly, 60% reported acne, 51% facial itch, and 36% rash from N95 mask use.[24] A COVID-19 survey of healthcare workers in Singapore found that 128/158 (81%) developed *de novo* PPE-associated headaches, increasing with duration of use (> 4 hours).[37]

One study (2x2 factorial design) examined the potential of mask fit test and training to mitigate discomfort, but found no clinically or statistically important differences between arms.[38]

Six observational studies reported either general discomfort[33] or spontaneously reported events among participants who wore facemasks.[14 16 27 35 40]

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Psychological

6 studies reported on psychological impacts from wearing face masks (4 RCTs, 2 observational).

Fear

A 3-arm RCT in a household setting found significantly higher risk perception scores in the mask group (38/60) than non-mask groups (30/60) ($p < 0.001$); participants in the mask group were more fearful that they and their family would get sick from the flu.[23 28]

In an observational study, children in a paediatric emergency department waiting room (n = 80) were shown pictures of clinicians wearing either a surgical mask or a clear face shield; 18 children (22.5%) reported surgical masks to be more frightening due to an inability to see clinicians faces, and 14 children (17.5%) reported face shields to be more frightening. However, 47 children (59%) reported that neither were frightening.[25]

Stigma

In a 2-arm cluster-RCT, 15 (29%) patients wearing masks reported they did not like being seen wearing a mask.[17] In a 3-arm RCT, more children reported disliking their parents wearing a P2 mask than a surgical mask (8/92 vs 6/94), however the difference was not significant.[31]

Loneliness

One observational study reported on the loneliness outcome. In a survey investigating the psychosocial effects associated with working in a hospital during the SARS outbreak, 222 (13%) respondents reported a sense of isolation as one reason masks were perceived as bothersome.[36]

Empathy

One RCT reported that the wearing of a face mask by doctors had a negative effect on patient perceptions of the doctors' empathy during consultations, with a mean Consultation and Relational Empathy (CARE) score in the mask group of 33.93 (SD = 7.65, n = 514), and 34.91 (SD = 7.84, n = 516) in the no mask group ($p = 0.04$).[50]

Dyspnoea and other physiological consequences

Studies of physiological impacts were generally done on masks designed for dust, vapours, and other non-transmission purposes; few studied surgical or N95 masks.

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A French crossover study (44 subjects) found surgical masks had no impact on 6-minute walking time, but subjects had an increased sense of dyspneal with a mask: 5.6 vs 4.6cm on a 10cm visual analogue scale (p<0.001),[39] which may come from the increased effort required. A study in 14 adults found that N95 masks increased respiratory resistance in 30 seconds of breathing by over 100%, resulting in average reduction in nasal spirometry of 37%.[29] A study of 20 subjects on a treadmill found the surgical masks increased respiratory rate by 1.6 breaths/minute (p = 0.02), heart rate by 9.5 beats/minute (p<0.001), and transcutaneous CO2 levels of 2.2 mmHg (p<0.001).[43] Finally, a Chinese study of 15 subjects wearing a monitoring garment for respiratory signals found that N95 masks increased both subjective breathing resistance (from none to mild) and increased respiratory rate, the muscle activity of sternomastoid, scalene, diaphragm and abdominal, the fatigue of scalene and intercostal.[18]

Communication

Nine studies (2 RCT, 7 observational) reported on communication difficulties while wearing facemasks.

A trial comparing the use of surgical and N95 masks by healthcare workers to prevent influenza, found more workers in the N95 mask group than the surgical mask group reported mask causing trouble with patient communication (8% vs 3%).[33] Another household-based trial of 15 participants who wore a surgical mask for approximately 60 minutes while performing various tasks,[23 28] found that participants did not report any interference with communication while answering the phone.

In a cross-over trial of 27 healthcare workers,[40] more participants in the surgical mask group reported diminished communication acuity (visual, auditory or vocal) as the reason for discontinuing mask use before the end of an 8 hour shift (7 complaints compared to 4 complaints among N95 mask wearers).

Of 2001 healthcare workers in Toronto responding to a survey during the SARS outbreak, difficulty communicating (47%), and difficulty recognizing people (24%) were identified as key reasons masks (surgical or N95) were perceived as being particularly bothersome.[36] In a survey of 149 healthcare workers,[14] 41 (27.5%) of respondents reported a difficulty 'always'/'most of time' in verbally communicating with patients while wearing a mask.

In another Canadian survey (115 healthcare workers),[35] 26 (23%) respondents reported that wearing masks interfered with their relationships with their patients. Among 148 healthcare workers asked to wear a mask during a 6-8 hour shift, 11 (7.4%) reported trouble communicating with patients.[20]

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In a study of 3 participants evaluating the impact of wearing a surgical or N95 mask on radio reception, all participants were able to accurately record all pilot-recited words regardless of the type of mask worn by the pilot. However, when the aircraft engine was turned on, the accuracy decreased for the N95 mask, compared to surgical or no mask.[48]

In another lab-based study, the performance or absence of fit testing prior to mask use did not affect communication, as 2 participants (out of 21) in each group reported ease of talking to be unsatisfactory.[38]

Mask contamination and other issues

One concern about mask use is the potential for contamination of the mask surface, and subsequent self-inoculation to the wearer's eyes or when demasking. No studies examined that directly, but one study of the health care workers found on average 10% of masks had viral contamination after usage, and that was higher for masks worn > 6 hours (OR 7.9, 95% CI 1.01-61.99) or > 25 patients seen (OR 5.02, 95% CI 1.35-18.60).[20] Given the rates of misuse (see Misuse section above) this contamination raises concerns about self-inoculation.

Several authors have raised concerns about "risk compensation" – non-adherence to other precautions because of the sense of protection – but we found no studies that quantify its extent.

Discussion

We identified 37 studies reporting downsides, harms and adverse events associated with the wearing of facemasks – 15 RCTs and 22 observational studies. The largest number of studies reported on the discomfort and irritation outcome (20 studies), fewest on misuse of mask (4 studies), with no studies directly investigating or quantifying mask contamination or risk compensation behaviour. The only meta-analysable outcome was adherence to facemask wear (17 studies, 11 meta-analysed). 47% more people wore facemasks in the facemask group compared to control, although the percentage of people wearing facemasks in the control group was non-zero in 5 studies; facemask wear adherence was also significantly higher (26%) in the surgical/medical mask group than the N95/P2 group. Risk of bias was generally high for blinding of participants and personnel, and selection bias, and low for attrition and reporting biases.

This is the first systematic review to investigate the downsides of wearing facemasks and forms an important step as a bridge between research and action. The review aligns with the aims of the BESSI (behavioural, environmental, social and systems interventions) collaboration, and addresses

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important evidence gaps in the appropriate use of facemasks. The review's strength lies in its inclusion of non-randomised study designs in addition to RCTs, as trials frequently underreport or fail to report harms.[52] Additionally, the inclusion and exclusion criteria were tested and refined on a test library of 98 references, prior to screening the full search results. The key limitation includes the hospital-setting of most of the included studies: as hospital workers are accustomed to wearing masks, the conclusions may not be fully generalisable to the community. Although this varies among the studies that reported mask use in hospital setting, as there are different confounding factors that may contribute to increased reporting of irritation (e.g. length of shift, air-conditioning on the wards and whether the staff were wearing the full PPE which adds to the full discomfort). We report two differences between the protocol and the review: first, the comparison of facemask to control in the adherence outcome was reported using risk difference (rather than pre-planned odds ratio) to more clearly convey the differences between the two groups (odds ratio for compliance with facemask wear was reported for the facemask versus facemask comparison, however). Second, not having anticipated data availability, we did not prespecify a subgroup analysis of the intervention (facemask wearing) by studies which evaluated facemask wear alone, and studies evaluating facemask with handwashing. Potential behavioural or cultural bias may have been introduced into the findings due to some cultural groups already being more accustomed to wearing face masks when coughing or sick. However, among our included studies only 4 were conducted in the community in Asian countries and of these studies 3 reported adherence to face mask use as an issue and 1 study reported that 264 out of the 624 of the participants who have failed N95 fit test have used masks previously. Furthermore, the use of face masks/coverings in the various study settings compared to their use in a real-life pandemic may differ from a behavioural perspective and this may affect findings related to adherence. However, other reported harms, such as physiological changes, may be less affected by the differences in circumstance.

Several recent systematic reviews have focused on the effectiveness of masks in preventing or reducing viral transmission; some of these reviews reported on harms in the included studies.[8 18 42 53] However, none specifically focused on the wider set of studies examining the physiological, psychological and other adverse effects addressed in this review. The Cochrane review on physical barriers noted the impact of masks on discomfort and communication in some of the randomised trials, and its findings are consistent with this review, but did not extend to studies with outcomes other than viral transmission or non-randomised study design.[3]

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The downsides identified in this review should aid in designing strategies to mitigate problems, and guide the situations where the benefits of masks might outweigh the downsides. As suggested by the higher adherence to surgical masks than to the N95 masks, mitigation of discomforts may also increase adherence to facemask wear, and hence their effectiveness – whether for preventing transmission of the virus by the wearer (e.g. surgical masks) or for preventing inhalation of viral particles in the environment (e.g. N95 masks). Mitigation might be achieved by considering of the when, where, and how of mask wearing (including the fitting process required for some masks like FFP and N95), as the choice of alternative would be dependent on the specific context i.e. it may not be appropriate to use surgical masks or other face masks interchangeably with respirator masks in situations where the goal is to prevent inhalation of aerosolised viral particles as they are not designed for that purpose unlike respirator masks, or by mask redesign or substitution with alternatives (e.g. face shields).

Potential mitigation strategies

Limiting circumstances: Use of facemasks should be restricted to higher risk circumstances, including crowded, indoor spaces, where physical distancing is not possible, e.g. public transport. This recommendation corresponds with suggestions by Chu et.al. who reported that optimum face mask use in public settings could result in a large reduction in infection.[5] Conversely, exercising outdoors is both low risk and has higher downside of wearing masks, because of the increased perceived dyspnoea.

Limiting duration of facemask wear: Duration increases both discomfort and non-adherence. Duration might be decreased by demasking during breaks or scheduling mask breaks. Changing masks more often will help with adherence and the contamination risks, but will increase costs and environmental problems with waste disposal, as well as lead to potential contamination/transmission risks if not performed appropriately.

Modification for specific groups. Some groups are likely to have greater difficulty with mask wearing adherence and correct usage – including children, some patients with mental illnesses, those with cognitive impairment, or respiratory disorders such as asthma or chronic airways disease, and patients with recent facial trauma or oromaxillofacial surgery.[54-56]

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Substitution. Face shields may provide an alternative to facemasks, which may mitigate several of the downsides (e.g. reducing the communication difficulties and breathing resistance), while also providing eye protection. However, there is little evidence on the discomforts of wearing face shields, and on the degree of protection provided. Other innovative mask designs currently being developed, require discomfort and adherence evaluations in addition to the droplet penetration.

Currently, existing research does not allow firm conclusions as there are insufficient data to quantify all of the adverse effects that might reduce the acceptability, adherence, and effectiveness of facemasks. Any new research on facemasks should assess and report the harms and downsides, including behavioural issues (i.e. risk compensation behaviour) and the psychological impact of mandated facemask wear. There is an urgent need for priority funding for high quality research on methods and designs to mitigate downsides of facemask wearing, particularly the assessment of possible alternatives.

Statements

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Contributors

CDM, and PG conceived the study. MB, AMS, NK, PG, and CDM drafted the protocol. JC designed the search strategy and performed the literature search. MB, NK, AMS and JC screened studies for eligibility, and assessed the risk of bias. All authors performed data extraction. AMS performed the meta-analysis. MB, NK, AMS, PG and CDM performed the narrative analysis for non-meta-analysed studies. All authors critically reviewed the manuscript, contributed to its revision, and approved the final version submitted. The corresponding author confirms that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted. CDM is the guarantor.

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Competing interests

All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: no support from any organisation for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

Ethical approval

Not needed.

Data sharing

No additional data available.

Transparency

The corresponding author affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

This manuscript has been deposited as a preprint. Link: https://www.medrxiv.org/content/10.1101/2020.06.16.20133207v1

Patient and public involvement

No patient or public involvement in the study.

Figure legends/captions

Figure 1. PRISMA flow diagram
Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.
Figure 3. Comparison of adherence to face masks vs control
Figure 4. Comparison of adherence to surgical/medical face masks vs N95/P2 masks

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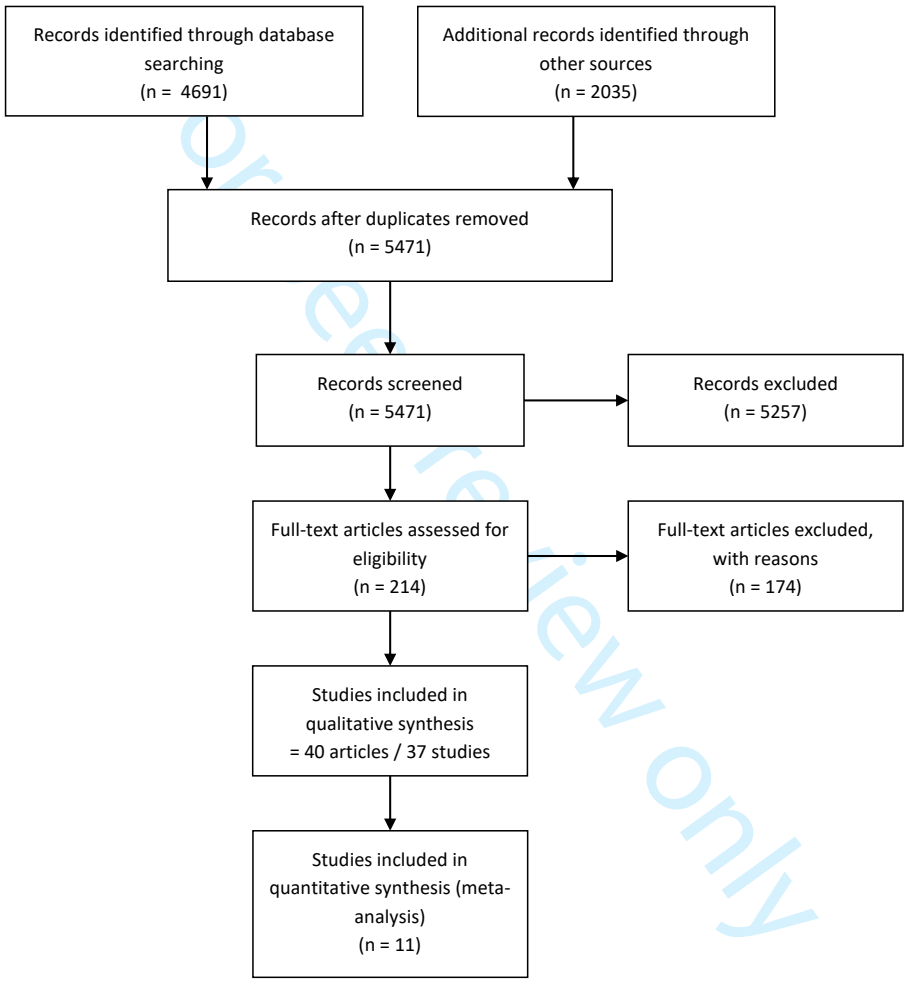


Figure 1. PRISMA flow diagram

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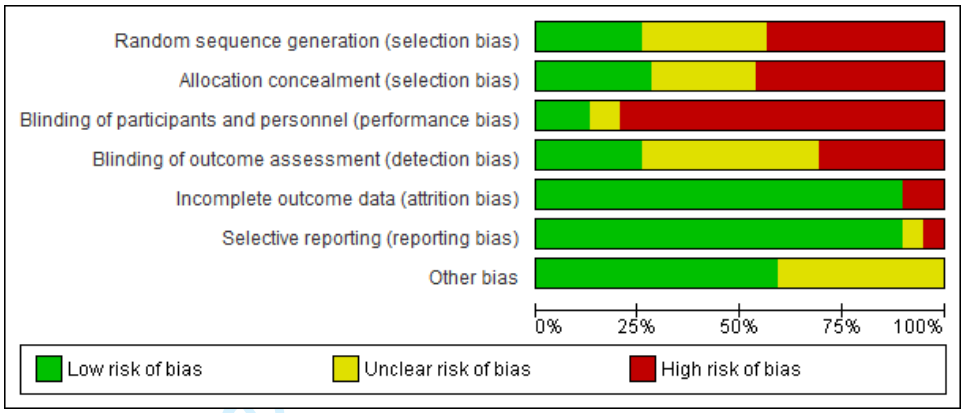


Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

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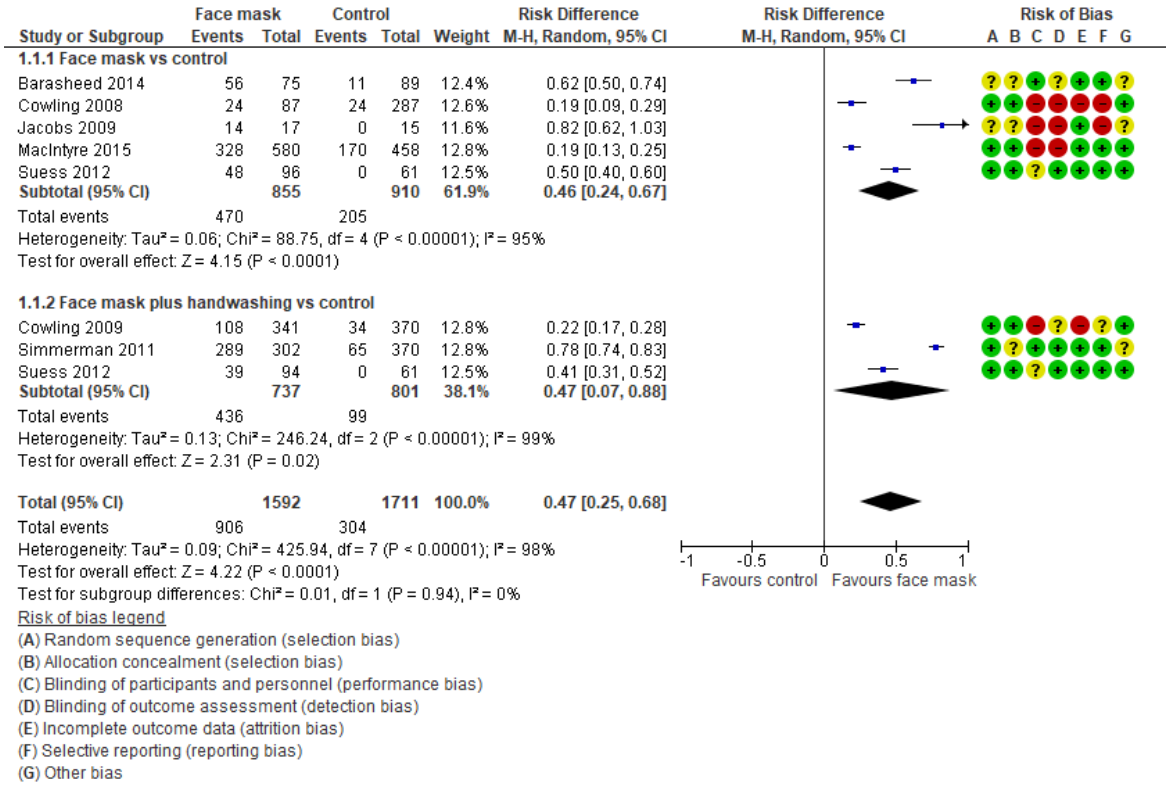


Figure 3. Comparison of adherence to face masks vs control

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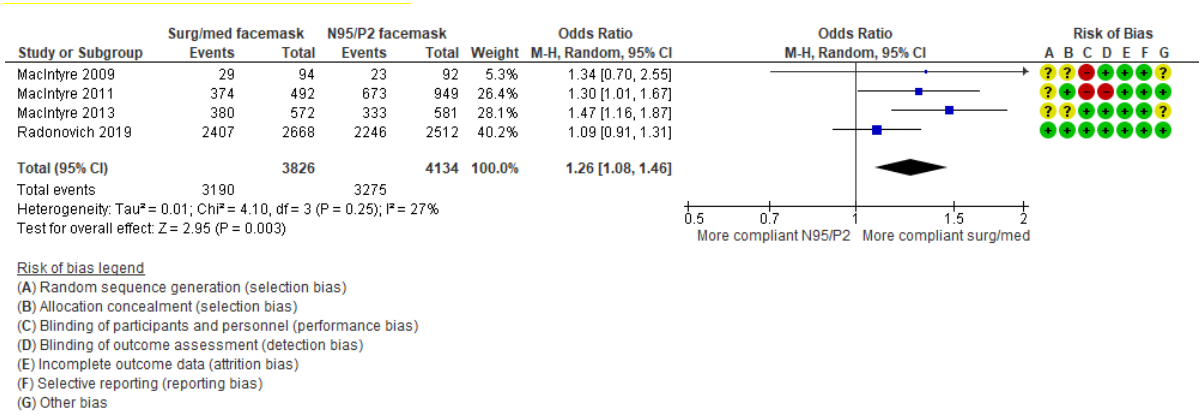


Figure 4. Comparison of adherence to surgical/medical face masks vs N95/P2 masks

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Appendix 1. Summary of the different types of facemasks included in the review and their intended uses

Type of face mask/covering	Description and Intended use
Surgical mask[1, 2]	- Loose-fitting, disposable device, made in different thicknesses - Creates a physical, protective barrier between the mouth and nose of the wearer and potential contaminants in the immediate environment i.e. splashes, sprays, large droplets or splatter. - Primarily intended to prevent the wearer from spreading respiratory droplets - Intended for single use - Does not filter or block very small particles, gases or chemicals in the air that may be transmitted by coughs, sneezes, or certain medical procedures. - Do not provide complete protection from germs and other contaminants because of the loose fit between the surface of the mask and the face.
N95 mask[2, 3]	- Respiratory protective device designed to achieve a very close facial fit. The edges of the respirator are designed to form a seal around the nose and mouth. - Require fit-testing and must be adjusted to the face. - Intended to help reduce wearer exposure to airborne particles as well as larger droplets and protect the wearer from splashes and sprays. - Filters out very fine particles. - Commonly used in healthcare settings. - Should not be shared or reused.
Non-medical face masks/covering[4]	- Cloth face coverings are only intended to help contain the wearer's respiratory droplets from being spread. - At least 2-3 layers needed to block respiratory droplets. - Not intended for use in a clinical setting or explicitly to prevent the transmission of disease between persons. - Some are reusable, and need to be washed after each use.

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Appendix 2. List of excluded face masks

- Oxygen face masks
- OxyMask
- Goldman nasal masks
- Patil-Syracuse Mask
- Ventilation masks
- non-invasive ventilation masks
- Venti-Mask
- Venturi mask
- Mechanical ventilation masks
- Facemask anaesthesia (Anaesthesia induction using face masks)
- Sports protective masks (Hockey masks)
- Diving masks, Scuba masks
- Thermoplastic masks (protective masks for irradiation fields and radiotherapy)
- Bag-valve-mask
- Laryngeal mask
- Oronasal masks
- Sleep masks
- N95 masks with valves (as it does not reduce the spread of virus transmission)

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Appendix 3. Database search strings

Searches conducted 18 May 2020:

PubMed

(("Masks/adverse effects"[MAJR])

OR

((Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR “Face shield”[tiab] OR “Face shields”[tiab] OR Visors[tiab] OR “Face covering”[tiab])

AND

("Adverse effects"[sh] OR “Adverse effect”[tiab] OR “Adverse effects”[tiab] OR “Adversely affect”[tiab] OR “Adverse event”[tiab] OR “Adverse events”[tiab] OR Discomfort[tiab] OR Fatigue[tiab] OR Pain[tiab] OR “Difficulty breathing”[tiab] OR “Breathing difficulty”[tiab] OR “Shortness of breath”[tiab] OR “Risk compensation”[tiab] OR Fomite[tiab] OR Fomites[tiab] OR Dislike[tiab] OR “Physiological impacts”[tiab] OR “Physiological impact”[tiab] OR Sweating[tiab] OR Irritation[tiab] OR “Psychological reactions”[tiab] OR Fear[tiab] OR Communication[tiab] OR “Deaf persons”[tiab] OR Lip-reading[tiab] OR “Lip reading”[tiab] OR Vitiligo[tiab] OR Humidity[tiab] OR “Face dermatitis”[tiab] OR Acne[tiab] OR Redness[tiab] OR Ulcer[tiab] OR “Speech understanding”[tiab] OR “Pinching of skin”[tiab] OR “Moisture buildup”[tiab] OR “Facial warmth”[tiab])

AND

(Relieve[tiab] OR Attitudes[tiab] OR Induced[tiab] OR Modified[tiab] OR Decreased[tiab] OR Decrease[tiab] OR Associated[tiab] OR Impair[tiab] OR Reduce[tiab] OR Reducing[tiab] OR Reduced[tiab] OR Duration[tiab])

AND

(“randomized controlled trial”[pt] OR “controlled clinical trial”[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR “case-control studies”[Mesh] OR “Cohort Studies”[Mesh] OR “case control”[tiab] OR Cohort[tiab] OR “Follow up”[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR Retrospective[tiab] OR “cross sectional”[tiab] OR “Cross-Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh] OR Etiology[sh] OR Effect[tiab] OR Effects[tiab] OR Affect[tiab])))

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR

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"Oxygen mask"[tiab] OR OxyMask[tiab] OR "Nasal masks"[tiab] OR "Patil-Syracuse Mask"[tiab] OR "Ventilation masks"[tiab] OR "Venti-Mask"[tiab] OR "Venturi mask"[tiab] OR "Diving masks"[tiab] OR "Thermoplastic masks"[tiab] OR Bag-valve-mask[tiab] OR "Laryngeal mask"[tiab] OR "Laryngeal masks"[tiab] OR "Oronasal masks"[tiab] OR "Sleep mask"[tiab] OR Editorial[pt] OR Letter[pt] OR "Case reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

OR

(Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR "Face shield"[tiab] OR "Face shields"[tiab] OR Visors[tiab] OR "Face covering"[tiab])

AND

("Respiratory Tract Infections"[Mesh] OR Influenza[tiab] OR "Respiratory Tract Diseases"[Mesh] OR Influenzas[tiab] OR "Influenza-like"[tiab] OR ILI[tiab] OR "common cold"[tiab] OR colds[tiab] OR coryza[tiab] OR coronavirus[Mesh] OR "sars virus"[Mesh] OR coronavirus[tiab] OR Coronaviruses[tiab] OR "coronavirus infections"[Mesh] OR "severe acute respiratory syndrome"[Mesh] OR "severe acute respiratory syndrome"[tiab] OR "severe acute respiratory syndromes"[tiab] OR sars[tiab] OR "respiratory syncytial viruses"[Mesh] OR "respiratory syncytial virus, human"[Mesh] OR "Respiratory Syncytial Virus Infections"[Mesh] OR "respiratory syncytial virus"[tiab] OR "respiratory syncytial viruses"[tiab] OR rsv[tiab] OR parainfluenza[tiab] OR "Respiratory illness"[tiab] OR ((Transmission[tiab]) AND (Coughing[tiab] OR Sneezing[tiab])) OR ((respiratory[tiab] AND Tract[tiab]) AND (infection[tiab] OR Infections[tiab] OR illness[tiab])) OR "Communicable Disease Control"[Mesh] OR "Disease Outbreaks"[Mesh] OR "Disease Transmission, Infectious"[Mesh] OR "Infection Control"[Mesh] OR Transmission[sh] OR "Communicable Disease Control"[tiab] OR "Secondary transmission"[tiab] OR ((Reduced[tiab] OR Reduce[tiab] OR Reduction[tiab] OR Reducing[tiab] OR Lower[tiab]) AND (Incidence[tiab] OR Occurrence[tiab] OR Transmission[tiab] OR Secondary[tiab])) OR Physiological[tiab])

AND

("Adverse effects"[sh] OR "Adverse effect"[tiab] OR "Adverse effects"[tiab] OR "Adverse event"[tiab] OR "Adverse events"[tiab] OR "Index cases"[tiab] OR "Attack rates"[tiab] OR ((Reduce[tiab] OR Reduction[tiab]) AND (Rate[tiab] OR Rates[tiab])) OR Perception[tiab] OR Prevent[tiab] OR Adherence[tiab])

AND

("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR "case-control studies"[Mesh] OR "Cohort Studies"[Mesh] OR "case control"[tiab] OR Cohort[tiab] OR "Follow up"[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR retrospective[tiab] OR "cross sectional"[tiab] OR "Cross-

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Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh])

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR “Oxygen mask”[tiab] OR OxyMask[tiab] OR “Goldman nasal masks”[tiab] OR “Patil-Syracuse Mask”[tiab] OR “Ventilation masks”[tiab] OR “Venti-Mask”[tiab] OR “Venturi mask”[tiab] OR “Hockey mask”[tiab] OR “Diving masks”[tiab] OR “Thermoplastic masks”[tiab] OR Bag-valve-mask[tiab] OR “Laryngeal mask”[tiab] OR “Laryngeal masks”[tiab] OR “Oronasal masks”[tiab] OR “Sleep mask”[tiab] OR "Case Reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

Cochrane CENTRAL

(([mh Masks/AE])

OR

(([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)))

NOT

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([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Nasal masks":ti,ab OR "Patil Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

OR

([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

([mh "Respiratory Tract Infections"] OR Influenza:ti,ab OR [mh "Respiratory Tract Diseases"] OR Influenzas:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR [mh coronavirus] OR [mh "sars virus"] OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR [mh "coronavirus infections"] OR [mh "severe acute respiratory syndrome"] OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR [mh "respiratory syncytial viruses"] OR [mh "respiratory syncytial virus, human"] OR [mh "Respiratory Syncytial Virus Infections"] OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR [mh "Communicable Disease Control"] OR [mh "Disease Outbreaks"] OR [mh "Disease Transmission, Infectious"] OR [mh "Infection Control"] OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

NOT

([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

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Embase

('Masks'/exp OR 'Respiratory protective devices'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR Lip-reading:ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab OR Etiology:ti,ab OR Effect:ti,ab OR Effects:ti,ab OR Affect:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR 'letter'/exp OR 'editorial'/exp OR 'case report'/exp)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

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[embase]/lim

OR

('Mask'/exp OR 'gas mask'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

('respiratory tract infection'/exp OR Influenza:ti,ab OR 'respiratory tract disease'/exp OR Influenzas:ti,ab OR Influenza-like:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR 'Coronavirinae'/exp OR 'SARS coronavirus'/exp OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR 'Coronavirus infection'/exp OR 'severe acute respiratory syndrome'/exp OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR 'Pneumovirus'/exp OR 'Human respiratory syncytial virus'/exp OR 'respiratory syncytial virus infection'/exp OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR 'communicable disease control'/exp OR 'epidemic'/exp OR 'disease transmission'/exp OR 'Infection Control'/exp OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

('adverse event'/exp OR "Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal

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mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR "Case Reports":it)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

[embase]/lim

Europe PMC

https://europepmc.org/search?query=%28TITLE%3A"masks"%20AND%20%28%222019-nCoV%22%20OR%20%222019nCoV%22%20OR%20%22COVID-19%22%20OR%20%22SARS-CoV-2%22%20OR%20%28%22wuhan%22%20AND%20%22coronavirus%22%29%20OR%20%22Coronavirus%22%20OR%20%22Corona%20virus%22%20OR%20%22corona-virus%22%20OR%20%22corona%20viruses%22%20OR%20%22coronaviruses%22%20OR%20%22SARS-CoV%22%20OR%20%22Orthocoronavirinae%22%20OR%20%22MERS-CoV%22%20OR%20%22Severe%20Acute%20Respiratory%20Syndrome%22%20OR%20%22Middle%20East%20Respiratory%20Syndrome%22%20OR%20%28%22SARS%22%20AND%20%22virus%22%29%20OR%20%22soluble%20ACE2%22%20OR%20%28%22ACE2%22%20AND%20%22virus%22%29%20OR%20%28%22ARDS%22%20AND%20%22virus%22%29%20or%20%28%22angiotensin-converting%20enzyme%202%22%20AND%20%22virus%22%29%29%29%20AND%20%28SRC%3APPR%29&page=1

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Appendix 4. List of excluded studies

Excluded studies from database search

No.	Reference	Reason for exclusion
1.	Ahmad, E. F., et al. (2001). "The effect of wearing the veil by Saudi ladies on the occurrence of respiratory diseases." J Asthma 38(5): 423-426.	Study type
2.	Ahmad, I., et al. (2017). "Assessment of knowledge, practice and barrier in use of facemask among university students." Pakistan Journal of Medical and Health Sciences 11(4): 1657-1658.	Reported outcomes
3.	Aiello, A. E., et al. (2010). "A randomized intervention trial of mask use and hand hygiene to reduce seasonal influenza-like illness and influenza infections among young adults in a university setting." International Journal of Infectious Diseases 14: e320.	Duplicate
4.	Aiello, A. E., et al. (2010). "Mask use, hand hygiene, and seasonal influenza-like illness among young adults: a randomized intervention trial." J Infect Dis 201(4): 491-498.	No extractable data
5.	Akyuz, N., et al. (2014). "How and how much do endoscopy professionals protect themselves against infection?" Int J Surg 12(7): 720-724.	Reported outcomes
6.	Albarrak, A. I., et al. (2019). "Middle East Respiratory Syndrome (MERS): Comparing the knowledge, attitude and practices of different health care workers." J Infect Public Health.	Reported outcomes
7.	AlGhamri, A. A., et al. (2013). "The effects of wearing respirators on human fine motor, visual, and cognitive performance." Ergonomics 56(5): 791-802.	Intervention
8.	Allsopp, J., et al. (1997). "Survey of the use of personal protective equipment and prevalence of work related symptoms among dental staff." Occup Environ Med 54(2): 125-134.	Reported outcomes
9.	Alqahtani, A. S., et al. (2016). "Exploring barriers to and facilitators of preventive measures against infectious diseases among Australian Hajj pilgrims: cross-sectional studies before and after Hajj." Int J Infect Dis 47: 53-59.	No extractable data
10.	Alsarheed, M. (2011). "Children's Perception of Their Dentists." Eur J Dent 5(2): 186-190.	Study type
11.	Aydin, O., et al. (24 April 2020). "Performance of fabrics for home-	Reported

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	made masks against spread of respiratory infection through droplets: a quantitative mechanistic study." PREPRINT (Version 1 - 2020.04.19.20071779), Available at: medRxix: https://doi.org/10.1101/2020.04.19.20071779.	outcomes
12.	Badin, A. (1969). "[The surgeon's fatigue and discomfort behind the mask]." Gynaecologia 167(5): 430-433.	Study type
13.	Bansal, S., et al. (2009). "Respirator physiological effects under simulated work conditions." J Occup Environ Hyg 6(4): 221-227.	Intervention
14.	Beuvelet, M., et al. (2017). "Healthcare workers' perception towards the systematic use of mask during a seasonal influenza outbreak in a French University Hospital: A descriptive study." Antimicrobial Resistance and Infection Control 6.	Unable to access full text
15.	Bhoyrul, B., et al. (2019). "A review of non-glove personal protective equipment-related occupational dermatoses reported to EPIDERM between 1993 and 2013." Contact Dermatitis 80(4): 217-221.	Study type
16.	Bischoff, W., et al. (2017). "Evaluation of a novel powered air-purifying respirator (PAPR) vs. a N95 respirator mask for the protection against influenza in a human exposure model." Open Forum Infectious Diseases 4: S168.	Reported outcomes
17.	Brandt, C., et al. (2011). "Attitudes of influenza-vaccinated health care workers toward masks to prevent nosocomial transmission of influenza." Influenza Other Respir Viruses 5(1): 61-66.	Study type
18.	Buitrago, F. (2010). "Hygienic measures and barriers are effective strategies to prevent the transmission of viral infections." FMC Formacion Medica Continuada en Atencion Primaria 17(5): 361.	Study type
19.	Caretti, D. M. and P. D. Gardner (1999). "Respirator fit factor performance while sweating." American Industrial Hygiene Association Journal 60(1): 84-88.	Intervention
20.	Chan, E. Y., et al. (2015). "Knowledge, attitudes, and practices of Hong Kong population towards human A/H7N9 influenza pandemic preparedness, China, 2014." BMC Public Health 15: 943.	Reported outcomes
21.	Chan, M. F., et al. (2008). "Investigating the knowledge, attitudes and practice patterns of operating room staff towards standard and transmission-based precautions: results of a cluster analysis." J Clin Nurs 17(8): 1051-1062.	Reported outcomes
22.	Chan, M. T., et al. (2013). "Mask ventilation and dispersion of exhaled air." Am J Respir Crit Care Med 187(7): e12-14.	Study type

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23.	Cherrie, J. W., et al. (2019). "In-mask temperature and humidity can validate respirator wear-time and indicate lung health status." J Expo Sci Environ Epidemiol 29(4): 578-583.	Intervention
24.	Chung, P. K., et al. (2017). "The process by which perceived autonomy support predicts motivation, intention, and behavior for seasonal influenza prevention in Hong Kong older adults." BMC Public Health 18(1): 65.	Study type
25.	Coca, A., et al. (2017). "Physiological Evaluation of Personal Protective Ensembles Recommended for Use in West Africa." Disaster Med Public Health Prep 11(5): 580-586.	Intervention
26.	Collyer, S. and P. J. Davis (2006). "Effect of facemask use on respiratory patterns of women in speech and singing." J Speech Lang Hear Res 49(2): 412-423.	Reported outcomes
27.	Condon, B. J. and T. Sinha (2010). "Who is that masked person: the use of face masks on Mexico City public transportation during the Influenza A (H1N1) outbreak." Health Policy 95(1): 50-56.	Reported outcomes
28.	Davies, A., et al. (2013). "Testing the efficacy of homemade masks: would they protect in an influenza pandemic?" Disaster Med Public Health Prep 7(4): 413-418.	Reported outcomes
29.	Davis, P. J., et al. (2007). "A survey of Alberta physicians' use of and attitudes toward face masks and face shields in the operating room setting." Am J Infect Control 35(7): 455-459.	Reported outcomes
30.	Del Ferraro, S., et al. (2018). "Hot environment: Effects of a power assisted filtering device incorporating a full face mask on newton manikin thermophysiological response." Occupational and Environmental Medicine 75: A503-A504.	Study type
31.	Del Ferraro, S., et al. (2017). "Thermophysiological response of Newton manikin equipped with power-assisted filtering device incorporating a full-face mask in hot environment." Int J Hyperthermia 33(7): 717-723.	Population
32.	Deris, Z. Z., et al. (2010). "The prevalence of acute respiratory symptoms and role of protective measures among Malaysian Hajj pilgrims." Journal of Travel Medicine 17(2): 82-88.	Study type
33.	Derrick, J. L. and C. D. Gomersall (2005). "Protecting healthcare staff from severe acute respiratory syndrome: filtration capacity of multiple surgical masks." J Hosp Infect 59(4): 365-368.	Reported outcomes
34.	Ding, S., et al. (1998). "[Heat strain for personnel wearing three levels	Intervention

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	of protective ensembles at moderate and heavy workloads]." Wei Sheng Yan Jiu 27(6): 361-364.	
35.	DiLeo T, Roberge RJ, Kim JH. Effect of wearing an N95 filtering facepiece respirator on superomedial orbital infrared indirect brain temperature measurements. J Clin Monit Comput. 2017;31(1):67-73.	Reported outcomes
36.	Dong, J. and Y. Chen (2010). "Impact of the bag exchange procedure on risk of peritonitis." Perit Dial Int 30(4): 440-447.	Reported outcomes
37.	Ellison, I. R. T. (2009). "Surgical masks or N95 respirators for the flu?" Medicine Today 10(12): 69-70.	Reported outcomes
38.	Emanuel, R., et al. (2017). "A study of special care dental patient preference toward choice of mask and visor use by dental clinicians." Spec Care Dentist 37(4): 164-167.	Reported outcomes
39.	Fabian, P., et al. (2011). "Influenza virus aerosols in human exhaled breath: Particle size, culturability, and effect of surgical masks." Epidemiology 22(1): S51.	Reported outcomes
40.	Field, M. H., et al. (2020). "Hydrocolloid dressing strip over bridge of nose to relieve pain and pressure from Filtered Face Piece (FFP) masks during the coronavirus (COVID-19) pandemic." Ann R Coll Surg Engl 102(5): 394-396.	Study type
41.	Fierro, J. L., et al. (2015). "Barriers to the Use of PPE to Prevent Pertussis Exposures in a Pediatric Primary Care Network." J Pediatric Infect Dis Soc 4(1): 49-56.	Reported outcomes
42.	Flury, D., et al. (2019). "Health-care associated Influenza Prevention Project (HAIP): First multicenter surveillance results (nosocomial influenza and prevention measures)." Antimicrobial Resistance and Infection Control 8.	Reported outcomes
43.	Fox, S. H. and A. B. DuBois (1993). "The effect of evaporative cooling of respiratory protective devices on skin temperature, thermal sensation, and comfort." Am Ind Hyg Assoc J 54(12): 705-710.	Intervention
44.	Ganczak, M. and Z. Szych (2007). "Surgical nurses and compliance with personal protective equipment." J Hosp Infect 66(4): 346-351.	Reported outcomes
45.	Gu, J., et al. (2015). "Preventive behaviors and mental distress in response to H1N1 among university students in Guangzhou, China." Asia Pac J Public Health 27(2): Np1867-1879.	Reported outcomes
46.	Guo, Y. P., et al. (2008). "Evaluation on masks with exhaust valves and with exhaust holes from physiological and subjective responses." J	Intervention

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	Physiol Anthropol 27(2): 93-102.	
47.	Gwosdow, A. R., et al. (1989). "Effect of thermal conditions on the acceptability of respiratory protective devices on humans at rest." Am Ind Hyg Assoc J 50(4): 188-195.	Intervention
48.	Hassaniazad, M., et al. (2016). "Preventive measures for Crimean-Congo hemorrhagic fever in healthcare workers; how high is the chance of transmission?" Acta Medica Mediterranea 32: 2017-1024.	Reported outcomes
49.	Harber, P., et al. (1989). "Relationship of subjective tolerance of respirator loads to physiologic effects and psychophysical load sensitivity." J Occup Med 31(8): 681-686.	Intervention
50.	Hardis, K. E., et al. (1983). "Correlation of qualitative and quantitative results from testing respirator fit." Am Ind Hyg Assoc J 44(2): 78-87.	Intervention
51.	Hickey, J., et al. (2014). "Pandemic preparedness: perceptions of vulnerable migrants in Thailand towards WHO-recommended non-pharmaceutical interventions: a cross-sectional study." BMC Public Health 14: 665.	Reported outcomes
52.	Hines, S. E., et al. (2019). "User acceptance of reusable respirators in health care." American Journal of Infection Control 47(6): 648-655.	Intervention
53.	Ho, R. C., et al. (2014). "Impact of 2013 south Asian haze crisis: study of physical and psychological symptoms and perceived dangerousness of pollution level." BMC Psychiatry 14: 81.	Reported outcomes
54.	Huang, W. and L. Morawska (2019). "Face masks could raise pollution risks." Nature 574(7776): 29-30.	Study type
55.	Huynh. TD (30 March 2020). "The more I fear about COVID-19, the more I wear medical masks: A survey on risk perception and medical masks uses." PREPRINT (Version 1 - 2020.03.26.20044388), Available at: medRxix: https://doi.org/10.1101/2020.03.26.20044388.	Reported outcomes
56.	Ioannidou, C., et al. (2010). "Promotion of hygiene measures to prevent pandemic influenza transmission in greek nursery schools: The teachers'a perspectives." Acta Paediatrica, International Journal of Paediatrics 99: 88.	Reported outcomes
57.	Jaeger, J. L., et al. (2011). "Transmission of 2009 pandemic influenza A (H1N1) virus among healthcare personnel-Southern California, 2009." Infect Control Hosp Epidemiol 32(12): 1149-1157.	Reported outcomes
58.	James, R., et al. (1984). "Effects of respirators under heat/work conditions." Am Ind Hyg Assoc J 45(6): 399-404.	Intervention

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59.	Jencsik, T., et al. (2017). "Influenza-related attitudes of healthcare workers at institutions for acute and chronic diseases." Value in Health 20(9): A791-A792.	Study type
60.	Johnson, A. T., et al. (2000). "Effect of external dead volume on performance while wearing a respirator." Aihaj 61(5): 678-684.	Intervention
61.	Johnson, A. T., et al. (2000). "Communication using a telephone while wearing a respirator." Aihaj 61(2): 264-267.	Intervention
62.	Kawanami, S., et al. (2018). "Physiological effects of wearing powered airpurifying respirators with different blower system." Occupational and Environmental Medicine 75: A201.	Intervention
63.	Kim, J. H., et al. (2015). "Effect of wearing an N95 respirator on infrared tympanic membrane temperature measurements." J Clin Monit Comput 29(6): 691-695.	Reported outcomes
64.	Kishimoto, K., et al. (2016). "[Whether Visual Information Loss of Expression by Community Pharmacist Masks to Effect a Change in Feeling of Trust among Help-seeker?]." Yakugaku Zasshi 136(10): 1401-1413.	Study type
65.	Krishna, S., et al. (2019). "The gloves are off: The state of personal protective equipment (PPE) use in contact, droplet, and standard precautions at a major teaching hospital." Open Forum Infectious Diseases 6: S429-S430.	Study type
66.	Kroo, L., et al. (29 April 2020). "Pneumask: Modified Full-Face Snorkel Masks as Reusable Personal Protective Equipment for Hospital Personnel." PREPRINT (Version 1 - 2020.04.24.20078907), Available at: medRxix: https://doi.org/10.1101/2020.04.24.20078907.	Intervention
67.	Latheef, F., et al. (2015). "Personal protective equipment-associated dermatoses reported to EPIDERM: A 20-year review." British Journal of Dermatology 173: 172-173.	Duplicate
68.	Lau, J. T., et al. (2003). "Monitoring community responses to the SARS epidemic in Hong Kong: from day 10 to day 62." J Epidemiol Community Health 57(11): 864-870.	Reported outcomes
69.	Leung, N. H., et al. (06 March 2020). "Respiratory Virus Shedding in Exhaled Breath and Efficacy of Face Masks." PREPRINT (Version 1), Available at: Research Square [+https://doi.org/10.21203/rs.3.rs-16836/v1+],.	Reported outcomes
70.	Lee, H. P. and Y. Wang de (2011). "Objective assessment of increase in breathing resistance of N95 respirators on human subjects." Ann	Reported outcomes

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	Occup Hyg 55(8): 917-921.	
71.	Lee, J. A., et al. (2017). "Compliance of personal protective equipment in a hospital." Antimicrobial Resistance and Infection Control 6.	Reported outcomes
72.	Li, Y., et al. (2006). "Antimicrobial effect of surgical masks coated with nanoparticles." J Hosp Infect 62(1): 58-63.	Intervention
73.	Li, C., et al. (2020). "[Management and prevention of common skin problems during epidemic prevention and control of COVID-19]." Nan Fang Yi Ke Da Xue Xue Bao 40(2): 168-170.	Reported outcomes
74.	Loeb, M., et al. (2009). "Surgical mask vs N95 respirator for preventing influenza among health care workers: a randomized trial." Jama 302(17): 1865-1871.	Reported outcomes
75.	Loibner, M., et al. (2019). "Limiting factors for wearing personal protective equipment (PPE) in a health care environment evaluated in a randomised study." PLoS One 14(1): e0210775.	Intervention
76.	Lonnroth, E. C. and H. Shahnavaz (1998). "Adverse health reactions in skin, eyes, and respiratory tract among dental personnel in Sweden." Swed Dent J 22(1): 33-45.	Reported outcomes
77.	Lopes, A. C., et al. (2008). "[Adherence to standard precautions by the public pre-hospital health team in Belo Horizonte, Minas Gerais State, Brazil]." Cad Saude Publica 24(6): 1387-1396.	Reported outcomes
78.	Lunding, S., et al. (1998). "[Poor compliance with universal precautions among Danish physicians]." Ugeskr Laeger 160(12): 1789-1793.	Reported outcomes
79.	Lutz, M., et al. (2019). "Physiological effects of a spit sock." Am J Emerg Med 37(2): 291-293.	Intervention
80.	Ma, X., et al. (2014). "Knowledge, attitudes and practices relating to influenza A(H7N9) risk among live poultry traders in Guangzhou City, China." BMC Infect Dis 14: 554.	Reported outcomes
81.	MacIntyre, C. R., et al. (2016). "Cluster randomised controlled trial to examine medical mask use as source control for people with respiratory illness." BMJ Open 6(12): e012330.	Reported outcomes
82.	Martins, R. S., et al. (2013). "Adequacy to the use of personal protective equipment in the isolations of emergency." Antimicrobial Resistance and Infection Control 2.	Reported outcomes
83.	Maschmeyer, G., et al. (2009). "A prospective, randomised study on the use of well-fitting masks for prevention of invasive aspergillosis in	Intervention

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	high-risk patients." Ann Oncol 20(9): 1560-1564.	
84.	Mathews, R., et al. (2008). "Provision and use of personal protective equipment and safety devices in the National Study to Prevent Blood Exposure in Paramedics." Am J Infect Control 36(10): 743-749.	Study type
85.	Mauritzson-Sandberg, E. (1995). "An evaluation of respiratory protective devices used in children's evacuation." Ergonomics 38(4): 707-713.	Purpose of study
86.	Meyer, J. P., et al. (1997). "Field study of subjective assessment of negative pressure half-masks. Influence of the work conditions on comfort and efficiency." Appl Ergon 28(5-6): 331-338.	Purpose of study
87.	Millqvist, E., et al. (1995). "Prevention of asthma induced by cold air by cellulose-fabric face mask." Allergy 50(3): 221-224.	Reported outcomes
88.	Morishima, M. and K. Kishida (2018). "Understanding attitudes toward hygiene mask use in Japanese daily life by using a repeated cross-sectional survey." Work 61(2): 303-311.	Study type
89.	Morishima, M., et al. (2011). "An investigation of the use of hay fever masks for the youth." J Hum Ergol (Tokyo) 40(1): 151-156.	Intervention
90.	Moser, A., et al. (2015). "Rates and determinants of vaccination against seasonal and pandemic influenza in Swiss prehospital emergency medical services workers." Praxis 104: 142.	Study type
91.	Moser, A., et al. (2016). "Vaccination Against Seasonal or Pandemic Influenza in Emergency Medical Services." Prehosp Disaster Med 31(2): 155-162.	Study type
92.	Motoyama, Y. L., et al. (2016). "Airflow-Restricting Mask Reduces Acute Performance in Resistance Exercise." Sports (Basel) 4(4).	Intervention
93.	Nichol, K., et al. (2008). "The individual, environmental, and organizational factors that influence nurses' use of facial protection to prevent occupational transmission of communicable respiratory illness in acute care hospitals." Am J Infect Control 36(7): 481-487.	Study type
94.	Nichol, K., et al. (2013). "Behind the mask: Determinants of nurse's adherence to facial protective equipment." Am J Infect Control 41(1): 8-13.	Study type
95.	Nielsen, R., et al. (1987). "The effect of temperature and humidity levels in a protective mask on user acceptability during exercise." Am Ind Hyg Assoc J 48(7): 639-645.	Intervention

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96.	Palmiero, A. J., et al. (2016). "Speech intelligibility assessment of protective facemasks and air-purifying respirators." J Occup Environ Hyg 13(12): 960-968.	Study type
97.	Parveen, S. N. A., et al. (2018). "Identifying acceptable and feasible infection control interventions for nipah encephalitis outbreaks in bangladesh." American Journal of Infection Control 46(6): S24.	Study type
98.	Peponis, T., et al. (2017). "Trauma team utilization of universal precautions: if you see something, say something." Eur J Trauma Emerg Surg 43(1): 145-150.	Reported outcomes
99.	Perez, V., et al. (2012). "Stress, adherence to preventive measures for reducing influenza transmission and influenza-like illness." J Epidemiol Community Health 66(7): 605-610.	No extractable data
100.	Radonovich Jr, L. J., et al. (2009). "Diminished speech intelligibility associated with certain types of respirators worn by healthcare workers." Journal of Occupational and Environmental Hygiene 7(1): 63-70.	Intervention
101.	Radonovich, L., et al. (2018). "Results of the respiratory protection effectiveness clinical trial (respect)." Open Forum Infectious Diseases 5: S51.	Duplicate
102.	Ramirez, J. A. and P. T. O'Shaughnessy (2016). "The effect of simulated air conditions on N95 filtering facepiece respirators performance." J Occup Environ Hyg 13(7): 491-500.	Study type
103.	Raven, P. B., et al. (1977). "Maximal stress test performance while wearing a self-contained breathing apparatus." J Occup Med 19(12): 802-806.	Intervention
104.	Rengasamy, S. and B. C. Eimer (2013). "N95-companion measurement of cout/cin ratios for two n95 filtering facepiece respirators and one surgical mask." J Occup Environ Hyg 10(10): 527-532.	Population
105.	Roberge, R. (2011). "Facemask use by children during infectious disease outbreaks." Biosecur Bioterror 9(3): 225-231.	Study type
106.	Roberge, R. J., et al. (2010). "Surgical mask placement over N95 filtering facepiece respirators: physiological effects on healthcare workers." Respirology 15(3): 516-521.	Intervention
107.	Roberge, R. J., et al. (2010). "Reusable elastomeric air-purifying respirators: physiologic impact on health care workers." Am J Infect Control 38(5): 381-386.	Intervention

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108.	Rozzi, T., et al. (2012). "Pilot study of aromatic hydrocarbon adsorption characteristics of disposable filtering facepiece respirators that contain activated carbon." J Occup Environ Hyg 9(11): 624-629.	Purpose of study
109.	van der Sande, M., et al. (2008). "Professional and home-made face masks reduce exposure to respiratory infections among the general population." PLoS ONE 3(7): e2618.	Reported outcomes
110.	Santos, P. B., et al. (2013). "Adherence to biosafety practices by nursing staff in the face of situations of occupational risk." Antimicrobial Resistance and Infection Control 2.	Unable to access full text
111.	Schmitt, C. H. (2004). "Secrets behind the mask." US News World Rep 137(4): 37-41.	Study type
112.	Seliga, R., et al. (1991). "Effect of work load and respirator wear on postural stability, heart rate, and perceived exertion." Am Ind Hyg Assoc J 52(10): 417-422.	Intervention
113.	Seng, M., et al. (2018). "Comfort and exertion while using filtering facepiece respirators with exhalation valve and an active venting system among male military personnel." Singapore Med J 59(6): 327-334.	Purpose of study
114.	Smith, C. L., et al. (2013). "Carbon dioxide rebreathing in respiratory protective devices: influence of speech and work rate in full-face masks." Ergonomics 56(5): 781-790.	Intervention
115.	Sokol, K. A., et al. (2016). "Masks for prevention of respiratory viruses on the BMT unit: results of a quality initiative." Transpl Infect Dis 18(6): 965-967.	Reported outcomes
116.	Stebbins, S., et al. (2009). "Using nonpharmaceutical interventions to prevent influenza transmission in elementary school children: parent and teacher perspectives." J Public Health Manag Pract 15(2): 112-117.	Reported outcomes
117.	Stockwell, R. E., et al. (2018). "Prolonged wear of face masks prevents the release of pseudomonas aeruginosa aerosols during coughing in people with cystic fibrosis." Pediatric Pulmonology 53: 269.	Unable to access full text
118.	Strauch, A. L., et al. (2016). "Assessing the efficacy of tabs on filtering facepiece respirator straps to increase proper doffing techniques while reducing contact transmission of pathogens." J Occup Environ Hyg 13(10): 794-801.	Intervention
119.	Sullivan, K. (1993). "Surgical face masks: protection of self or patient?" N Z Dent J 89(398): 137.	Study type

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120.	Suruda, A., et al. (2003). "Fatal injuries in the United States involving respirators, 1984-1995." Appl Occup Environ Hyg 18(4): 289-292.	Study type
121.	Tak, C. N., et al. (2009). "Preventing healthcare workers from acquiring influenza." Infection Control and Hospital Epidemiology 30(3): 292-295.	Reported outcomes
122.	Tan, K. T. and M. W. Greaves (2004). "N95 acne." Int J Dermatol 43(7): 522-523.	Study type
123.	Tang, C. S. and C. Y. Wong (2004). "Factors influencing the wearing of facemasks to prevent the severe acute respiratory syndrome among adult Chinese in Hong Kong." Prev Med 39(6): 1187-1193.	Reported outcomes
124.	Tataris, K., et al. (2010). "Patient attitudes and compliance while wearing surgical masks to prevent disease transmission." Academic Emergency Medicine 17: S58.	Unable to access full text
125.	Vanjak, D., et al. (2006). "[Cases of pertussis among healthcare workers in a maternity ward: management of a health alert]." Med Mal Infect 36(3): 151-156.	Duplicate reference
126.	Vijaya, K., et al. (2005). "Behaviour of Singaporeans during the SARS outbreak: The impact of anxiety and public satisfaction with media information." International Journal of Health Promotion and Education 43(1): 17-22.	Reported outcomes
127.	Wong, S. Y., et al. (2012). "Primary care physicians' response to pandemic influenza in Hong Kong: a mixed quantitative and qualitative study." Int J Infect Dis 16(9): e687-691.	Reported outcomes
128.	Yang, P., et al. (2011). "Mask-wearing and respiratory infection in healthcare workers in Beijing, China." Braz J Infect Dis 15(2): 102-108.	Study type

Excluded studies identified from Forward & Backward citation analysis

129.	Barratt, R., et al. (2019). "Enablers of, and barriers to, optimal glove and mask use for routine care in the emergency department: an ethnographic study of Australian clinicians." Australasian Emergency Care.	Study type
130.	Caretti, D. M. and L. C. Strickler (2003). "Speech intelligibility during respirator wear: Influences of respirator speech diaphragm size and background noise." American Industrial Hygiene Association Journal 64(6): 846-850.	Intervention
131.	Chughtai, A. A., et al. (2015). "Practices around the use of masks and	Reported

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	respirators among hospital health care workers in 3 diverse populations." American Journal of Infection Control 43(10): 1116-1118.	outcomes
132.	Enerson, D. M., et al. (1967). "Heat and moisture trapping beneath surgical face masks: a consideration of factors affecting the surgeon's discomfort and performance." Surgery 62(6): 1007-1016.	Reported outcomes
133.	Goh, D. Y. T., et al. (2019). "A randomised clinical trial to evaluate the safety, fit, comfort of a novel N95 mask in children." Scientific Reports 9(1).	Purpose of study
134.	Harber P, Bansal S, Santiago S, et al. Multidomain subjective response to respirator use during simulated work. Journal of Occupational and Environmental Medicine. 2009;51(1):38-45.	Reported outcomes
135.	Harber, P., et al. (2010). "Respirator physiologic impact in persons with mild respiratory disease." Journal of Occupational and Environmental Medicine 52(2): 155-162.	Intervention
136.	Harber, P., et al. (2011). "Respirator impact on work task performance." Journal of Occupational and Environmental Medicine 53(1): 22-26.	Reported outcomes
137.	Honarbakhsh, M., et al. (2018). "Knowledge, perceptions and practices of healthcare workers regarding the use of respiratory protection equipment at Iran hospitals." Journal of Infection Prevention 19(1): 29-36.	Reported outcomes
138.	Jacobs, J. L., et al. (2009). "Use of surgical face masks to reduce the incidence of the common cold among health care workers in Japan: A randomized controlled trial." American Journal of Infection Control 37(5): 417-419.	Duplicate
139.	Jones, J. G. (1991). "The physiological cost of wearing a disposable respirator." American Industrial Hygiene Association Journal 52(6): 219-225.	Intervention
140.	Kim, J. H., et al. (2013). "Pulmonary and heart rate responses to wearing N95 filtering facepiece respirators." American Journal of Infection Control 41(1): 24-27.	Reported outcomes
141.	Kim, J. H., et al. (2015). "Pressure drop of filtering facepiece respirators: How low should we go?" International Journal of Occupational Medicine and Environmental Health 28(1): 71-80.	Intervention
142.	Kim, J. H., et al. (2016). "Physiologic and fit factor profiles of N95 and P100 filtering facepiece respirators for use in hot, humid	Reported

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	environments." American Journal of Infection Control 44(2): 194-198.	outcomes
143.	Lim, E. C. H., et al. (2007). "Headaches and the N95 face-mask amongst healthcare providers [2]." Acta Neurologica Scandinavica 116(1): 73.	Duplicate
144.	Li Y, Tokura H, Guo YP, et al. Effects of wearing N95 and surgical facemasks on heart rate, thermal stress and subjective sensations. International Archives of Occupational and Environmental Health. 2005;78(6):501-509.	No extractable data
145.	Lin, Y. C. and C. P. Chen (2019). "Thermoregulation and thermal sensation in response to wearing tight-fitting respirators and exercising in hot-and-humid indoor environment." Building and Environment 160.	Intervention
146.	Locatelli, S. M., et al. (2014). "Health care workers' reported discomfort while wearing filtering face-piece respirators." Workplace Health and Safety 62(9): 362-368.	Reported outcomes
147.	Luximon, Y., et al. (2016). "Time dependent infrared thermographic evaluation of facemasks." Work 54(4): 825-835.	Reported outcomes
148.	MacIntyre, C. R., et al. (2014). "Efficacy of face masks and respirators in preventing upper respiratory tract bacterial colonization and co-infection in hospital healthcare workers." Preventive Medicine 62: 1-7.	Reported outcomes
149.	MacIntyre, C. R., et al. (2013). "A randomized clinical trial of three options for N95 respirators and medical masks in health workers." American Journal of Respiratory and Critical Care Medicine 187(9): 960-966.	Duplicate
150.	Mauritzson-Sandberg, E. V. A. (1991). "Psychological effects on prolonged use of respiratory protective devices in children." Ergonomics 34(3): 313-319.	Reported outcomes
151.	McDiarmid, M., et al. (2020). "N95 Respirators vs Medical Masks in Outpatient Settings." JAMA - Journal of the American Medical Association 323(8): 789.	Study type
152.	Morishima, M., et al. (2014). "Experiences and problems with hygiene masks reported by japanese hay fever sufferers." International Journal of Clothing Science and Technology 26(4): 262-273.	Intervention
153.	Morishima, M., et al. (2017). "Problems related to mask use in hay fever sufferers by repeated surveys." International Journal of Clothing Science and Technology 29(1): 123-134.	Purpose of study

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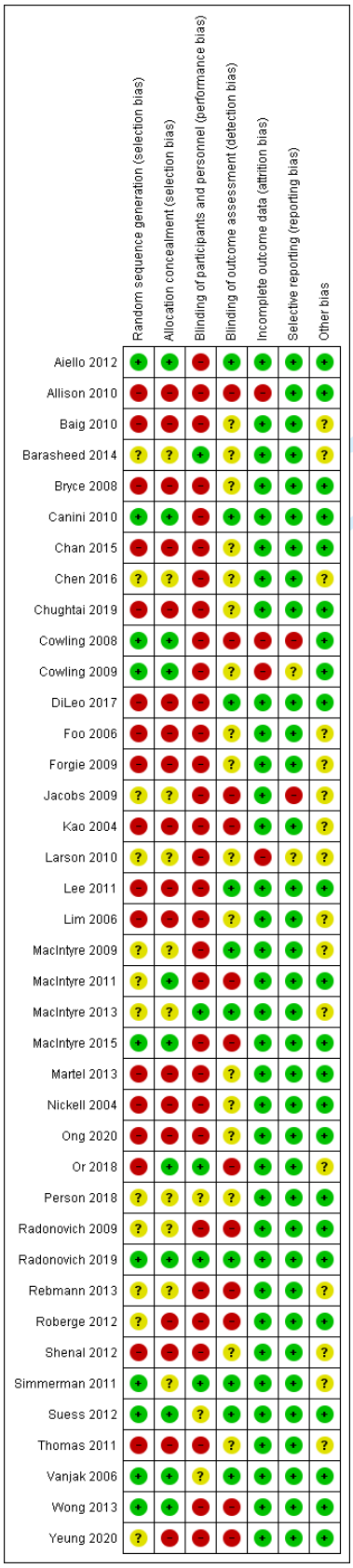
154.	Ong, J. J. Y., et al. (2020). "Headaches Associated With Personal Protective Equipment – A Cross-Sectional Study Among Frontline Healthcare Workers During COVID-19." Headache 60(5): 864-877.	Duplicate
155.	Phin NF, Rylands AJ, Allan J, Edwards C, Enstone JE, Nguyen-Van-Tam JS. Personal protective equipment in an influenza pandemic: a UK simulation exercise. Journal of Hospital Infection. 2009;71(1):15-21.	Study design
156.	Radonovich Jr, L. J., et al. (2010). "Diminished speech intelligibility associated with certain types of respirators worn by healthcare workers." Journal of Occupational and Environmental Hygiene 7(1): 63-70.	Intervention
157.	Radonovich, L. J., et al. (2019). "A tolerability assessment of new respiratory protective devices developed for health care personnel: A randomized simulated clinical study." PLoS ONE 14(1).	No extractable data
158.	Roberge, R., et al. (2012). "Thermal burden of N95 filtering facepiece respirators." Annals of Occupational Hygiene 56(7): 808-814.	Reported outcomes
159.	Roberge, R. J. (2008). "Effect of surgical masks worn concurrently over N95 filtering facepiece respirators: Extended service life versus increased user burden." Journal of Public Health Management and Practice 14(2): E19-E26.	Study type
160.	Roberge RJ, Coca A, Williams WJ, Powell JB, Palmiero AJ. Physiological impact of the n95 filtering facepiece respirator on healthcare workers. Respiratory Care. 2010;55(5):569-577.	Intervention
161.	Roberge, R. J., et al. (2010). "Effect of exhaled moisture on breathing resistance of N95 filtering facepiece respirators." Annals of Occupational Hygiene 54(6): 671-677.	Study type
162.	Roberge, R. J., et al. (2012). "N95 filtering facepiece respirator deadspace temperature and humidity." Journal of Occupational and Environmental Hygiene 9(3): 166-171.	Reported outcomes
163.	Roberge, R. J., et al. (2013). "Impact of low filter resistances on subjective and physiological responses to filtering facepiece respirators." PLoS ONE 8(12).	Intervention
164.	Roberge RJ, Kim JH, Powell JB. N95 respirator use during advanced pregnancy. American Journal of Infection Control. 2014;42(10):1097-1100.	No extractable data
165.	Sakaguchi, H., et al. (2010). "Maintenance of influenza virus infectivity on the surfaces of personal protective equipment and clothing used in healthcare settings." Environmental Health and Preventive Medicine	Purpose of study

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	15(6): 344-349.	
166.	Shi, J., et al. (2017). "Cardiovascular benefits of wearing particulate-filtering respirators: A randomized crossover trial." Environmental Health Perspectives 125(2): 175-180.	Purpose of study
167.	Shigayeva, A., et al. (2007). "Factors associated with critical-care healthcare workers' adherence to recommended barrier precautions during the Toronto severe acute respiratory syndrome outbreak." Infection Control and Hospital Epidemiology 28(11): 1275-1283.	No extractable data
168.	Skaria, S. D. and G. C. Smaldone (2014). "Respiratory source Control using surgical masks with nanofiber media." Annals of Occupational Hygiene 58(6): 771-781.	Purpose of study
169.	Tamura, E., et al. (2013). "Effect of pharmacists wearing a mask on patient consultation intention." Yakugaku Zasshi 133(6): 737-745.	Study type
170.	Tong, P. S. Y., et al. (2015). "Respiratory consequences of N95-type Mask usage in pregnant healthcare workers-A controlled clinical study." Antimicrobial Resistance and Infection Control 4(1).	Intervention
171.	Turnberg, W., et al. (2009). "Personal healthcare worker (HCW) and work-site characteristics that affect HCWs' use of respiratory-infection control measures in ambulatory healthcare settings." Infection Control and Hospital Epidemiology 30(1): 47-52.	Reported outcomes
172.	Wada, K., et al. (2012). "Wearing face masks in public during the influenza season may reflect other positive hygiene practices in Japan." BMC Public Health 12(1).	Reported outcomes
173.	Wu, S., et al. (2011). "Anxiety during respirator use: Comparison of two respirator types." Journal of Occupational and Environmental Hygiene 8(3): 123-128.	Study type
174.	Yip, W. L., et al. (2005). "The effect of wearing a face mask on body temperature." Hong Kong Journal of Emergency Medicine 12(1): 23-27.	Reported outcomes

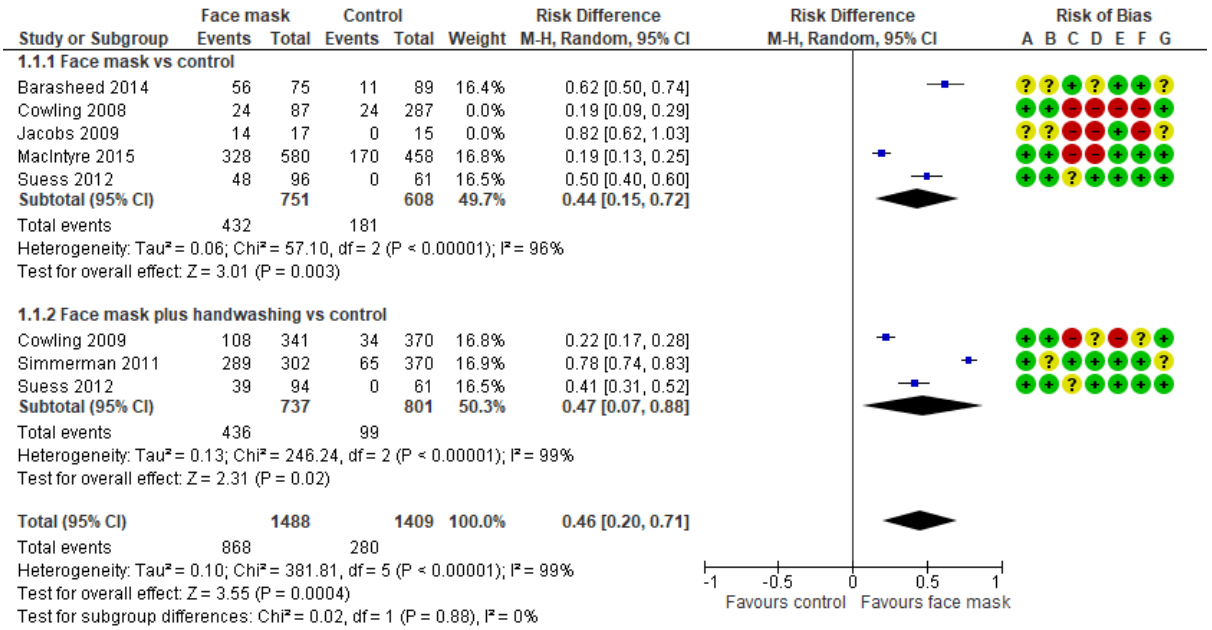
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Appendix 5. Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

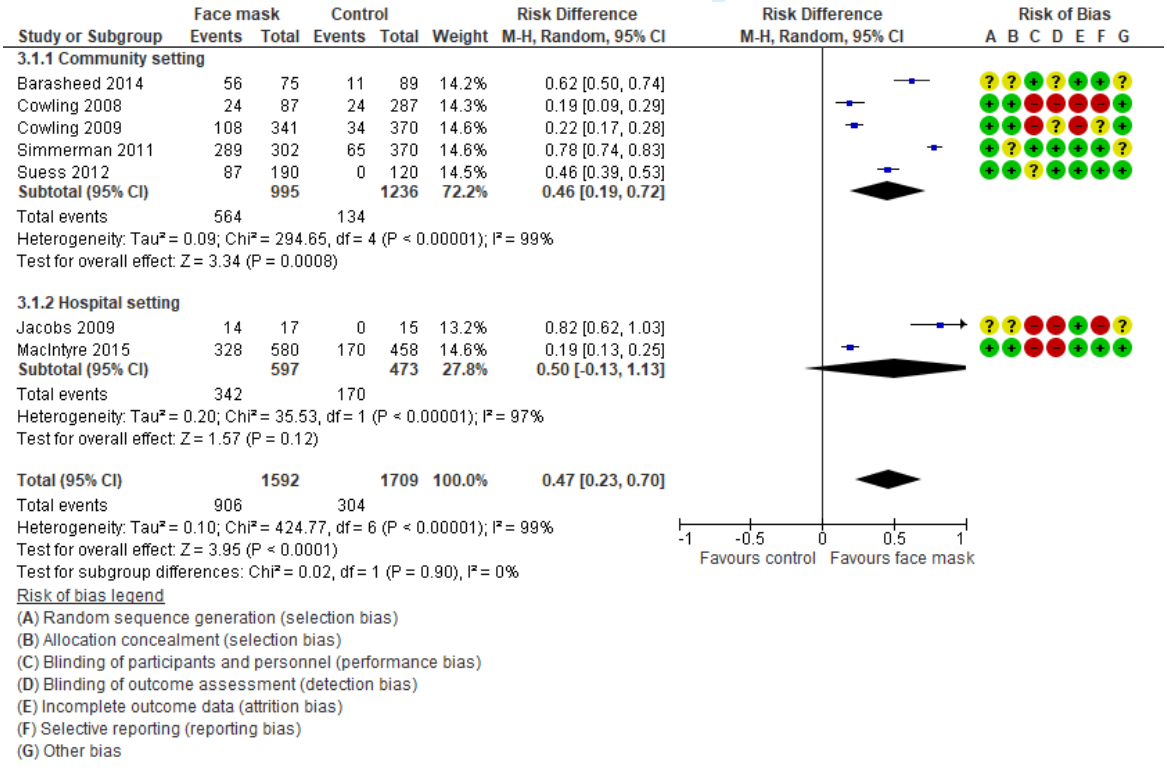


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Appendix 5. Figure 4. Analysis with studies rated at high risk of bias in more than 2 domains excluded



Appendix 5. Figure 5. Analysis with studies sub-grouped by setting (community setting, hospital setting)





PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5, 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-17
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11, 12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11, 12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Title page

Title

Downsides of face masks and possible mitigation strategies: a systematic review and meta-analysis

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Abstract

Objective

To identify, appraise, and synthesise studies evaluating the downsides of wearing facemasks in any setting. We also discuss potential strategies to mitigate these downsides.

Design

Systematic review and meta-analysis

Data sources

PubMed, Embase, CENTRAL, EuropePMC were searched (inception-18/5/2020), and clinical registries were searched via CENTRAL. We also did a forward-backward citation search of the included studies.

Inclusion criteria

We included randomised controlled trials and observational studies comparing facemask use to any active intervention or to control.

Data extraction and analysis

Two author pairs independently screened articles for inclusion, extracted data and assessed the quality of included studies. The primary outcomes were compliance, discomforts, harms, and adverse events of wearing facemasks.

Results

We screened 5471 articles, including 37 (40 references); 11 were meta-analysed. For mask wear adherence, 47% (95%CI 25%-68%, P< 0.0001) more people wore facemasks in the facemask group compared to control; adherence was significantly higher (26%, 95%CI 8%-46%, P< 0.01) in the surgical/medical mask group than in N95/P2 group. The largest number of studies reported on the discomfort and irritation outcome (20-studies); fewest reported on the misuse of masks, and none reported on mask contamination or risk compensation behaviour. Risk of bias was generally high for blinding of participants and personnel and low for attrition and reporting biases.

Conclusions

There are insufficient data to quantify all of the adverse effects that might reduce the acceptability, adherence, and effectiveness of face masks. New research on facemasks should assess and report

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the harms and downsides. Urgent research is also needed on methods and designs to mitigate the downsides of facemask wearing, particularly the assessment of possible alternatives.

Systematic review registration

Open Science Framework website https://osf.io/sa6kf/ (Timestamp 20-05-2020)

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1 Strengths and Limitations of this study

- This is the first systematic review to investigate the downsides of wearing facemasks.
- We included non-randomised study designs in addition to RCTs, as trials frequently underreported or failed to report harms.
- We reported two deviations from the study protocol, relating to the measurement of effect and the sub-group analysis.

2

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Introduction

Respiratory viruses are predominantly transmitted by aerosol, droplets and fomites.[1] Facemasks – such as surgical masks, N95 masks, and face shields, and substitutes for surgical masks such as home-made cloth masks – are a physical barrier to aerosol and droplet transmission. During the COVID-19 pandemic, some jurisdictions have implemented policies mandating the use of masks in public places, on public transport or in other crowded environments, to prevent people becoming infected, or infecting others.

While most health organisations mandate the use of facemasks by health workers when caring for patients during a pandemic, recommendations for mask wear in the community vary widely – and include use by all, use only in certain situations (e.g. on public transport, or in crowded places where social distancing is not possible), and no specific recommendations about mask use.

Several trials have evaluated the impact on respiratory infections by use of surgical and N95 masks, which may, at best, modestly reduce acute respiratory infection transmission.[2-4] Population observational studies suggest that masks have a more substantial effect.[5] However, the downsides of mask-wearing were either not considered or not reported in most studies. Most trials have focussed on facemasks protecting the wearer, rather than others in the community, are often low powered and include confounding factors resulting in the current evidence for the efficacy of facemasks being less than adequate.[6]

The current controversies, mixed messaging and debate over the use of face masks to prevent the spread of COVID-19 warrants further investigation into their use. Anecdotal evidence, and some studies, suggest that there may be a variety of downsides arising from mask use, including: discomfort, sense of difficulty breathing, and communication problems particularly for those who use lip reading.[6 7] Our aim is to systematically identify and summarise these downsides, to assist policymakers when formulating mask-wearing policies in public settings. We also discuss potential strategies to mitigate downsides of mask-wearing.

Methods

This systematic review is reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[8] We followed the "2 week systematic review" (2weekSR)

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processes for the review.[9] The review protocol was developed prospectively and uploaded to the Open Science Framework data on 20-05-2020 (see: https://osf.io/sa6kf/). Deviations from the protocol are reported in the Methods section and highlighted as a limitation in the Discussion section.

Inclusion criteria

We included studies of people of any age or gender, in any setting. We included studies of any face covering aimed at reducing virus transmission, including surgical masks, N95 masks, cloth masks (both homemade and commercially available). (See Appendix 1 for a summary of included facemasks and their intended purpose). Studies evaluating the use of masks for non-virus transmission purposes (e.g. valved masks for preventing inhalation of airborne contaminants such as particles, gases, or vapours) were excluded (see Appendix 2 for a complete list of excluded facemasks).

We included studies comparing the use of facemask to any active intervention (e.g. another mask, or another intervention such as hand-washing etc.), and studies comparing the use of facemask to control (comprising standard practice i.e. face masks not explicitly provided to study group) in situations where their use was not mandatory.

We included only primary studies – i.e., randomised controlled trials (RCTs) and observational studies of any design. We intentionally included this range of study types to identify appropriate data that corresponded with the listed outcomes. For example, surveys are appropriate for identifying frequency of discomfort, and breathing trials are appropriate for understanding physiological changes. We excluded studies that could not provide a quantitative estimate of the size or frequency of adverse effects such as case reports, case series, as well as qualitative studies, and reviews.

Search strategies

We searched PubMed, Embase, Cochrane CENTRAL, EuropePMC (inception-18/5/2020). The search string was designed for PubMed and translated for use in other databases using the Polyglot Search Translator (Appendix 3).[10] Clinical trial registries were searched via Cochrane CENTRAL, which includes the WHO ICTRP and clinicaltrials.gov. On 22/5/2020, we conducted a backwards and forwards citation analysis in Scopus, on all of the included studies.

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No restrictions by language or publication date were imposed. We included publications that were published in full; abstract only publications were included if they had an accompanying record (e.g. trial registry record, or another public report), with additional information.

Study selection and screening

Four authors (MB&NK, AMS&JC) independently screened the titles and abstracts against the inclusion criteria. One author (JC) retrieved full-text, and four authors (MB&NK, AMS&JC) screened the full-texts. Disagreements were resolved by discussion, or a third author (PG or CDM).

Data extraction

A data extraction form was piloted on three studies. Four authors (MB&NK, AMS&JC) extracted the data. (See Box 1)

Box 1. List of extracted information

- **General information:** study authors, location, study design, duration, aim, setting
- **Participants:** health status, disease (if applicable), sample size, age, gender, smoker status, co-morbidities
- **Intervention and Comparator(s):** number of participants, type of face covering, adjunct interventions, number of face coverings used, duration of use, disposal
- **Outcomes:** definition, measurement instrument, number of adverse events or harms reported (The outcomes were discomforts, harms, and adverse events of wearing face masks, Adherence to facemask wearing, Misuse of masks, Discomfort and other physical irritation from masks, Psychological outcomes (e.g. fear, etc.), Dyspnoea (difficulty breathing, shortness of breath) and other physiological impacts, Communication impacts, and mask contamination).

Assessment of risk of bias in included studies

Four authors (MB&NK, AMS&JC) independently assessed the risk of bias using the Cochrane Risk of Bias Tool.[11] Disagreements were resolved by discussion or a third author (CDM or PG). Each potential source of bias was graded as low, high or unclear, and judgements were supported by a quote from the study.

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Measurement of effect and data synthesis

Where feasible (≥2 studies reporting the same outcome), we expressed outcome measures as odds ratios (OR) with 95% confidence intervals, analysed using Review Manager (RevMan 5). Anticipating considerable heterogeneity among the included studies, we used a random effects model. We reported the adherence to facemask wear using risk difference rather than odds ratio, to more clearly convey the differences in adherence between the intervention and control group (not pre-specified in the protocol).

When meta-analysis was not possible or appropriate, we followed the guidance of the Cochrane Collaboration (Cochrane Handbook Section 12.2).[11] When narrative synthesis was required, we reported the results separately for each harm or adverse outcome.

The individual was used as the unit of analysis, where possible; otherwise, we extracted the information as it was presented, e.g. the number of harms in each group. We attempted to contact investigators or study sponsors to provide missing data.

We used the I^2 statistic to measure heterogeneity. Because we included fewer than 10 trials, we did not create a funnel plot.

We did not pre-specify subgroup or sensitivity analyses. However, as data was available, we conducted a subgroup analysis of adherence to mask-wearing by studies which evaluated facemask wear alone, and those evaluating facemask together with handwashing.

Patient and public involvement

Patients and the public were not involved in the design and conduct of this review.

Results

Database searches identified 4,691 publications, supplemented with 2,035 references from forward and backward citation searches and other sources, totalling 6,726. After de-duplication, 5,471 references were screened by title and abstract; we full-text screened 214 references, excluding 174 (see Appendix 4 for a list of excluded studies with reasons). We included 40 articles [12-51] corresponding to 37 studies (Table 1 reports the characteristics of included studies), and meta-analysed 11 studies.[15 21 22 26 31-34 41 45 47] (Figure 1).

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Table 1. Characteristics of included studies

First author (country, year)	Study design	Setting	Study duration	Type of participants				Number of participants	Type of Face Mask intervention**				Type of reported outcomes (adverse effects) analysed						
				Healthy		Sick*													
				A	Ch	A	Ch		Surgical	N95	Any	Other	Discomfort and irritation	Dyspnoea & other physiological	Misuse	Adherence	Psychological impacts	Communication impacts	Mask contamination
Aiello (USA, 2012)	C-RCT	RH	3M	✓				1188; 37 RH	✓				✓			✓			
Allison (USA, 2010)	Single arm study	Sch	1M	✓	✓			20 A, 503 Ch	✓							✓			
Baig (USA, 2010)	S	H	1M	✓				149[1]		✓			✓					✓	
Barasheed (Australia, 2014)	C-RCT	Tents	1W	✓	✓	✓		164; 22 Tents	✓							✓			
Bryce (Canada, 2008)	S	H	1W S	✓				137		✓			✓			✓			
Canini (France, 2010)	C-RCT	HH	1Inf-S	✓	✓		✓	306; 105 HH	✓				✓			✓	✓		
Chen (China, 2016)	Multiple crossover lab-based trials	Lab	NR	✓				15		✓				✓					
Chugtai (China, 2019)[1]	Prevalence	H	2M	✓				148	✓				✓			✓		✓	✓
Cowling (Hong Kong, 2008)	C-RCT	HH	7M	✓	✓	✓	✓	370; 128 HH[3]	✓							✓			
Cowling (Hong Kong, 2009)	C-RCT	HH	8M	✓	✓	✓	✓	1015; 322 HH[3]	✓							✓			
Foo (Singapore 2006)	S	H	2S	✓				340			✓		✓						
Forgie (Canada, 2009)[4]	S	ED	2M	✓	✓			80[5]	✓			✓					✓		
Jacobs (Japan, 2009)	RCT	H	2M	✓				33	✓							✓			
Kao (Taiwan, 2004)[6]	Before-after	H	2M	✓				39		✓			✓						
Larson (USA, 2010) with	C-RCT	HH	19M	✓	✓			2708; 617 HH[3]	✓				✓			✓	✓	✓	

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additional data from (Ferng et al. 2011)																			
Lee (Singapore, 2011)[7]	Single arm study	Lab	NR	✓				14		✓				✓					
Lim (Singapore, 2006)	S	H	1Y	✓				212		✓			✓						
MacIntyre (Australia 2009)	C-RCT	HH	2WS	✓			✓	290 A; 145 HH	✓			✓	✓			✓	✓		
MacIntyre (China, 2011)	C-RCT	H	3M	✓				1441; 15 H	✓	✓			✓			✓		✓	
MacIntyre (China, 2013)	C-RCT	H	3M	✓				1669; 19 H	✓	✓			✓			✓			
MacIntyre (Vietnam, 2015) with additional data from (Chughtai et al. 2016)	C-RCT	H	5W	✓				1607; 14 H	✓			✓	✓			✓			
Martel (Canada, 2013)[8]	Direct observation study	ED	1M	✓				115			✓		✓		✓			✓	
Nickell (Canada, 2004)	S	H	1M	✓				2001	✓	✓			✓				✓	✓	
Ong (Singapore, 2020)	S	H	2M	✓				158	✓				✓						
Or (Hong Kong, 2018)	Lab-based study	Lab	NR	✓				84		✓			✓					✓	
Person (France, 2018)	Randomised crossover	NC[4]	NR	✓				44	✓					✓					
Radonovich (USA, 2009)	Multiple crossover Trial	H	NC	✓				27	✓	✓		✓	✓					✓	
Radonovich (USA, 2019)	C-RCT	H	3.7Y	✓				2862	✓	✓						✓			
Rebmann (USA, 2013)	Multiple Crossover Trial	H	2D	✓				10		✓					✓				
Roberge (USA, 2012)	Crossover	Lab	3M	✓				20	✓					✓					
Shenal (USA, 2012)	Multiple crossover field trial	H	NR	✓				27	✓			✓	✓						
Simmerman (Thailand, 2011)	RCT	HH	16M	✓			✓	465	✓							✓			
Suess (Germany, 2012) with additional data from (Suess et al. 2011)	C-RCT	HH	2Inf-S	✓			✓	111 HH	✓				✓			✓			

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Thomas (USA, 2011)	NC[9]	HB-E	NR	✓				3	✓	✓								✓	
Vanjak (France, 2006)	S	H	2M	✓				238 patients & 210 staff			✓				✓				
Wong (Hong Kong, 2013)	RCT	PCC	2M	✓				1031 patients & 9 doctors			✓						✓		
Yeung (Singapore, 2020)	S	HH	1W	✓				2231 HH		✓					✓				

C-RCT: cluster randomized controlled trial, **RCT**: randomized controlled trial, **S:** Survey, **Sch:** Schools, **RH:** residence halls, **HH:** Households, **H:** Hospitals, **Tel:** Telephone survey of home residents, **Lab:** Laboratory, **ED:** Emergency departments, **HB-E:** Helicopter Bell (used for helicopter emergency medical services), **PCC:** Primary care clinics, **N/A:** Not applicable, **NR:** Not reported, **NC:** Not clear, **Y:** Year, **M:** Months, **W:** Weeks, **D:** Days, **Inf-S:** Influenza season, **WS:** Winter season, **A:** Healthy Adults, **Ch:** Children.

*Sick adults or children, with influenza, influenza-like illness, or upper respiratory tract infections. **Only Face masks arms of the included studies. [1] Authors state 159 surveys returned but 10 were excluded from analysis as ineligible. [2] reports 3 studies: pilot study 1 lab testing; pilot study 2 (small clinical testing); study 3 main study which is extracted here. [3] number of households that were allocated. [4] Participants in the waiting room of ED, could be sick due to an infection. [5] Pairs of adults and children. [6] Patients with End stage renal disease without any influenza or ILI. [7] Data reported here is for the N95 group. [8] 2-part study: part 1) direct observation study [data in this row] + part 2) survey, 4 "in a covered, flat, straight corridor, with a length of 30 meters and marked every 3 meters". [9] In the first part of the study, the pilot recited each word with the aircraft engine off when wearing and not wearing either the surgical facemask or one of the N95 respirators. In the second part of the study, the same procedure occurred, except that the 407-helicopter engine was running at idle.

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[Insert Figure 1 here]

Risk of bias

Inclusion of both observational studies and RCTs could explain the high risk of bias across the included studies. Reporting of sequence generation, allocation concealment and blinding of outcome assessment was poor in 20% - 30% of studies. Due to the nature of interventions, blinding of participants was rare. For RCTs, we reported blinding of outcome assessment for the main trial (not the adverse events) – as some outcomes were lab-confirmed and were considered of low risk. We found no evidence of incomplete outcome data or selective reporting of outcomes. Funding statement, funder's role and authors' conflict of interests were adequately reported in most studies (see Figure 2, and figure 3 in Appendix 5).

[Insert Figure 2 here]

Adherence to facemask wearing

17 studies (14 RCTs, 3 observational [10,13,17]) reported on adherence to facemask wearing; of these 11 had sufficient data to pool and were meta-analysed.

Face mask versus control comparison

Comparison of face masks to control was sub-grouped into studies comparing facemask alone vs control; and studies of facemask plus handwashing vs control. (Figure 3)

Facemasks alone vs control (5 studies) showed the facemask group had a significantly higher facemask wear compared to control (Risk Difference (RD): 0.46, 95% CI 0.24 to 0.67, P< 0.0001). Studies evaluating facemask plus handwashing vs control (n= 3), similarly showed significantly higher facemask wear in the facemask group (RD: 0.47 (95% CI 0.07 to 0.88, P< 0.0001).
Overall, 7 studies (3303 participants) compared facemasks to control. Facemask wear was 47% higher in the facemask group, although heterogeneity was very high (RD: 0.47, 95% CI 0.25 to 0.68, P< 0.0001, I^2 = 98%.)

[Insert Figure 3 here]

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We explored the possible sources of heterogeneity. Excluding studies with 3 or more domains at high risk of bias did not decrease heterogeneity (I^2 = 96% for facemask vs control; 99% for facemask plus handwashing versus control) (Figure 4, Appendix 5). We excluded study population as the source of heterogeneity, because subgrouping studies by those in a community/household settings (which included both index cases and their contacts) versus those in a hospital setting (which included healthy healthcare workers) likewise did not decrease heterogeneity (I^2 = 99% for community/household studies, and 97% for hospital studies) (Figure 5, Appendix 5). We excluded intervention and control as sources of heterogeneity, since all studies compared medical/surgical masks to control (no mask) – although some mask-wear did occur in the control groups.

We consider the outcome measurement to be the most likely other source of heterogeneity. All studies relied on self-reporting of the outcome; only one verified this by counting the number of masks used.[21] What was considered 'wearing a facemask' varied: it was unclear,[15 21 45] involved wearing a facemask "always or mostly/often",[22 47] included wearing a facemask whilst on hospital property, performing a healthcare worker role,[26] and included facemask wear during a shift for 70% of time or more.[32] The follow-up was very short (5 - 9 days) for 4 studies;[15 21 22 47] it was longer for 3 studies (21 - 77 days).[26 32 45]

Facemask (surgical/medical) vs facemask (N92/P2 mask)

Four studies (7960 participants) compared adherence for different facemasks.[31 33 34 41] Facemask wear was significantly higher in the surgical/medical facemask group than in the N95/P2 group, OR= 1.26 (95% CI 1.08 to 1.46, P< 0.01). Heterogeneity was very low (I^2 = 27%). (Figure 4)

[Insert Figure 4 here]

Studies not included in the meta-analysis

RCTs (n = 3)

One study in a residential hall setting reported similar duration of facemask wear per day in the facemask alone group vs facemask plus handwashing group (mean of 5.08 and 5.04 hours/day, respectively).[12] Another in a household setting reported that within the facemask group, there were no significant differences between individuals with ILI among contacts vs no ILI among contacts, for facemask use.[17] Finally, 22 of 44 households randomised to the 'education with sanitiser and facemasks' arm, reported having used a mask within 48 hours of episode onset.[28]

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Observational studies (n = 3)

In elementary school setting, approximately twice as many teachers as students wore face masks.[13] A mean compliance score with N95 use guidelines was 21.2 (on a 25-point scale) among frequent users of N95 respirators in a hospital setting.[16] Another study among healthcare workers found that majority of survey respondents (91%) wore 1-2 masks per day (range 1 - 4).[20]

Misuse

Mask misuse appears less studied than other harms and discomforts. A study of 10 nurses observed for 10 minutes/hour over 2 shifts found that they touched their face 2-3 times per hour, their mask 5 times per hour, and their eyes once per 2 hours, when observed by students.[42] In a study of health workers, 13 of the 53 who responded (25%), reported wearing masks only covering their mouth, not their nose.[49] A study in two hospitals,[35] observed triage nurse behaviour with 118 patients with fever and cough, found that in only 18% of cases the nurses informed patients of the need to wear mask, and in half of those, gave instruction on the need to cover both mouth and nose. A cross-sectional study evaluating the proficiency of the Singaporean public in wearing N95 masks found only 90/714 subjects passed the visual mask fit test; the most common criteria performed incorrectly were: strap placement, leaving a visible gap between the mask and skin, and tightening the nose-clip.[51]

Discomfort and irritation

Several RCTs of specifically measured mask wear discomfort,[17 31-33 47] but most only recorded spontaneously reported events [12 28 34] or did not report any.[15 21 22 26 41 45 50] A trial of household index influenza cases allocated to wear masks or no mask, found the 51 allocated to masks wore them on average 3.8 hours/day and 38 (76%) reported discomfort (Table 2).[17] A study of health care workers in Beijing asked to wear masks for their full shift, found 84% complained of at least one problem (Table 2).[20] In a German household study, 65/172 participants reported problems with mask wearing – most commonly warmth, pain and shortness of breath.[47]

In a trial of healthcare workers comparing surgical and N95 masks to prevent influenza, more workers found the N95 uncomfortable (42%) than the medical mask (10%) when worn an average of 5 hours per day,[33] with significant differences in headaches, difficulty breathing, and pressure on the nose (Table 2). A trial of cloth versus medical masks in healthcare workers found similar rates of discomfort.[32] A community trial comparing surgical and P2 (N95) masks found >50% reporting concerns, primarily discomfort, with similar rates (15% vs. 17%) across groups.[31]

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Discomfort increases with duration of mask wearing. A crossover field trial of 27 healthcare workers found increased discomfort over time; half the subjects were unwilling to wear a medical mask for the full 8-hour shift despite regularly wearing them for short periods.[44]

Table 2. Types of discomfort assessed in trials of face masks used to prevent viral transmission.

First author, year (Type of mask)	Population, Number	Difficulty Breathing	Facial irritation or discomfort	Headache	Other
Canini, 2010[17] (surgical masks)	Household, 105 index cases	34%	14%	-	46% warmth
Chughtai 2019[20] (surgical masks)	Health care workers, 148	12%	17%	6%	-
MacIntyre 2011[33] (Surgical)	Health Care workers, 492	12%	11%	4%	-
MacIntyre 2011[33] (N95)	Health Care workers, 949	19%	52%	13%	-
MacIntyre, 2015[32] (Cloth v Medical)	Health Care workers, 1130	18%	35%	-	-

Two surveys of healthcare staff in Singapore during the SARS epidemic assessed headache and skin reactions.[24 30] In one, 79/212 (37%) reported face-mask-associated headaches, 26 (33%) reported headache frequency exceeding six times/month, and 6 had taken sick leave.[30] Another survey of healthcare workers in Singapore, found that of the 307 staff who used masks regularly, 60% reported acne, 51% facial itch, and 36% rash from N95 mask use.[24] A COVID-19 survey of healthcare workers in Singapore found that 128/158 (81%) developed *de novo* PPE-associated headaches, increasing with duration of use (> 4 hours).[37]

One study (2x2 factorial design) examined the potential of mask fit test and training to mitigate discomfort, but found no clinically or statistically important differences between arms.[38]

Six observational studies reported either general discomfort[33] or spontaneously reported events among participants who wore facemasks.[14 16 27 35 40]

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Psychological

6 studies reported on psychological impacts from wearing face masks (4 RCTs, 2 observational).

Fear

A 3-arm RCT in a household setting found significantly higher risk perception scores in the mask group (38/60) than non-mask groups (30/60) ($p < 0.001$); participants in the mask group were more fearful that they and their family would get sick from the flu.[23 28]

In an observational study, children in a paediatric emergency department waiting room (n = 80) were shown pictures of clinicians wearing either a surgical mask or a clear face shield; 18 children (22.5%) reported surgical masks to be more frightening due to an inability to see clinicians faces, and 14 children (17.5%) reported face shields to be more frightening. However, 47 children (59%) reported that neither were frightening.[25]

Stigma

In a 2-arm cluster-RCT, 15 (29%) patients wearing masks reported they did not like being seen wearing a mask.[17] In a 3-arm RCT, more children reported disliking their parents wearing a P2 mask than a surgical mask (8/92 vs 6/94), however the difference was not significant.[31]

Loneliness

One observational study reported on the loneliness outcome. In a survey investigating the psychosocial effects associated with working in a hospital during the SARS outbreak, 222 (13%) respondents reported a sense of isolation as one reason masks were perceived as bothersome.[36]

Empathy

One RCT reported that the wearing of a face mask by doctors had a negative effect on patient perceptions of the doctors' empathy during consultations, with a mean Consultation and Relational Empathy (CARE) score in the mask group of 33.93 (SD = 7.65, n = 514), and 34.91 (SD = 7.84, n = 516) in the no mask group ($p = 0.04$).[50]

Dyspnoea and other physiological consequences

Studies of physiological impacts were generally done on masks designed for dust, vapours, and other non-transmission purposes; few studied surgical or N95 masks.

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A French crossover study (44 subjects) found surgical masks had no impact on 6-minute walking time, but subjects had an increased sense of dyspneal with a mask: 5.6 vs 4.6cm on a 10cm visual analogue scale (p<0.001),[39] which may come from the increased effort required. A study in 14 adults found that N95 masks increased respiratory resistance in 30 seconds of breathing by over 100%, resulting in average reduction in nasal spirometry of 37%.[29] A study of 20 subjects on a treadmill found the surgical masks increased respiratory rate by 1.6 breaths/minute (p = 0.02), heart rate by 9.5 beats/minute (p<0.001), and transcutaneous CO2 levels of 2.2 mmHg (p<0.001).[43] Finally, a Chinese study of 15 subjects wearing a monitoring garment for respiratory signals found that N95 masks increased both subjective breathing resistance (from none to mild) and increased respiratory rate, the muscle activity of sternomastoid, scalene, diaphragm and abdominal, the fatigue of scalene and intercostal.[18]

Communication

Nine studies (2 RCT, 7 observational) reported on communication difficulties while wearing facemasks.

A trial comparing the use of surgical and N95 masks by healthcare workers to prevent influenza, found more workers in the N95 mask group than the surgical mask group reported mask causing trouble with patient communication (8% vs 3%).[33] Another household-based trial of 15 participants who wore a surgical mask for approximately 60 minutes while performing various tasks,[23 28] found that participants did not report any interference with communication while answering the phone.

In a cross-over trial of 27 healthcare workers,[40] more participants in the surgical mask group reported diminished communication acuity (visual, auditory or vocal) as the reason for discontinuing mask use before the end of an 8 hour shift (7 complaints compared to 4 complaints among N95 mask wearers).

Of 2001 healthcare workers in Toronto responding to a survey during the SARS outbreak, difficulty communicating (47%), and difficulty recognizing people (24%) were identified as key reasons masks (surgical or N95) were perceived as being particularly bothersome.[36] In a survey of 149 healthcare workers,[14] 41 (27.5%) of respondents reported a difficulty 'always'/'most of time' in verbally communicating with patients while wearing a mask.

In another Canadian survey (115 healthcare workers),[35] 26 (23%) respondents reported that wearing masks interfered with their relationships with their patients. Among 148 healthcare workers asked to wear a mask during a 6-8 hour shift, 11 (7.4%) reported trouble communicating with patients.[20]

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In a study of 3 participants evaluating the impact of wearing a surgical or N95 mask on radio reception, all participants were able to accurately record all pilot-recited words regardless of the type of mask worn by the pilot. However, when the aircraft engine was turned on, the accuracy decreased for the N95 mask, compared to surgical or no mask.[48]

In another lab-based study, the performance or absence of fit testing prior to mask use did not affect communication, as 2 participants (out of 21) in each group reported ease of talking to be unsatisfactory.[38]

Mask contamination and other issues

One concern about mask use is the potential for contamination of the mask surface, and subsequent self-inoculation to the wearer's eyes or when demasking. No studies examined that directly, but one study of the health care workers found on average 10% of masks had viral contamination after usage, and that was higher for masks worn > 6 hours (OR 7.9, 95% CI 1.01-61.99) or > 25 patients seen (OR 5.02, 95% CI 1.35-18.60).[20] Given the rates of misuse (see Misuse section above) this contamination raises concerns about self-inoculation.

Several authors have raised concerns about "risk compensation" – non-adherence to other precautions because of the sense of protection – but we found no studies that quantify its extent.

Discussion

We identified 37 studies reporting downsides, harms and adverse events associated with the wearing of facemasks – 15 RCTs and 22 observational studies. The largest number of studies reported on the discomfort and irritation outcome (20 studies), fewest on misuse of mask (4 studies), with no studies directly investigating or quantifying mask contamination or risk compensation behaviour. The only meta-analysable outcome was adherence to facemask wear (17 studies, 11 meta-analysed). 47% more people wore facemasks in the facemask group compared to control, although the percentage of people wearing facemasks in the control group was non-zero in 5 studies; facemask wear adherence was also significantly higher (26%) in the surgical/medical mask group than the N95/P2 group. Risk of bias was generally high for blinding of participants and personnel, and selection bias, and low for attrition and reporting biases.

This is the first systematic review to investigate the downsides of wearing facemasks and forms an important step as a bridge between research and action. The review aligns with the aims of the BESSI (behavioural, environmental, social and systems interventions) collaboration, and addresses

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important evidence gaps in the appropriate use of facemasks. The review's strength lies in its inclusion of non-randomised study designs in addition to RCTs, as trials frequently underreport or fail to report harms.[52] Additionally, the inclusion and exclusion criteria were tested and refined on a test library of 98 references, prior to screening the full search results. The key limitation includes the hospital-setting of most of the included studies: as hospital workers are accustomed to wearing masks, the conclusions may not be fully generalisable to the community. Although this varies among the studies that reported mask use in hospital setting, as there are different confounding factors that may contribute to increased reporting of irritation (e.g. length of shift, air-conditioning on the wards and whether the staff were wearing the full PPE which adds to the full discomfort). We report two differences between the protocol and the review: first, the comparison of facemask to control in the adherence outcome was reported using risk difference (rather than pre-planned odds ratio) to more clearly convey the differences between the two groups (odds ratio for compliance with facemask wear was reported for the facemask versus facemask comparison, however). Second, not having anticipated data availability, we did not prespecify a subgroup analysis of the intervention (facemask wearing) by studies which evaluated facemask wear alone, and studies evaluating facemask with handwashing. Potential behavioural or cultural bias may have been introduced into the findings due to some cultural groups already being more accustomed to wearing face masks when coughing or sick. However, among our included studies only 4 were conducted in the community in Asian countries and of these studies 3 reported adherence to face mask use as an issue and 1 study reported that 264 out of the 624 of the participants who have failed N95 fit test have used masks previously. Furthermore, the use of face masks/coverings in the various study settings compared to their use in a real-life pandemic may differ from a behavioural perspective and this may affect findings related to adherence. However, other reported harms, such as physiological changes, may be less affected by the differences in circumstance.

Several recent systematic reviews have focused on the effectiveness of masks in preventing or reducing viral transmission; some of these reviews reported on harms in the included studies.[8 18 42 53] However, none specifically focused on the wider set of studies examining the physiological, psychological and other adverse effects addressed in this review. The Cochrane review on physical barriers noted the impact of masks on discomfort and communication in some of the randomised trials, and its findings are consistent with this review, but did not extend to studies with outcomes other than viral transmission or non-randomised study design.[3]

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The downsides identified in this review should aid in designing strategies to mitigate problems, and guide the situations where the benefits of masks might outweigh the downsides. As suggested by the higher adherence to surgical masks than to the N95 masks, mitigation of discomforts may also increase adherence to facemask wear, and hence their effectiveness – whether for preventing transmission of the virus by the wearer (e.g. surgical masks) or for preventing inhalation of viral particles in the environment (e.g. N95 masks). Mitigation might be achieved by considering of the when, where, and how of mask wearing (including the fitting process required for some masks like FFP and N95), as the choice of alternative would be dependent on the specific context i.e. it may not be appropriate to use surgical masks or other face masks interchangeably with respirator masks in situations where the goal is to prevent inhalation of aerosolised viral particles as they are not designed for that purpose unlike respirator masks, or by mask redesign or substitution with alternatives (e.g. face shields).

Potential mitigation strategies

Limiting circumstances: Use of facemasks should be restricted to higher risk circumstances, including crowded, indoor spaces, where physical distancing is not possible, e.g. public transport. This recommendation corresponds with suggestions by Chu et.al. who reported that optimum face mask use in public settings could result in a large reduction in infection.[5] Conversely, exercising outdoors is both low risk and has higher downside of wearing masks, because of the increased perceived dyspnoea.

Limiting duration of facemask wear: Duration increases both discomfort and non-adherence. Duration might be decreased by demasking during breaks or scheduling mask breaks. Changing masks more often will help with adherence and the contamination risks, but will increase costs and environmental problems with waste disposal, as well as lead to potential contamination/transmission risks if not performed appropriately.

Modification for specific groups. Some groups are likely to have greater difficulty with mask wearing adherence and correct usage – including children, some patients with mental illnesses, those with cognitive impairment, or respiratory disorders such as asthma or chronic airways disease, and patients with recent facial trauma or oromaxillofacial surgery.[54-56]

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Substitution. Face shields may provide an alternative to facemasks, which may mitigate several of the downsides (e.g. reducing the communication difficulties and breathing resistance), while also providing eye protection. However, there is little evidence on the discomforts of wearing face shields, and on the degree of protection provided, as airborne particles could escape through the upward and downwards jet.[57 58] Other innovative mask designs currently being developed, require discomfort and adherence evaluations in addition to the droplet penetration.

Currently, existing research does not allow firm conclusions as there are insufficient data to quantify all of the adverse effects that might reduce the acceptability, adherence, and effectiveness of facemasks. Any new research on facemasks should assess and report the harms and downsides, including behavioural issues (i.e. risk compensation behaviour) and the psychological impact of mandated facemask wear. There is an urgent need for priority funding for high quality research on methods and designs to mitigate downsides of facemask wearing, particularly the assessment of possible alternatives.

Statements

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Contributors

CDM, and PG conceived the study. MB, AMS, NK, PG, and CDM drafted the protocol. JC designed the search strategy and performed the literature search. MB, NK, AMS and JC screened studies for eligibility, and assessed the risk of bias. All authors performed data extraction. AMS performed the meta-analysis. MB, NK, AMS, PG and CDM performed the narrative analysis for non-meta-analysed studies. All authors critically reviewed the manuscript, contributed to its revision, and approved the final version submitted. The corresponding author confirms that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted. CDM is the guarantor.

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Competing interests

All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: no support from any organisation for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

Ethical approval

Not needed.

Data availability

All data relevant to the study are included in the article or uploaded as supplementary information

Transparency

The corresponding author affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

This manuscript has been deposited as a preprint. Link: https://www.medrxiv.org/content/10.1101/2020.06.16.20133207v1

Patient and public involvement

No patient or public involvement in the study.

Figure legends/captions

Figure 1. PRISMA flow diagram

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

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1 **Figure 3.** Comparison of adherence to face masks vs control

2 **Figure 4.** Comparison of adherence to surgical/medical face masks vs N95/P2 masks

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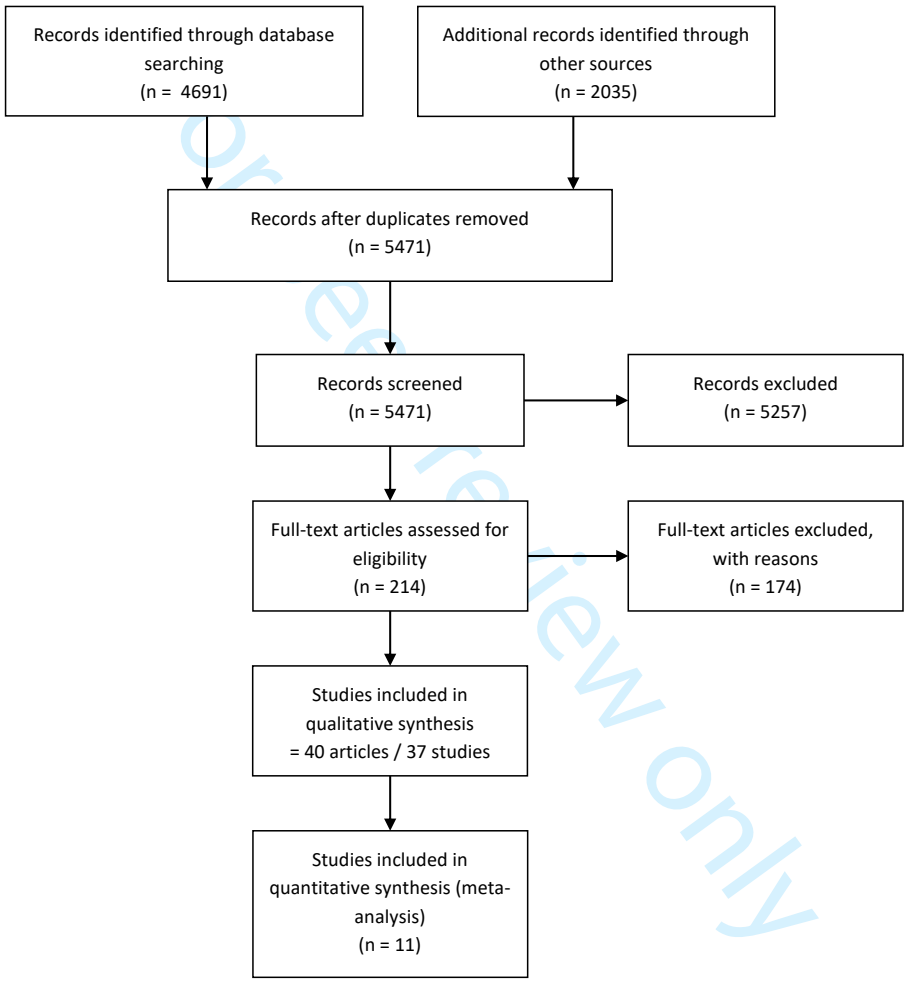


Figure 1. PRISMA flow diagram

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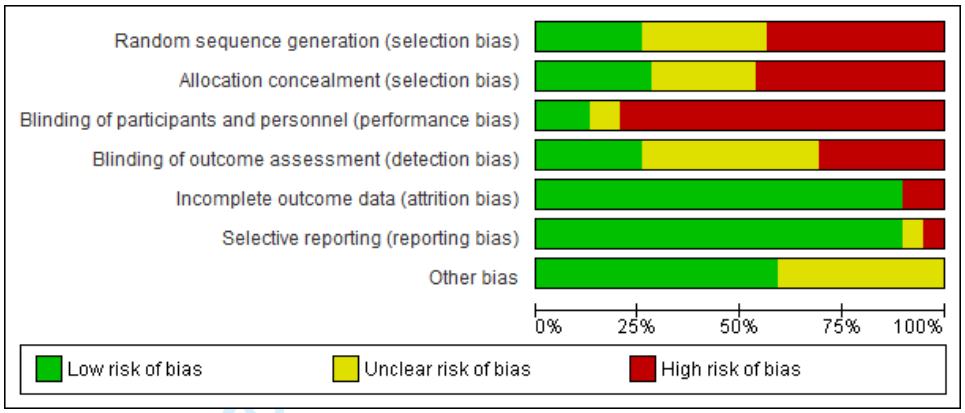


Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

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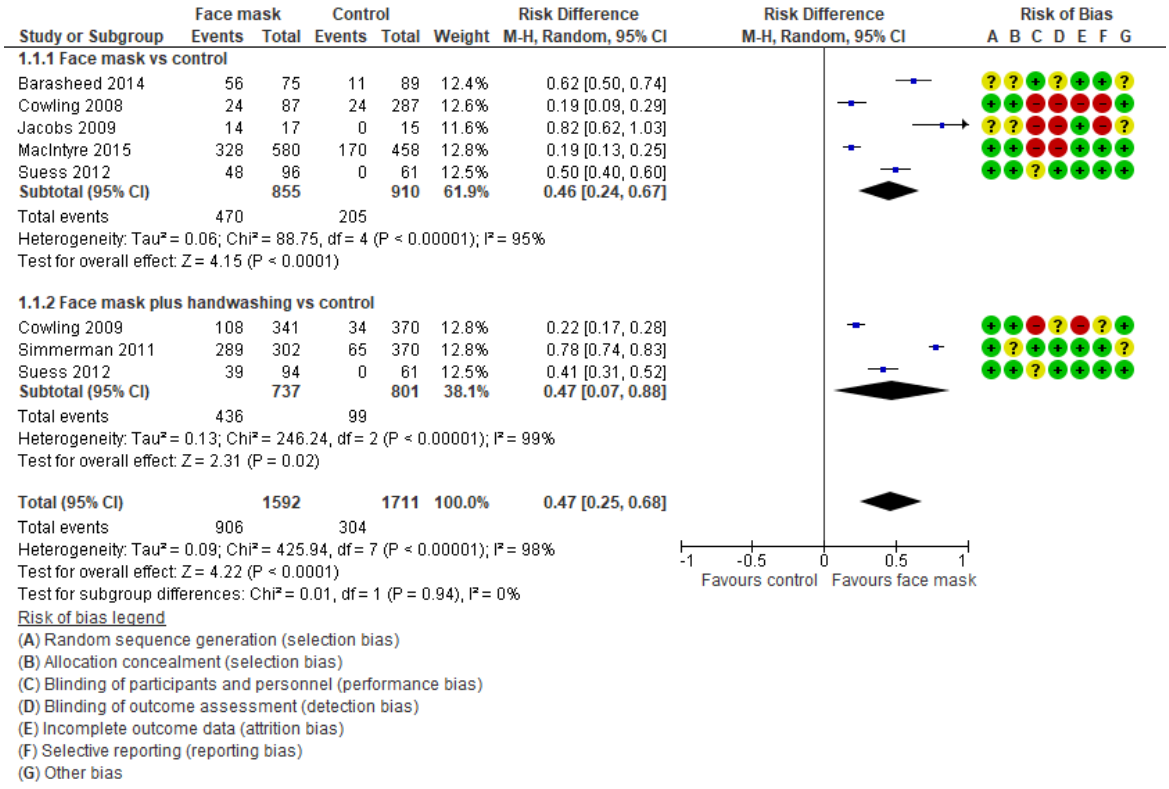


Figure 3. Comparison of adherence to face masks vs control

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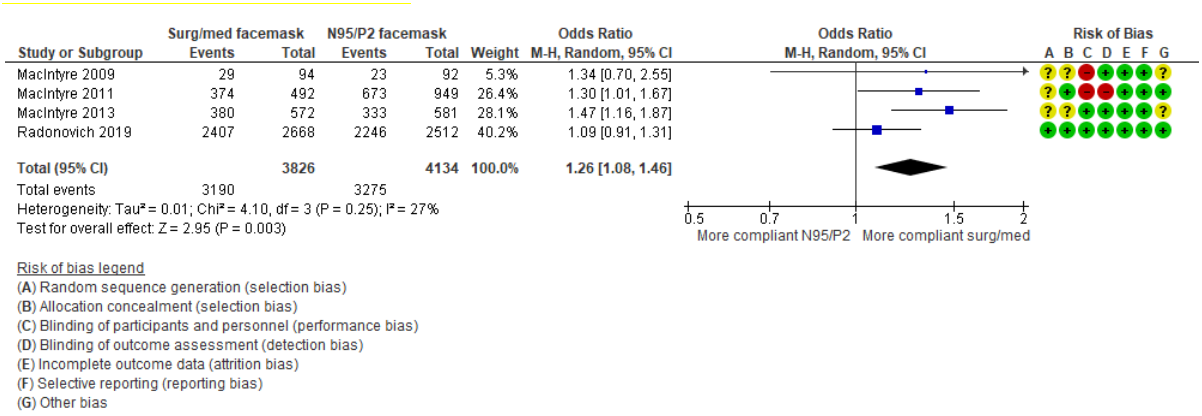


Figure 4. Comparison of adherence to surgical/medical face masks vs N95/P2 masks

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Appendix 1. Summary of the different types of facemasks included in the review and their intended uses

Type of face mask/covering	Description and Intended use
Surgical mask[1, 2]	- Loose-fitting, disposable device, made in different thicknesses - Creates a physical, protective barrier between the mouth and nose of the wearer and potential contaminants in the immediate environment i.e. splashes, sprays, large droplets or splatter. - Primarily intended to prevent the wearer from spreading respiratory droplets - Intended for single use - Does not filter or block very small particles, gases or chemicals in the air that may be transmitted by coughs, sneezes, or certain medical procedures. - Do not provide complete protection from germs and other contaminants because of the loose fit between the surface of the mask and the face.
N95 mask[2, 3]	- Respiratory protective device designed to achieve a very close facial fit. The edges of the respirator are designed to form a seal around the nose and mouth. - Require fit-testing and must be adjusted to the face. - Intended to help reduce wearer exposure to airborne particles as well as larger droplets and protect the wearer from splashes and sprays. - Filters out very fine particles. - Commonly used in healthcare settings. - Should not be shared or reused.
Non-medical face masks/covering[4]	- Cloth face coverings are only intended to help contain the wearer's respiratory droplets from being spread. - At least 2-3 layers needed to block respiratory droplets. - Not intended for use in a clinical setting or explicitly to prevent the transmission of disease between persons. - Some are reusable, and need to be washed after each use.

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Appendix 2. List of excluded face masks

- Oxygen face masks
- OxyMask
- Goldman nasal masks
- Patil-Syracuse Mask
- Ventilation masks
- non-invasive ventilation masks
- Venti-Mask
- Venturi mask
- Mechanical ventilation masks
- Facemask anaesthesia (Anaesthesia induction using face masks)
- Sports protective masks (Hockey masks)
- Diving masks, Scuba masks
- Thermoplastic masks (protective masks for irradiation fields and radiotherapy)
- Bag-valve-mask
- Laryngeal mask
- Oronasal masks
- Sleep masks
- N95 masks with valves (as it does not reduce the spread of virus transmission)

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Appendix 3. Database search strings

Searches conducted 18 May 2020:

PubMed

(("Masks/adverse effects"[MAJR])

OR

((Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR “Face shield”[tiab] OR “Face shields”[tiab] OR Visors[tiab] OR “Face covering”[tiab])

AND

("Adverse effects"[sh] OR “Adverse effect”[tiab] OR “Adverse effects”[tiab] OR “Adversely affect”[tiab] OR “Adverse event”[tiab] OR “Adverse events”[tiab] OR Discomfort[tiab] OR Fatigue[tiab] OR Pain[tiab] OR “Difficulty breathing”[tiab] OR “Breathing difficulty”[tiab] OR “Shortness of breath”[tiab] OR “Risk compensation”[tiab] OR Fomite[tiab] OR Fomites[tiab] OR Dislike[tiab] OR “Physiological impacts”[tiab] OR “Physiological impact”[tiab] OR Sweating[tiab] OR Irritation[tiab] OR “Psychological reactions”[tiab] OR Fear[tiab] OR Communication[tiab] OR “Deaf persons”[tiab] OR Lip-reading[tiab] OR “Lip reading”[tiab] OR Vitiligo[tiab] OR Humidity[tiab] OR “Face dermatitis”[tiab] OR Acne[tiab] OR Redness[tiab] OR Ulcer[tiab] OR “Speech understanding”[tiab] OR “Pinching of skin”[tiab] OR “Moisture buildup”[tiab] OR “Facial warmth”[tiab])

AND

(Relieve[tiab] OR Attitudes[tiab] OR Induced[tiab] OR Modified[tiab] OR Decreased[tiab] OR Decrease[tiab] OR Associated[tiab] OR Impair[tiab] OR Reduce[tiab] OR Reducing[tiab] OR Reduced[tiab] OR Duration[tiab])

AND

(“randomized controlled trial”[pt] OR “controlled clinical trial”[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR “case-control studies”[Mesh] OR “Cohort Studies”[Mesh] OR “case control”[tiab] OR Cohort[tiab] OR “Follow up”[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR Retrospective[tiab] OR “cross sectional”[tiab] OR “Cross-Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh] OR Etiology[sh] OR Effect[tiab] OR Effects[tiab] OR Affect[tiab])))

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR

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"Oxygen mask"[tiab] OR OxyMask[tiab] OR "Nasal masks"[tiab] OR "Patil-Syracuse Mask"[tiab] OR "Ventilation masks"[tiab] OR "Venti-Mask"[tiab] OR "Venturi mask"[tiab] OR "Diving masks"[tiab] OR "Thermoplastic masks"[tiab] OR Bag-valve-mask[tiab] OR "Laryngeal mask"[tiab] OR "Laryngeal masks"[tiab] OR "Oronasal masks"[tiab] OR "Sleep mask"[tiab] OR Editorial[pt] OR Letter[pt] OR "Case reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

OR

(Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR "Face shield"[tiab] OR "Face shields"[tiab] OR Visors[tiab] OR "Face covering"[tiab])

AND

("Respiratory Tract Infections"[Mesh] OR Influenza[tiab] OR "Respiratory Tract Diseases"[Mesh] OR Influenzas[tiab] OR "Influenza-like"[tiab] OR ILI[tiab] OR "common cold"[tiab] OR colds[tiab] OR coryza[tiab] OR coronavirus[Mesh] OR "sars virus"[Mesh] OR coronavirus[tiab] OR Coronaviruses[tiab] OR "coronavirus infections"[Mesh] OR "severe acute respiratory syndrome"[Mesh] OR "severe acute respiratory syndrome"[tiab] OR "severe acute respiratory syndromes"[tiab] OR sars[tiab] OR "respiratory syncytial viruses"[Mesh] OR "respiratory syncytial virus, human"[Mesh] OR "Respiratory Syncytial Virus Infections"[Mesh] OR "respiratory syncytial virus"[tiab] OR "respiratory syncytial viruses"[tiab] OR rsv[tiab] OR parainfluenza[tiab] OR "Respiratory illness"[tiab] OR ((Transmission[tiab]) AND (Coughing[tiab] OR Sneezing[tiab])) OR ((respiratory[tiab] AND Tract[tiab]) AND (infection[tiab] OR Infections[tiab] OR illness[tiab])) OR "Communicable Disease Control"[Mesh] OR "Disease Outbreaks"[Mesh] OR "Disease Transmission, Infectious"[Mesh] OR "Infection Control"[Mesh] OR Transmission[sh] OR "Communicable Disease Control"[tiab] OR "Secondary transmission"[tiab] OR ((Reduced[tiab] OR Reduce[tiab] OR Reduction[tiab] OR Reducing[tiab] OR Lower[tiab]) AND (Incidence[tiab] OR Occurrence[tiab] OR Transmission[tiab] OR Secondary[tiab])) OR Physiological[tiab])

AND

("Adverse effects"[sh] OR "Adverse effect"[tiab] OR "Adverse effects"[tiab] OR "Adverse event"[tiab] OR "Adverse events"[tiab] OR "Index cases"[tiab] OR "Attack rates"[tiab] OR ((Reduce[tiab] OR Reduction[tiab]) AND (Rate[tiab] OR Rates[tiab])) OR Perception[tiab] OR Prevent[tiab] OR Adherence[tiab])

AND

("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR "case-control studies"[Mesh] OR "Cohort Studies"[Mesh] OR "case control"[tiab] OR Cohort[tiab] OR "Follow up"[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR retrospective[tiab] OR "cross sectional"[tiab] OR "Cross-

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Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh])

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR “Oxygen mask”[tiab] OR OxyMask[tiab] OR “Goldman nasal masks”[tiab] OR “Patil-Syracuse Mask”[tiab] OR “Ventilation masks”[tiab] OR “Venti-Mask”[tiab] OR “Venturi mask”[tiab] OR “Hockey mask”[tiab] OR “Diving masks”[tiab] OR “Thermoplastic masks”[tiab] OR Bag-valve-mask[tiab] OR “Laryngeal mask”[tiab] OR “Laryngeal masks”[tiab] OR “Oronasal masks”[tiab] OR “Sleep mask”[tiab] OR "Case Reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

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(([mh Masks/AE])

OR

(([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)))

NOT

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([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Nasal masks":ti,ab OR "Patil Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

OR

([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

([mh "Respiratory Tract Infections"] OR Influenza:ti,ab OR [mh "Respiratory Tract Diseases"] OR Influenzas:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR [mh coronavirus] OR [mh "sars virus"] OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR [mh "coronavirus infections"] OR [mh "severe acute respiratory syndrome"] OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR [mh "respiratory syncytial viruses"] OR [mh "respiratory syncytial virus, human"] OR [mh "Respiratory Syncytial Virus Infections"] OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR [mh "Communicable Disease Control"] OR [mh "Disease Outbreaks"] OR [mh "Disease Transmission, Infectious"] OR [mh "Infection Control"] OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

NOT

([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

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Embase

('Masks'/exp OR 'Respiratory protective devices'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR Lip-reading:ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab OR Etiology:ti,ab OR Effect:ti,ab OR Effects:ti,ab OR Affect:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR 'letter'/exp OR 'editorial'/exp OR 'case report'/exp)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

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[embase]/lim

OR

('Mask'/exp OR 'gas mask'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

('respiratory tract infection'/exp OR Influenza:ti,ab OR 'respiratory tract disease'/exp OR Influenzas:ti,ab OR Influenza-like:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR 'Coronavirinae'/exp OR 'SARS coronavirus'/exp OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR 'Coronavirus infection'/exp OR 'severe acute respiratory syndrome'/exp OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR 'Pneumovirus'/exp OR 'Human respiratory syncytial virus'/exp OR 'respiratory syncytial virus infection'/exp OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR 'communicable disease control'/exp OR 'epidemic'/exp OR 'disease transmission'/exp OR 'Infection Control'/exp OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

(‘adverse event’/exp OR "Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal

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mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR "Case Reports":it)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

[embase]/lim

Europe PMC

https://europepmc.org/search?query=%28TITLE%3A"masks"%20AND%20%28%222019-nCoV%22%20OR%20%222019nCoV%22%20OR%20%22COVID-19%22%20OR%20%22SARS-CoV-2%22%20OR%20%28%22wuhan%22%20AND%20%22coronavirus%22%29%20OR%20%22Coronavirus%22%20OR%20%22Corona%20virus%22%20OR%20%22corona-virus%22%20OR%20%22corona%20viruses%22%20OR%20%22coronaviruses%22%20OR%20%22SARS-CoV%22%20OR%20%22Orthocoronavirinae%22%20OR%20%22MERS-CoV%22%20OR%20%22Severe%20Acute%20Respiratory%20Syndrome%22%20OR%20%22Middle%20East%20Respiratory%20Syndrome%22%20OR%20%28%22SARS%22%20AND%20%22virus%22%29%20OR%20%22soluble%20ACE2%22%20OR%20%28%22ACE2%22%20AND%20%22virus%22%29%20OR%20%28%22ARDS%22%20AND%20%22virus%22%29%20or%20%28%22angiotensin-converting%20enzyme%202%22%20AND%20%22virus%22%29%29%29%20AND%20%28SRC%3APPR%29&page=1

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Appendix 4. List of excluded studies

Excluded studies from database search

No.	Reference	Reason for exclusion
1.	Ahmad, E. F., et al. (2001). "The effect of wearing the veil by Saudi ladies on the occurrence of respiratory diseases." J Asthma 38(5): 423-426.	Study type
2.	Ahmad, I., et al. (2017). "Assessment of knowledge, practice and barrier in use of facemask among university students." Pakistan Journal of Medical and Health Sciences 11(4): 1657-1658.	Reported outcomes
3.	Aiello, A. E., et al. (2010). "A randomized intervention trial of mask use and hand hygiene to reduce seasonal influenza-like illness and influenza infections among young adults in a university setting." International Journal of Infectious Diseases 14: e320.	Duplicate
4.	Aiello, A. E., et al. (2010). "Mask use, hand hygiene, and seasonal influenza-like illness among young adults: a randomized intervention trial." J Infect Dis 201(4): 491-498.	No extractable data
5.	Akyuz, N., et al. (2014). "How and how much do endoscopy professionals protect themselves against infection?" Int J Surg 12(7): 720-724.	Reported outcomes
6.	Albarrak, A. I., et al. (2019). "Middle East Respiratory Syndrome (MERS): Comparing the knowledge, attitude and practices of different health care workers." J Infect Public Health.	Reported outcomes
7.	AlGhamri, A. A., et al. (2013). "The effects of wearing respirators on human fine motor, visual, and cognitive performance." Ergonomics 56(5): 791-802.	Intervention
8.	Allsopp, J., et al. (1997). "Survey of the use of personal protective equipment and prevalence of work related symptoms among dental staff." Occup Environ Med 54(2): 125-134.	Reported outcomes
9.	Alqahtani, A. S., et al. (2016). "Exploring barriers to and facilitators of preventive measures against infectious diseases among Australian Hajj pilgrims: cross-sectional studies before and after Hajj." Int J Infect Dis 47: 53-59.	No extractable data
10.	Alsarheed, M. (2011). "Children's Perception of Their Dentists." Eur J Dent 5(2): 186-190.	Study type
11.	Aydin, O., et al. (24 April 2020). "Performance of fabrics for home-	Reported

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	made masks against spread of respiratory infection through droplets: a quantitative mechanistic study." PREPRINT (Version 1 - 2020.04.19.20071779), Available at: medRxix: https://doi.org/10.1101/2020.04.19.20071779.	outcomes
12.	Badin, A. (1969). "[The surgeon's fatigue and discomfort behind the mask]." Gynaecologia 167(5): 430-433.	Study type
13.	Bansal, S., et al. (2009). "Respirator physiological effects under simulated work conditions." J Occup Environ Hyg 6(4): 221-227.	Intervention
14.	Beuvelet, M., et al. (2017). "Healthcare workers' perception towards the systematic use of mask during a seasonal influenza outbreak in a French University Hospital: A descriptive study." Antimicrobial Resistance and Infection Control 6.	Unable to access full text
15.	Bhoyrul, B., et al. (2019). "A review of non-glove personal protective equipment-related occupational dermatoses reported to EPIDERM between 1993 and 2013." Contact Dermatitis 80(4): 217-221.	Study type
16.	Bischoff, W., et al. (2017). "Evaluation of a novel powered air-purifying respirator (PAPR) vs. a N95 respirator mask for the protection against influenza in a human exposure model." Open Forum Infectious Diseases 4: S168.	Reported outcomes
17.	Brandt, C., et al. (2011). "Attitudes of influenza-vaccinated health care workers toward masks to prevent nosocomial transmission of influenza." Influenza Other Respir Viruses 5(1): 61-66.	Study type
18.	Buitrago, F. (2010). "Hygienic measures and barriers are effective strategies to prevent the transmission of viral infections." FMC Formacion Medica Continuada en Atencion Primaria 17(5): 361.	Study type
19.	Caretti, D. M. and P. D. Gardner (1999). "Respirator fit factor performance while sweating." American Industrial Hygiene Association Journal 60(1): 84-88.	Intervention
20.	Chan, E. Y., et al. (2015). "Knowledge, attitudes, and practices of Hong Kong population towards human A/H7N9 influenza pandemic preparedness, China, 2014." BMC Public Health 15: 943.	Reported outcomes
21.	Chan, M. F., et al. (2008). "Investigating the knowledge, attitudes and practice patterns of operating room staff towards standard and transmission-based precautions: results of a cluster analysis." J Clin Nurs 17(8): 1051-1062.	Reported outcomes
22.	Chan, M. T., et al. (2013). "Mask ventilation and dispersion of exhaled air." Am J Respir Crit Care Med 187(7): e12-14.	Study type

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23.	Cherrie, J. W., et al. (2019). "In-mask temperature and humidity can validate respirator wear-time and indicate lung health status." J Expo Sci Environ Epidemiol 29(4): 578-583.	Intervention
24.	Chung, P. K., et al. (2017). "The process by which perceived autonomy support predicts motivation, intention, and behavior for seasonal influenza prevention in Hong Kong older adults." BMC Public Health 18(1): 65.	Study type
25.	Coca, A., et al. (2017). "Physiological Evaluation of Personal Protective Ensembles Recommended for Use in West Africa." Disaster Med Public Health Prep 11(5): 580-586.	Intervention
26.	Collyer, S. and P. J. Davis (2006). "Effect of facemask use on respiratory patterns of women in speech and singing." J Speech Lang Hear Res 49(2): 412-423.	Reported outcomes
27.	Condon, B. J. and T. Sinha (2010). "Who is that masked person: the use of face masks on Mexico City public transportation during the Influenza A (H1N1) outbreak." Health Policy 95(1): 50-56.	Reported outcomes
28.	Davies, A., et al. (2013). "Testing the efficacy of homemade masks: would they protect in an influenza pandemic?" Disaster Med Public Health Prep 7(4): 413-418.	Reported outcomes
29.	Davis, P. J., et al. (2007). "A survey of Alberta physicians' use of and attitudes toward face masks and face shields in the operating room setting." Am J Infect Control 35(7): 455-459.	Reported outcomes
30.	Del Ferraro, S., et al. (2018). "Hot environment: Effects of a power assisted filtering device incorporating a full face mask on newton manikin thermophysiological response." Occupational and Environmental Medicine 75: A503-A504.	Study type
31.	Del Ferraro, S., et al. (2017). "Thermophysiological response of Newton manikin equipped with power-assisted filtering device incorporating a full-face mask in hot environment." Int J Hyperthermia 33(7): 717-723.	Population
32.	Deris, Z. Z., et al. (2010). "The prevalence of acute respiratory symptoms and role of protective measures among Malaysian Hajj pilgrims." Journal of Travel Medicine 17(2): 82-88.	Study type
33.	Derrick, J. L. and C. D. Gomersall (2005). "Protecting healthcare staff from severe acute respiratory syndrome: filtration capacity of multiple surgical masks." J Hosp Infect 59(4): 365-368.	Reported outcomes
34.	Ding, S., et al. (1998). "[Heat strain for personnel wearing three levels	Intervention

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	of protective ensembles at moderate and heavy workloads]." Wei Sheng Yan Jiu 27(6): 361-364.	
35.	DiLeo T, Roberge RJ, Kim JH. Effect of wearing an N95 filtering facepiece respirator on superomedial orbital infrared indirect brain temperature measurements. J Clin Monit Comput. 2017;31(1):67-73.	Reported outcomes
36.	Dong, J. and Y. Chen (2010). "Impact of the bag exchange procedure on risk of peritonitis." Perit Dial Int 30(4): 440-447.	Reported outcomes
37.	Ellison, I. R. T. (2009). "Surgical masks or N95 respirators for the flu?" Medicine Today 10(12): 69-70.	Reported outcomes
38.	Emanuel, R., et al. (2017). "A study of special care dental patient preference toward choice of mask and visor use by dental clinicians." Spec Care Dentist 37(4): 164-167.	Reported outcomes
39.	Fabian, P., et al. (2011). "Influenza virus aerosols in human exhaled breath: Particle size, culturability, and effect of surgical masks." Epidemiology 22(1): S51.	Reported outcomes
40.	Field, M. H., et al. (2020). "Hydrocolloid dressing strip over bridge of nose to relieve pain and pressure from Filtered Face Piece (FFP) masks during the coronavirus (COVID-19) pandemic." Ann R Coll Surg Engl 102(5): 394-396.	Study type
41.	Fierro, J. L., et al. (2015). "Barriers to the Use of PPE to Prevent Pertussis Exposures in a Pediatric Primary Care Network." J Pediatric Infect Dis Soc 4(1): 49-56.	Reported outcomes
42.	Flury, D., et al. (2019). "Health-care associated Influenza Prevention Project (HAIP): First multicenter surveillance results (nosocomial influenza and prevention measures)." Antimicrobial Resistance and Infection Control 8.	Reported outcomes
43.	Fox, S. H. and A. B. DuBois (1993). "The effect of evaporative cooling of respiratory protective devices on skin temperature, thermal sensation, and comfort." Am Ind Hyg Assoc J 54(12): 705-710.	Intervention
44.	Ganczak, M. and Z. Szych (2007). "Surgical nurses and compliance with personal protective equipment." J Hosp Infect 66(4): 346-351.	Reported outcomes
45.	Gu, J., et al. (2015). "Preventive behaviors and mental distress in response to H1N1 among university students in Guangzhou, China." Asia Pac J Public Health 27(2): Np1867-1879.	Reported outcomes
46.	Guo, Y. P., et al. (2008). "Evaluation on masks with exhaust valves and with exhaust holes from physiological and subjective responses." J	Intervention

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	Physiol Anthropol 27(2): 93-102.	
47.	Gwosdow, A. R., et al. (1989). "Effect of thermal conditions on the acceptability of respiratory protective devices on humans at rest." Am Ind Hyg Assoc J 50(4): 188-195.	Intervention
48.	Hassaniazad, M., et al. (2016). "Preventive measures for Crimean-Congo hemorrhagic fever in healthcare workers; how high is the chance of transmission?" Acta Medica Mediterranea 32: 2017-1024.	Reported outcomes
49.	Harber, P., et al. (1989). "Relationship of subjective tolerance of respirator loads to physiologic effects and psychophysical load sensitivity." J Occup Med 31(8): 681-686.	Intervention
50.	Hardis, K. E., et al. (1983). "Correlation of qualitative and quantitative results from testing respirator fit." Am Ind Hyg Assoc J 44(2): 78-87.	Intervention
51.	Hickey, J., et al. (2014). "Pandemic preparedness: perceptions of vulnerable migrants in Thailand towards WHO-recommended non-pharmaceutical interventions: a cross-sectional study." BMC Public Health 14: 665.	Reported outcomes
52.	Hines, S. E., et al. (2019). "User acceptance of reusable respirators in health care." American Journal of Infection Control 47(6): 648-655.	Intervention
53.	Ho, R. C., et al. (2014). "Impact of 2013 south Asian haze crisis: study of physical and psychological symptoms and perceived dangerousness of pollution level." BMC Psychiatry 14: 81.	Reported outcomes
54.	Huang, W. and L. Morawska (2019). "Face masks could raise pollution risks." Nature 574(7776): 29-30.	Study type
55.	Huynh. TD (30 March 2020). "The more I fear about COVID-19, the more I wear medical masks: A survey on risk perception and medical masks uses." PREPRINT (Version 1 - 2020.03.26.20044388), Available at: medRxix: https://doi.org/10.1101/2020.03.26.20044388.	Reported outcomes
56.	Ioannidou, C., et al. (2010). "Promotion of hygiene measures to prevent pandemic influenza transmission in greek nursery schools: The teachers'a perspectives." Acta Paediatrica, International Journal of Paediatrics 99: 88.	Reported outcomes
57.	Jaeger, J. L., et al. (2011). "Transmission of 2009 pandemic influenza A (H1N1) virus among healthcare personnel-Southern California, 2009." Infect Control Hosp Epidemiol 32(12): 1149-1157.	Reported outcomes
58.	James, R., et al. (1984). "Effects of respirators under heat/work conditions." Am Ind Hyg Assoc J 45(6): 399-404.	Intervention

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59.	Jencsik, T., et al. (2017). "Influenza-related attitudes of healthcare workers at institutions for acute and chronic diseases." Value in Health 20(9): A791-A792.	Study type
60.	Johnson, A. T., et al. (2000). "Effect of external dead volume on performance while wearing a respirator." Aihaj 61(5): 678-684.	Intervention
61.	Johnson, A. T., et al. (2000). "Communication using a telephone while wearing a respirator." Aihaj 61(2): 264-267.	Intervention
62.	Kawanami, S., et al. (2018). "Physiological effects of wearing powered airpurifying respirators with different blower system." Occupational and Environmental Medicine 75: A201.	Intervention
63.	Kim, J. H., et al. (2015). "Effect of wearing an N95 respirator on infrared tympanic membrane temperature measurements." J Clin Monit Comput 29(6): 691-695.	Reported outcomes
64.	Kishimoto, K., et al. (2016). "[Whether Visual Information Loss of Expression by Community Pharmacist Masks to Effect a Change in Feeling of Trust among Help-seeker?]." Yakugaku Zasshi 136(10): 1401-1413.	Study type
65.	Krishna, S., et al. (2019). "The gloves are off: The state of personal protective equipment (PPE) use in contact, droplet, and standard precautions at a major teaching hospital." Open Forum Infectious Diseases 6: S429-S430.	Study type
66.	Kroo, L., et al. (29 April 2020). "Pneumask: Modified Full-Face Snorkel Masks as Reusable Personal Protective Equipment for Hospital Personnel." PREPRINT (Version 1 - 2020.04.24.20078907), Available at: medRxix: https://doi.org/10.1101/2020.04.24.20078907.	Intervention
67.	Latheef, F., et al. (2015). "Personal protective equipment-associated dermatoses reported to EPIDERM: A 20-year review." British Journal of Dermatology 173: 172-173.	Duplicate
68.	Lau, J. T., et al. (2003). "Monitoring community responses to the SARS epidemic in Hong Kong: from day 10 to day 62." J Epidemiol Community Health 57(11): 864-870.	Reported outcomes
69.	Leung, N. H., et al. (06 March 2020). "Respiratory Virus Shedding in Exhaled Breath and Efficacy of Face Masks." PREPRINT (Version 1), Available at: Research Square [+https://doi.org/10.21203/rs.3.rs-16836/v1+],.	Reported outcomes
70.	Lee, H. P. and Y. Wang de (2011). "Objective assessment of increase in breathing resistance of N95 respirators on human subjects." Ann	Reported outcomes

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	Occup Hyg 55(8): 917-921.	
71.	Lee, J. A., et al. (2017). "Compliance of personal protective equipment in a hospital." Antimicrobial Resistance and Infection Control 6.	Reported outcomes
72.	Li, Y., et al. (2006). "Antimicrobial effect of surgical masks coated with nanoparticles." J Hosp Infect 62(1): 58-63.	Intervention
73.	Li, C., et al. (2020). "[Management and prevention of common skin problems during epidemic prevention and control of COVID-19]." Nan Fang Yi Ke Da Xue Xue Bao 40(2): 168-170.	Reported outcomes
74.	Loeb, M., et al. (2009). "Surgical mask vs N95 respirator for preventing influenza among health care workers: a randomized trial." Jama 302(17): 1865-1871.	Reported outcomes
75.	Loibner, M., et al. (2019). "Limiting factors for wearing personal protective equipment (PPE) in a health care environment evaluated in a randomised study." PLoS One 14(1): e0210775.	Intervention
76.	Lonnroth, E. C. and H. Shahnavaz (1998). "Adverse health reactions in skin, eyes, and respiratory tract among dental personnel in Sweden." Swed Dent J 22(1): 33-45.	Reported outcomes
77.	Lopes, A. C., et al. (2008). "[Adherence to standard precautions by the public pre-hospital health team in Belo Horizonte, Minas Gerais State, Brazil]." Cad Saude Publica 24(6): 1387-1396.	Reported outcomes
78.	Lunding, S., et al. (1998). "[Poor compliance with universal precautions among Danish physicians]." Ugeskr Laeger 160(12): 1789-1793.	Reported outcomes
79.	Lutz, M., et al. (2019). "Physiological effects of a spit sock." Am J Emerg Med 37(2): 291-293.	Intervention
80.	Ma, X., et al. (2014). "Knowledge, attitudes and practices relating to influenza A(H7N9) risk among live poultry traders in Guangzhou City, China." BMC Infect Dis 14: 554.	Reported outcomes
81.	MacIntyre, C. R., et al. (2016). "Cluster randomised controlled trial to examine medical mask use as source control for people with respiratory illness." BMJ Open 6(12): e012330.	Reported outcomes
82.	Martins, R. S., et al. (2013). "Adequacy to the use of personal protective equipment in the isolations of emergency." Antimicrobial Resistance and Infection Control 2.	Reported outcomes
83.	Maschmeyer, G., et al. (2009). "A prospective, randomised study on the use of well-fitting masks for prevention of invasive aspergillosis in	Intervention

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	high-risk patients." Ann Oncol 20(9): 1560-1564.	
84.	Mathews, R., et al. (2008). "Provision and use of personal protective equipment and safety devices in the National Study to Prevent Blood Exposure in Paramedics." Am J Infect Control 36(10): 743-749.	Study type
85.	Mauritzson-Sandberg, E. (1995). "An evaluation of respiratory protective devices used in children's evacuation." Ergonomics 38(4): 707-713.	Purpose of study
86.	Meyer, J. P., et al. (1997). "Field study of subjective assessment of negative pressure half-masks. Influence of the work conditions on comfort and efficiency." Appl Ergon 28(5-6): 331-338.	Purpose of study
87.	Millqvist, E., et al. (1995). "Prevention of asthma induced by cold air by cellulose-fabric face mask." Allergy 50(3): 221-224.	Reported outcomes
88.	Morishima, M. and K. Kishida (2018). "Understanding attitudes toward hygiene mask use in Japanese daily life by using a repeated cross-sectional survey." Work 61(2): 303-311.	Study type
89.	Morishima, M., et al. (2011). "An investigation of the use of hay fever masks for the youth." J Hum Ergol (Tokyo) 40(1): 151-156.	Intervention
90.	Moser, A., et al. (2015). "Rates and determinants of vaccination against seasonal and pandemic influenza in Swiss prehospital emergency medical services workers." Praxis 104: 142.	Study type
91.	Moser, A., et al. (2016). "Vaccination Against Seasonal or Pandemic Influenza in Emergency Medical Services." Prehosp Disaster Med 31(2): 155-162.	Study type
92.	Motoyama, Y. L., et al. (2016). "Airflow-Restricting Mask Reduces Acute Performance in Resistance Exercise." Sports (Basel) 4(4).	Intervention
93.	Nichol, K., et al. (2008). "The individual, environmental, and organizational factors that influence nurses' use of facial protection to prevent occupational transmission of communicable respiratory illness in acute care hospitals." Am J Infect Control 36(7): 481-487.	Study type
94.	Nichol, K., et al. (2013). "Behind the mask: Determinants of nurse's adherence to facial protective equipment." Am J Infect Control 41(1): 8-13.	Study type
95.	Nielsen, R., et al. (1987). "The effect of temperature and humidity levels in a protective mask on user acceptability during exercise." Am Ind Hyg Assoc J 48(7): 639-645.	Intervention

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96.	Palmiero, A. J., et al. (2016). "Speech intelligibility assessment of protective facemasks and air-purifying respirators." J Occup Environ Hyg 13(12): 960-968.	Study type
97.	Parveen, S. N. A., et al. (2018). "Identifying acceptable and feasible infection control interventions for nipah encephalitis outbreaks in bangladesh." American Journal of Infection Control 46(6): S24.	Study type
98.	Peponis, T., et al. (2017). "Trauma team utilization of universal precautions: if you see something, say something." Eur J Trauma Emerg Surg 43(1): 145-150.	Reported outcomes
99.	Perez, V., et al. (2012). "Stress, adherence to preventive measures for reducing influenza transmission and influenza-like illness." J Epidemiol Community Health 66(7): 605-610.	No extractable data
100.	Radonovich Jr, L. J., et al. (2009). "Diminished speech intelligibility associated with certain types of respirators worn by healthcare workers." Journal of Occupational and Environmental Hygiene 7(1): 63-70.	Intervention
101.	Radonovich, L., et al. (2018). "Results of the respiratory protection effectiveness clinical trial (respect)." Open Forum Infectious Diseases 5: S51.	Duplicate
102.	Ramirez, J. A. and P. T. O'Shaughnessy (2016). "The effect of simulated air conditions on N95 filtering facepiece respirators performance." J Occup Environ Hyg 13(7): 491-500.	Study type
103.	Raven, P. B., et al. (1977). "Maximal stress test performance while wearing a self-contained breathing apparatus." J Occup Med 19(12): 802-806.	Intervention
104.	Rengasamy, S. and B. C. Eimer (2013). "N95-companion measurement of cout/cin ratios for two n95 filtering facepiece respirators and one surgical mask." J Occup Environ Hyg 10(10): 527-532.	Population
105.	Roberge, R. (2011). "Facemask use by children during infectious disease outbreaks." Biosecur Bioterror 9(3): 225-231.	Study type
106.	Roberge, R. J., et al. (2010). "Surgical mask placement over N95 filtering facepiece respirators: physiological effects on healthcare workers." Respirology 15(3): 516-521.	Intervention
107.	Roberge, R. J., et al. (2010). "Reusable elastomeric air-purifying respirators: physiologic impact on health care workers." Am J Infect Control 38(5): 381-386.	Intervention

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108.	Rozzi, T., et al. (2012). "Pilot study of aromatic hydrocarbon adsorption characteristics of disposable filtering facepiece respirators that contain activated carbon." J Occup Environ Hyg 9(11): 624-629.	Purpose of study
109.	van der Sande, M., et al. (2008). "Professional and home-made face masks reduce exposure to respiratory infections among the general population." PLoS ONE 3(7): e2618.	Reported outcomes
110.	Santos, P. B., et al. (2013). "Adherence to biosafety practices by nursing staff in the face of situations of occupational risk." Antimicrobial Resistance and Infection Control 2.	Unable to access full text
111.	Schmitt, C. H. (2004). "Secrets behind the mask." US News World Rep 137(4): 37-41.	Study type
112.	Seliga, R., et al. (1991). "Effect of work load and respirator wear on postural stability, heart rate, and perceived exertion." Am Ind Hyg Assoc J 52(10): 417-422.	Intervention
113.	Seng, M., et al. (2018). "Comfort and exertion while using filtering facepiece respirators with exhalation valve and an active venting system among male military personnel." Singapore Med J 59(6): 327-334.	Purpose of study
114.	Smith, C. L., et al. (2013). "Carbon dioxide rebreathing in respiratory protective devices: influence of speech and work rate in full-face masks." Ergonomics 56(5): 781-790.	Intervention
115.	Sokol, K. A., et al. (2016). "Masks for prevention of respiratory viruses on the BMT unit: results of a quality initiative." Transpl Infect Dis 18(6): 965-967.	Reported outcomes
116.	Stebbins, S., et al. (2009). "Using nonpharmaceutical interventions to prevent influenza transmission in elementary school children: parent and teacher perspectives." J Public Health Manag Pract 15(2): 112-117.	Reported outcomes
117.	Stockwell, R. E., et al. (2018). "Prolonged wear of face masks prevents the release of pseudomonas aeruginosa aerosols during coughing in people with cystic fibrosis." Pediatric Pulmonology 53: 269.	Unable to access full text
118.	Strauch, A. L., et al. (2016). "Assessing the efficacy of tabs on filtering facepiece respirator straps to increase proper doffing techniques while reducing contact transmission of pathogens." J Occup Environ Hyg 13(10): 794-801.	Intervention
119.	Sullivan, K. (1993). "Surgical face masks: protection of self or patient?" N Z Dent J 89(398): 137.	Study type

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120.	Suruda, A., et al. (2003). "Fatal injuries in the United States involving respirators, 1984-1995." Appl Occup Environ Hyg 18(4): 289-292.	Study type
121.	Tak, C. N., et al. (2009). "Preventing healthcare workers from acquiring influenza." Infection Control and Hospital Epidemiology 30(3): 292-295.	Reported outcomes
122.	Tan, K. T. and M. W. Greaves (2004). "N95 acne." Int J Dermatol 43(7): 522-523.	Study type
123.	Tang, C. S. and C. Y. Wong (2004). "Factors influencing the wearing of facemasks to prevent the severe acute respiratory syndrome among adult Chinese in Hong Kong." Prev Med 39(6): 1187-1193.	Reported outcomes
124.	Tataris, K., et al. (2010). "Patient attitudes and compliance while wearing surgical masks to prevent disease transmission." Academic Emergency Medicine 17: S58.	Unable to access full text
125.	Vanjak, D., et al. (2006). "[Cases of pertussis among healthcare workers in a maternity ward: management of a health alert]." Med Mal Infect 36(3): 151-156.	Duplicate reference
126.	Vijaya, K., et al. (2005). "Behaviour of Singaporeans during the SARS outbreak: The impact of anxiety and public satisfaction with media information." International Journal of Health Promotion and Education 43(1): 17-22.	Reported outcomes
127.	Wong, S. Y., et al. (2012). "Primary care physicians' response to pandemic influenza in Hong Kong: a mixed quantitative and qualitative study." Int J Infect Dis 16(9): e687-691.	Reported outcomes
128.	Yang, P., et al. (2011). "Mask-wearing and respiratory infection in healthcare workers in Beijing, China." Braz J Infect Dis 15(2): 102-108.	Study type

Excluded studies identified from Forward & Backward citation analysis

129.	Barratt, R., et al. (2019). "Enablers of, and barriers to, optimal glove and mask use for routine care in the emergency department: an ethnographic study of Australian clinicians." Australasian Emergency Care.	Study type
130.	Caretti, D. M. and L. C. Strickler (2003). "Speech intelligibility during respirator wear: Influences of respirator speech diaphragm size and background noise." American Industrial Hygiene Association Journal 64(6): 846-850.	Intervention
131.	Chughtai, A. A., et al. (2015). "Practices around the use of masks and	Reported

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	respirators among hospital health care workers in 3 diverse populations." American Journal of Infection Control 43(10): 1116-1118.	outcomes
132.	Enerson, D. M., et al. (1967). "Heat and moisture trapping beneath surgical face masks: a consideration of factors affecting the surgeon's discomfort and performance." Surgery 62(6): 1007-1016.	Reported outcomes
133.	Goh, D. Y. T., et al. (2019). "A randomised clinical trial to evaluate the safety, fit, comfort of a novel N95 mask in children." Scientific Reports 9(1).	Purpose of study
134.	Harber P, Bansal S, Santiago S, et al. Multidomain subjective response to respirator use during simulated work. Journal of Occupational and Environmental Medicine. 2009;51(1):38-45.	Reported outcomes
135.	Harber, P., et al. (2010). "Respirator physiologic impact in persons with mild respiratory disease." Journal of Occupational and Environmental Medicine 52(2): 155-162.	Intervention
136.	Harber, P., et al. (2011). "Respirator impact on work task performance." Journal of Occupational and Environmental Medicine 53(1): 22-26.	Reported outcomes
137.	Honarbakhsh, M., et al. (2018). "Knowledge, perceptions and practices of healthcare workers regarding the use of respiratory protection equipment at Iran hospitals." Journal of Infection Prevention 19(1): 29-36.	Reported outcomes
138.	Jacobs, J. L., et al. (2009). "Use of surgical face masks to reduce the incidence of the common cold among health care workers in Japan: A randomized controlled trial." American Journal of Infection Control 37(5): 417-419.	Duplicate
139.	Jones, J. G. (1991). "The physiological cost of wearing a disposable respirator." American Industrial Hygiene Association Journal 52(6): 219-225.	Intervention
140.	Kim, J. H., et al. (2013). "Pulmonary and heart rate responses to wearing N95 filtering facepiece respirators." American Journal of Infection Control 41(1): 24-27.	Reported outcomes
141.	Kim, J. H., et al. (2015). "Pressure drop of filtering facepiece respirators: How low should we go?" International Journal of Occupational Medicine and Environmental Health 28(1): 71-80.	Intervention
142.	Kim, J. H., et al. (2016). "Physiologic and fit factor profiles of N95 and P100 filtering facepiece respirators for use in hot, humid	Reported

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	environments." American Journal of Infection Control 44(2): 194-198.	outcomes
143.	Lim, E. C. H., et al. (2007). "Headaches and the N95 face-mask amongst healthcare providers [2]." Acta Neurologica Scandinavica 116(1): 73.	Duplicate
144.	Li Y, Tokura H, Guo YP, et al. Effects of wearing N95 and surgical facemasks on heart rate, thermal stress and subjective sensations. International Archives of Occupational and Environmental Health. 2005;78(6):501-509.	No extractable data
145.	Lin, Y. C. and C. P. Chen (2019). "Thermoregulation and thermal sensation in response to wearing tight-fitting respirators and exercising in hot-and-humid indoor environment." Building and Environment 160.	Intervention
146.	Locatelli, S. M., et al. (2014). "Health care workers' reported discomfort while wearing filtering face-piece respirators." Workplace Health and Safety 62(9): 362-368.	Reported outcomes
147.	Luximon, Y., et al. (2016). "Time dependent infrared thermographic evaluation of facemasks." Work 54(4): 825-835.	Reported outcomes
148.	MacIntyre, C. R., et al. (2014). "Efficacy of face masks and respirators in preventing upper respiratory tract bacterial colonization and co-infection in hospital healthcare workers." Preventive Medicine 62: 1-7.	Reported outcomes
149.	MacIntyre, C. R., et al. (2013). "A randomized clinical trial of three options for N95 respirators and medical masks in health workers." American Journal of Respiratory and Critical Care Medicine 187(9): 960-966.	Duplicate
150.	Mauritzson-Sandberg, E. V. A. (1991). "Psychological effects on prolonged use of respiratory protective devices in children." Ergonomics 34(3): 313-319.	Reported outcomes
151.	McDiarmid, M., et al. (2020). "N95 Respirators vs Medical Masks in Outpatient Settings." JAMA - Journal of the American Medical Association 323(8): 789.	Study type
152.	Morishima, M., et al. (2014). "Experiences and problems with hygiene masks reported by japanese hay fever sufferers." International Journal of Clothing Science and Technology 26(4): 262-273.	Intervention
153.	Morishima, M., et al. (2017). "Problems related to mask use in hay fever sufferers by repeated surveys." International Journal of Clothing Science and Technology 29(1): 123-134.	Purpose of study

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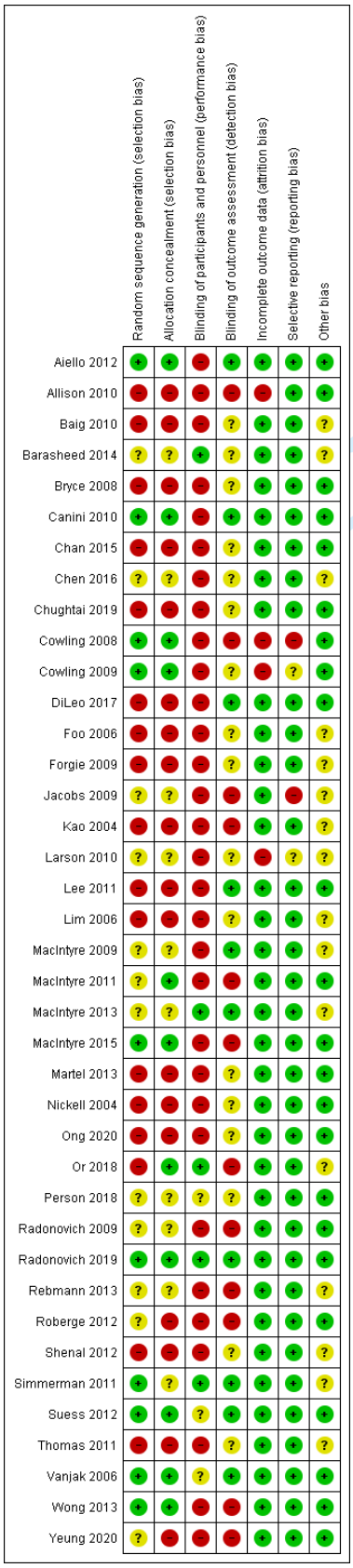
154.	Ong, J. J. Y., et al. (2020). "Headaches Associated With Personal Protective Equipment – A Cross-Sectional Study Among Frontline Healthcare Workers During COVID-19." Headache 60(5): 864-877.	Duplicate
155.	Phin NF, Rylands AJ, Allan J, Edwards C, Enstone JE, Nguyen-Van-Tam JS. Personal protective equipment in an influenza pandemic: a UK simulation exercise. Journal of Hospital Infection. 2009;71(1):15-21.	Study design
156.	Radonovich Jr, L. J., et al. (2010). "Diminished speech intelligibility associated with certain types of respirators worn by healthcare workers." Journal of Occupational and Environmental Hygiene 7(1): 63-70.	Intervention
157.	Radonovich, L. J., et al. (2019). "A tolerability assessment of new respiratory protective devices developed for health care personnel: A randomized simulated clinical study." PLoS ONE 14(1).	No extractable data
158.	Roberge, R., et al. (2012). "Thermal burden of N95 filtering facepiece respirators." Annals of Occupational Hygiene 56(7): 808-814.	Reported outcomes
159.	Roberge, R. J. (2008). "Effect of surgical masks worn concurrently over N95 filtering facepiece respirators: Extended service life versus increased user burden." Journal of Public Health Management and Practice 14(2): E19-E26.	Study type
160.	Roberge RJ, Coca A, Williams WJ, Powell JB, Palmiero AJ. Physiological impact of the n95 filtering facepiece respirator on healthcare workers. Respiratory Care. 2010;55(5):569-577.	Intervention
161.	Roberge, R. J., et al. (2010). "Effect of exhaled moisture on breathing resistance of N95 filtering facepiece respirators." Annals of Occupational Hygiene 54(6): 671-677.	Study type
162.	Roberge, R. J., et al. (2012). "N95 filtering facepiece respirator deadspace temperature and humidity." Journal of Occupational and Environmental Hygiene 9(3): 166-171.	Reported outcomes
163.	Roberge, R. J., et al. (2013). "Impact of low filter resistances on subjective and physiological responses to filtering facepiece respirators." PLoS ONE 8(12).	Intervention
164.	Roberge RJ, Kim JH, Powell JB. N95 respirator use during advanced pregnancy. American Journal of Infection Control. 2014;42(10):1097-1100.	No extractable data
165.	Sakaguchi, H., et al. (2010). "Maintenance of influenza virus infectivity on the surfaces of personal protective equipment and clothing used in healthcare settings." Environmental Health and Preventive Medicine	Purpose of study

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	15(6): 344-349.	
166.	Shi, J., et al. (2017). "Cardiovascular benefits of wearing particulate-filtering respirators: A randomized crossover trial." Environmental Health Perspectives 125(2): 175-180.	Purpose of study
167.	Shigayeva, A., et al. (2007). "Factors associated with critical-care healthcare workers' adherence to recommended barrier precautions during the Toronto severe acute respiratory syndrome outbreak." Infection Control and Hospital Epidemiology 28(11): 1275-1283.	No extractable data
168.	Skaria, S. D. and G. C. Smaldone (2014). "Respiratory source Control using surgical masks with nanofiber media." Annals of Occupational Hygiene 58(6): 771-781.	Purpose of study
169.	Tamura, E., et al. (2013). "Effect of pharmacists wearing a mask on patient consultation intention." Yakugaku Zasshi 133(6): 737-745.	Study type
170.	Tong, P. S. Y., et al. (2015). "Respiratory consequences of N95-type Mask usage in pregnant healthcare workers-A controlled clinical study." Antimicrobial Resistance and Infection Control 4(1).	Intervention
171.	Turnberg, W., et al. (2009). "Personal healthcare worker (HCW) and work-site characteristics that affect HCWs' use of respiratory-infection control measures in ambulatory healthcare settings." Infection Control and Hospital Epidemiology 30(1): 47-52.	Reported outcomes
172.	Wada, K., et al. (2012). "Wearing face masks in public during the influenza season may reflect other positive hygiene practices in Japan." BMC Public Health 12(1).	Reported outcomes
173.	Wu, S., et al. (2011). "Anxiety during respirator use: Comparison of two respirator types." Journal of Occupational and Environmental Hygiene 8(3): 123-128.	Study type
174.	Yip, W. L., et al. (2005). "The effect of wearing a face mask on body temperature." Hong Kong Journal of Emergency Medicine 12(1): 23-27.	Reported outcomes

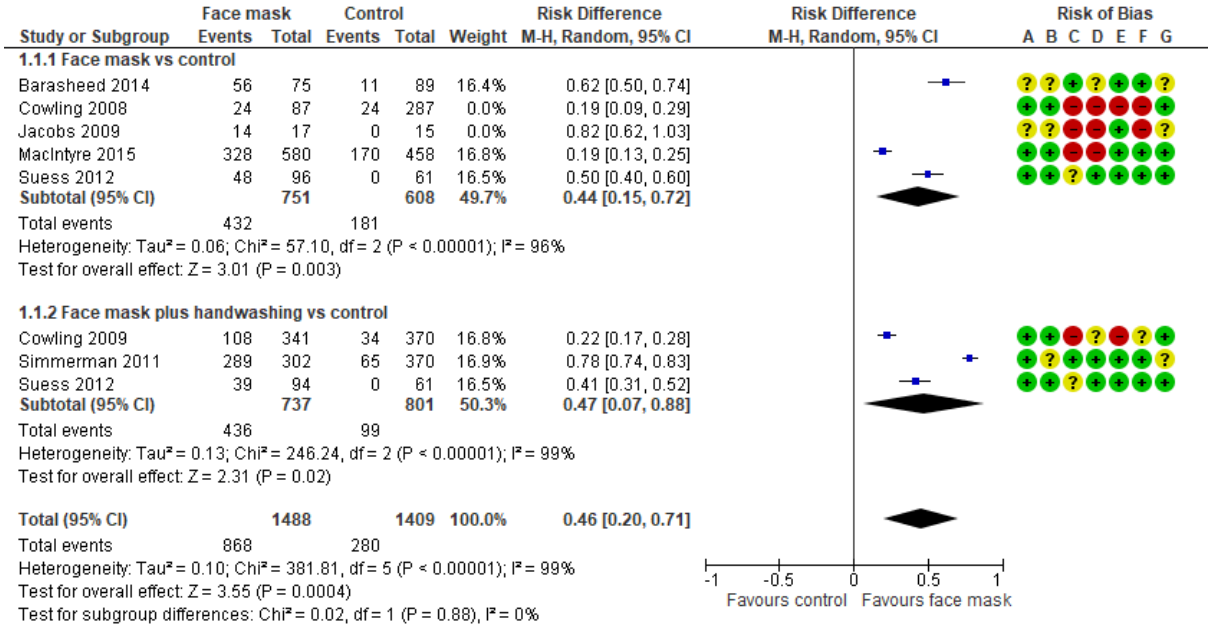
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Appendix 5. Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

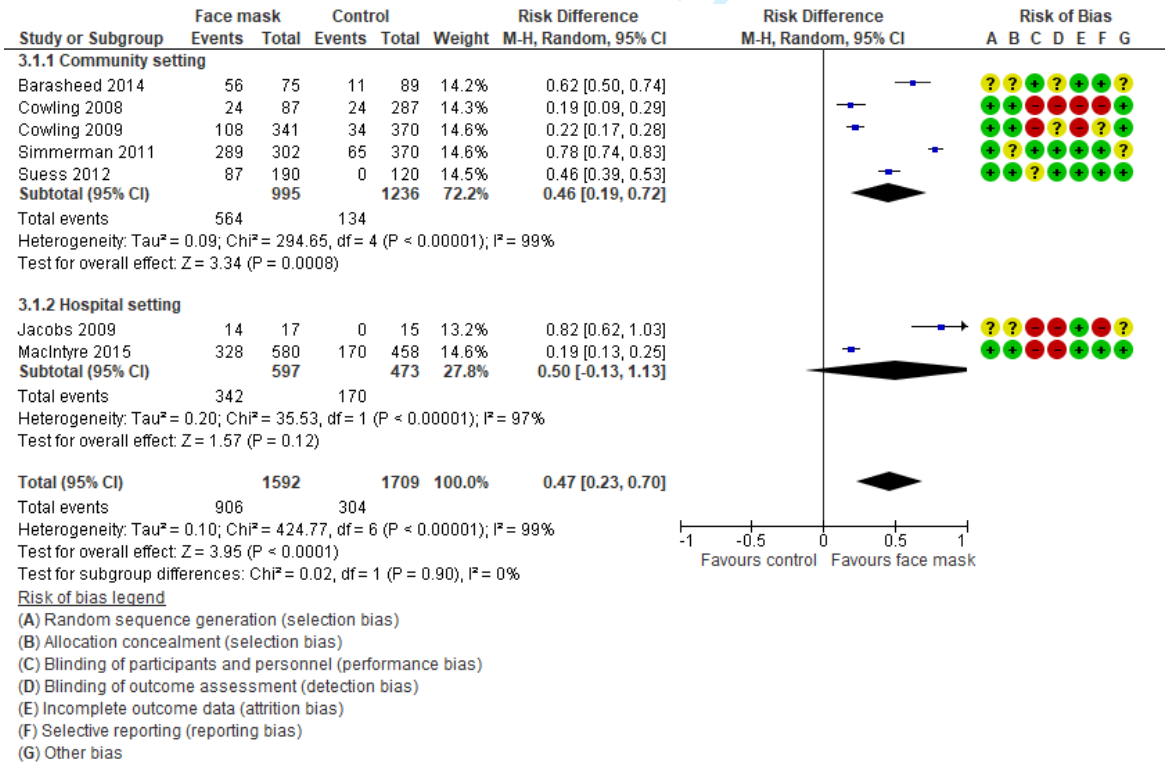


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Appendix 5. Figure 4. Analysis with studies rated at high risk of bias in more than 2 domains excluded



Appendix 5. Figure 5. Analysis with studies sub-grouped by setting (community setting, hospital setting)



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5, 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-17
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11, 12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11, 12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19

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